NEKHOROSHEV STABILITY FOR QUASI-CONVEX HAMILTONIANS: BLUEPRINT (TRACK N)

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Abstract. A proof plan, step by step, for the three frozen statements of track N of the KAM formalization: the exponential stability estimate for strictly quasi-convex nearly integrable Hamiltonians with a complex Hessian bound (N-HEAD-C, after Pöschel 1993, Theorem 1), the exact integrable dynamics for zero perturbation (N-ZERO) and the nonsingular isoenergetic derivative (N-QC). Every result is labelled and cites its pinned source. This version contains Part 0 only; the plan of the remaining parts is in `PLAN.md`.

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1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the three frozen theorems of track N (owner approval D20, 27 October 2026) and the eleven frozen definitions of `KAM/Frozen/Nekhoroshev/Defs/` together with the five track-K definitions they use. The Lean declarations are authoritative; the statements below are transcriptions of them, with the Lean names recorded under each. N-HEAD with real bounds only is *not* frozen (source gap F-N1, D17/D18/D20); the frozen headline is N-HEAD-C. The additions of D22 (N-PARTIAL-CONVEX, N-AVERAGING) are not yet frozen and are not part of this part. Results of track K's blueprint are cited as [K:`bp:p1:...`].

1.1. Conventions.

- $n \in \mathbb{N}$ is the number of degrees of freedom; N-HEAD-C assumes $n \geq 3$ (D17), N-ZERO and N-QC are stated for every n. A phase point is $z = (\theta, I)$ with angles θ and actions I, in this order.
- Ambient spaces are those of the Lean text: $\mathbb{R}^n = (\texttt{Fin}\ n \to \mathbb{R})$, $\mathbb{C}^n = (\texttt{Fin}\ n \to \mathbb{C})$, with the max norms $|v| = \max_i |v_i|$ on both. The embedding $\mathbb{R}^n \subset \mathbb{C}^n$ is componentwise. The norm $\|\cdot\|$ in every frozen statement is the max norm; on the phase space $\mathbb{R}^n \times \mathbb{R}^n$ it is $\|(\theta, I)\| = \max(|\theta|, |I|)$ (Lean product norm). The only Euclidean structure is the dot product $\sum_i \omega_i v_i$ used for orthogonality in quasi-convexity; Euclidean norms used later for comparison with the sources come with explicit inequalities ($|v| \leq \|v\|_2 \leq \sqrt{n}|v|$). For $n = 0$ all these spaces are $\{0\}$ and the norm is 0.
- Operator norms of linear and multilinear maps are the induced norms of these max norms. In particular $\|D^2 h(z)\|$ for complex h is the norm of $D^2 h(z) \in \mathcal{L}(\mathbb{C}^n, \mathcal{L}(\mathbb{C}^n, \mathbb{C}))$ (Lean `fderiv` applied twice), i.e. $\sup_{|u|,|v| \leq 1} |D^2 h(z)[u, v]|$.
- Angles are *not* reduced modulo $\mathbb{Z}^n$ in the statements: $\theta \in \mathbb{R}^n$ or $\theta \in T_s \subset \mathbb{C}^n$, and 1-periodicity is a hypothesis on f (Definition 1.12). So the torus is $\mathbb{R}^n/\mathbb{Z}^n$ and the Fourier characters are $e^{2\pi i k \cdot \theta}$ as in track K ([K:`bp:p1:decay`], [K:`bp:p1:summation`]). Pöschel's torus is $\mathbb{R}^n/2\pi\mathbb{Z}^n$ and his norms are Euclidean; the conversion is a separate step of the proof (Part 1 of `PLAN.md`).
- The complex strip is $T_s = \{\theta \in \mathbb{C}^n : |\operatorname{Im} \theta_i| < s\ \forall i\}$ (Lean `KAM.complexStrip`), as in track K.

1.2. Track-K definitions used.

Definition 1.1 (Complex action neighbourhood)**.** For $B \subset \mathbb{R}^n$ and $r \in \mathbb{R}$, $B_r = \{I \in \mathbb{C}^n : \exists J \in B,\ |I - J| < r\}$ (max norm, J viewed in $\mathbb{C}^n$). Thus $B_r = \emptyset$ for $r \leq 0$, and for B open B_r is open.
Frozen Lean declaration: `KAM.actionNbhd`
Source: BM11 p. 4 (D_s); track K, [K:`bp:p0:domains`].

Definition 1.2 (Frequency map)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and $I \in \mathbb{R}^n$, $\omega_h(I) = \nabla h(I) \in \mathbb{R}^n$ is the vector with $\omega_h(I)_i = D\big(J \mapsto \operatorname{Re} h(J)\big)(I)[e_i]$, the real partial derivatives of $\operatorname{Re} h$ on $\mathbb{R}^n$ at I. It is 0 if $\operatorname{Re} h|_{\mathbb{R}^n}$ is not differentiable at I (Lean totality, Remark 1.18).
Frozen Lean declaration: `KAM.frequencyMap`
Source: BM11 p. 1; Pos93 p. 190 ($\omega(I) = \partial_I h(I)$).

Definition 1.3 (Hamiltonian vector field)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ and $z = (\theta, I)$, $X_H(z) = \big((\partial_{I_i} H(z))_i, (-\partial_{\theta_i} H(z))_i\big)$, where $\partial_{I_i} H(z) = DH(z)[(0, e_i)]$ and $\partial_{\theta_i} H(z) = DH(z)[(e_i, 0)]$ are the Fréchet derivatives in the Lean sense (0 if H is not differentiable at z).
Frozen Lean declaration: `KAM.hamiltonianVectorField`
Source: BM11 p. 1; track K, [K:`bp:p0:hamiltonian`].

Definition 1.4 (Real Hamiltonian)**.** For $h : \mathbb{C}^n \to \mathbb{C}$, $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ and real $z = (\theta, I)$, $H_{h,f}(z) = \operatorname{Re}\big(h(I) + f(\theta, I)\big)$, with θ, I viewed in $\mathbb{C}^n$.
Frozen Lean declaration: `KAM.realHamiltonian`
Source: BM11 p. 5 ($H = h + f$).

Remark 1.5 (Strip and neighbourhood notation). `KAM.complexStrip` $n\, s = T_s$ as above; for $n > 0$ it is empty when $s \leq 0$, and for $n = 0$ it is the whole (one-point) space $\mathbb{C}^0$ for every s. No hypothesis $n > 0$ is added to N-ZERO or N-QC.
Frozen Lean declaration: `KAM.complexStrip`

1.3. Frozen track-N definitions.

Definition 1.6 (Action ball)**.** $B(0, R) = \{I \in \mathbb{R}^n : |I| < R\}$, the open ball of the max norm; written `actionBall` $n\, R$.

Frozen Lean declaration: `KAM.Nekhoroshev.actionBall`
Source: BM11 p. 4.

Definition 1.7 (Complex domain)**.** $D_s(R) = T_s \times B(0,R)_s \subset \mathbb{C}^n \times \mathbb{C}^n$; `complexDomain` $n\,R\,s$. It is the period-one lift of BM11's D_s, with a single width s for angles and actions. It is open.
Frozen Lean declaration: `KAM.Nekhoroshev.complexDomain`
Source: BM11 p. 4 (D_s); ours for the lift.

Definition 1.8 (Real action domain)**.** $\mathcal{B}_s(R) = \{I \in \mathbb{R}^n : I \in B(0,R)_s\}$; `realActionDomain` $n\,R\,s$.
Frozen Lean declaration: `KAM.Nekhoroshev.realActionDomain`
Source: BM11 p. 4 (B_s).

Remark 1.9 (Computing $\mathcal{B}_s(R)$). Since $|I-J|$ for real I, J is the real max norm, $\mathcal{B}_s(R) = \{I \in \mathbb{R}^n : \exists J,\ |J| < R,\ |I-J| < s\} = B(0, R+s)$ for $R, s > 0$. This identity is ours and is proved from the definitions (it is not stated in the source); it shows that $B(0,R) \subset \mathcal{B}_s(R)$ and that $\mathcal{B}_s(R)$ is open.

Definition 1.10 (Real part of the integrable part)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and $I \in \mathbb{R}^n$, $h_{\mathbb{R}}(I) = \operatorname{Re} h(I)$; `realPart` h.
Frozen Lean declaration: `KAM.Nekhoroshev.realPart`
Source: BM11 p. 4 ($h : B_s \to \mathbb{R}$).

Definition 1.11 (Analytic integrable part, $h \in A_s(D)$)**.** `IsAnalyticIntegrable` $R\,s\,h$ holds if

(i) h is complex differentiable on $B(0,R)_s$;
(ii) h is continuous on $\overline{B(0,R)_s}$ (closure in $\mathbb{C}^n$);
(iii) there is $A \geq 0$ with $|h(I)| \leq A$ for all $I \in B(0,R)_s$;
(iv) $\operatorname{Im} h(I) = 0$ for all $I \in \mathcal{B}_s(R)$;
(v) $h_{\mathbb{R}}$ is of class C^3 on $\mathcal{B}_s(R)$ (real Fréchet sense; the set is open).

No numerical bound on h is fixed. Clause (v) is redundant by holomorphy ([K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:real-restriction`]) and records the regularity that guards every real derivative.
Frozen Lean declaration: `KAM.Nekhoroshev.IsAnalyticIntegrable`
Source: BM11 p. 4 ($A_s(D)$, integrable case, with continuity on the closure).

Definition 1.12 (Analytic perturbation of size ε)**.** `IsAnalyticPerturbation` $R\,s\,\varepsilon\,f$ holds if $\varepsilon \geq 0$ and

(i) f is complex differentiable on $D_s(R)$ and continuous on $\overline{D_s(R)}$;
(ii) $f(\theta + k, I) = f(\theta, I)$ for all $(\theta, I) \in D_s(R)$ and $k \in \mathbb{Z}^n$;
(iii) $\operatorname{Im} f(\theta, I) = 0$ for all $\theta \in \mathbb{R}^n$ and $I \in \mathcal{B}_s(R)$;
(iv) $|f(z)| \leq \varepsilon$ for all $z \in D_s(R)$ (complex absolute value).

The size is measured by $\leq \varepsilon$, not the source's strict bound; the source is applied with 2ε. For $\varepsilon = 0$ this forces $f = 0$ on $D_s(R)$.
Frozen Lean declaration: `KAM.Nekhoroshev.IsAnalyticPerturbation`
Source: BM11 pp. 4–5 ($f \in A_s(D)$, $|f|_s < \varepsilon$); deviation: $\leq \varepsilon$ (ruling Q2, D18).

Definition 1.13 (Derivative bounds $B(M)$)**.** For $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$ and $M \in \mathbb{R}$, `HasDerivativeBounds` $U\,h\,M$ holds if $M > 0$, U is open, $h_{\mathbb{R}} \in C^3(U)$, and for all $I \in U$, all $k \in \{1,2,3\}$ and all $(i_1, \dots, i_k) \in \{1, \dots, n\}^k$,

$$\left|D^k h_{\mathbb{R}}(I)[e_{i_1}, \dots, e_{i_k}]\right| \leq M.$$

These are all ordered coordinate partials of orders one to three; for C^3 functions they coincide with the multi-index partials $\partial^k h$ of the source (symmetry, [K:`bp:p1:ext-schwarz`]). The bound is on coordinate partials, not on operator norms.
Frozen Lean declaration: `KAM.Nekhoroshev.HasDerivativeBounds`
Source: BM11 p. 4, $B(M)$.

Definition 1.14 (Strict quasi-convexity $QC(m)$)**.** For $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$, $m \in \mathbb{R}$, `IsStrictlyQuasiConvex` $U\, h\, m$ holds if $m > 0$, U is open, $h_{\mathbb{R}} \in C^2(U)$, and for every $I \in U$: $\omega_h(I) \neq 0$ and

$$\sum_i \omega_h(I)_i v_i = 0 \implies m|v|^2 \le D^2 h_{\mathbb{R}}(I)[v,v] \qquad (v \in \mathbb{R}^n),$$

with $|v|$ the max norm on the left and the Euclidean dot product in the hypothesis.
Frozen Lean declaration: `KAM.Nekhoroshev.IsStrictlyQuasiConvex`
Source: BM11 p. 4, $QC(m)$ (the source's norm on the right is the max norm, orthogonality is Euclidean).

Definition 1.15 (Complex Hessian bound)**.** `HasComplexHessianBound` $R\, s\, h\, M_{\mathbb{C}}$ holds if $M_{\mathbb{C}} > 0$, h is C^2 in the complex sense on $B(0,R)_s$ and $\|D^2 h(z)\| \le M_{\mathbb{C}}$ for all $z \in B(0,R)_s$, the norm being that of $D(Dh)(z) \in \mathcal{L}(\mathbb{C}^n, \mathcal{L}(\mathbb{C}^n, \mathbb{C}))$ for the max norm. Every entry $|\partial_i \partial_j h|$ is then $\le M_{\mathbb{C}}$, and the Euclidean operator norm of the Hessian matrix is $\le n M_{\mathbb{C}}$. This is not the spectral radius of a complex symmetric matrix.
Frozen Lean declaration: `KAM.Nekhoroshev.HasComplexHessianBound`
Source: Pos93 p. 190, "The Hessian of the integrable hamiltonian h" (conversion by n recorded in `docs/tracks/N/sources.md`, anchor POS190-HESS).

Definition 1.16 (Solutions on a time set)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $U \subset \mathbb{R}^n$, $J \subset \mathbb{R}$ and $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$, `IsSolutionOn` $H\, U\, J\, z$ holds if for every $t \in J$: $z(t)_2 \in U$, H is differentiable at $z(t)$, and z is differentiable at t (two-sided, also at endpoints of J, since z is defined on $\mathbb{R}$) with $z'(t) = X_H(z(t))$.
Frozen Lean declaration: `KAM.Nekhoroshev.IsSolutionOn`
Source: BM11 Theorem 2.1 p. 5 and the maximal-solution argument p. 13.

Definition 1.17 (The integrable orbit)**.** For $h : \mathbb{C}^n \to \mathbb{C}$, $\theta_0, I_0 \in \mathbb{R}^n$ and $t \in \mathbb{R}$: $\gamma_{\theta_0, I_0}(t) = (\theta_0 + t\,\omega_h(I_0),\ I_0)$.
Frozen Lean declaration: `KAM.Nekhoroshev.integrableOrbit`
Source: BM11 p. 1 ($H(\theta, I) = h(I)$).

Remark 1.18 (Lean totality). The frozen definitions and theorems use `fderiv` (which is 0 at a point of non-differentiability), `iteratedFDeriv`, the operator norm, `rpow` ($\varepsilon^{\pm 1/(2n)}$), `Real.exp` and the max norm; there is no matrix inverse, no Bochner integral and no division by a quantity that may vanish. The guards are:

(a) In N-HEAD-C and N-ZERO, ω_h and $D^2 h_{\mathbb{R}}$ are used only on the open set $\mathcal{B}_s(R)$ where $h_{\mathbb{R}}$ is C^3 (Definitions 1.11(v), 1.13, 1.14), so the derivatives are the genuine ones. N-QC uses them on its arbitrary open U, where its hypothesis gives C^2 (item (e)). The definitions `frequencyMap`, `integrableOrbit` themselves are total and keep Lean's default values outside the differentiability locus.
(b) $D^2 h$ for complex h is used only on $B(0,R)_s$, open, where h is C^2 (Definition 1.15).
(c) X_H is used only inside `IsSolutionOn`, which requires differentiability of H at every visited point; in N-ZERO and N-HEAD-C this is a proof obligation: $H_{h,f}$ is real-differentiable on $\mathbb{R}^n \times \mathcal{B}_s(R) \supseteq \mathbb{R}^n \times B(0,R)$ (angles first, actions restricted) because h, f are holomorphic on open sets containing the real points ([K:`bp:p1:hamiltonian-real`]).
(d) $\varepsilon^{-1/(2n)}$ and $\varepsilon^{1/(2n)}$ occur only with $\varepsilon > 0$ in N-HEAD-C, and $2n \neq 0$.
(e) In N-QC the map $\Psi(I', \lambda) = (h_{\mathbb{R}}(I'), \lambda \omega_h(I'))$ is differentiable at (I, λ) because $h_{\mathbb{R}} \in C^2$ on the open set $U \ni I$; `fderiv` is then the true derivative.

1.4. The frozen theorems.

Theorem 1.19 (N-HEAD-C: exponential stability, complex Hessian bound)**.** *Let $n \ge 3$ and $R, s, M, m, M_{\mathbb{C}}, \omega_{\min} > 0$. Then there are $C, T, c, \varepsilon_0 > 0$ with*

$$C\, \varepsilon_0^{1/(2n)} \le R/4$$

such that the following holds for every $h : \mathbb{C}^n \to \mathbb{C}$, $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ *and* $\varepsilon \in \mathbb{R}$ *with*

(H1) `IsAnalyticIntegrable` $R\, s\, h$ *(Definition 1.11);*
(H2) `HasComplexHessianBound` $R\, s\, h\, M_{\mathbb{C}}$ *(Definition 1.15);*
(H3) $\omega_{\min} \le |\omega_h(I)|$ *for every* $I \in B(0,R)$ *(max norm);*
(H4) `IsStrictlyQuasiConvex` $\mathcal{B}_s(R)\, h\, m$ *(Definition 1.14);*
(H5) `HasDerivativeBounds` $\mathcal{B}_s(R)\, h\, M$ *(Definition 1.13);*
(H6) `IsAnalyticPerturbation` $R\, s\, \varepsilon\, f$ *(Definition 1.12);*
(H7) $0 < \varepsilon \le \varepsilon_0$.

For every $\theta_0 \in \mathbb{R}^n$ *and* $I_0 \in B(0, R/2)$, *with* $\tau := T\exp\bigl(c\,\varepsilon^{-1/(2n)}\bigr)$ *and* $H = H_{h,f}$, *there is* $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$ *with*

(a) $z(0) = (\theta_0, I_0)$;
(b) `IsSolutionOn` $H\, B(0,R)\, [-\tau, \tau]\, z$;
(c) $|z(t)_2 - I_0| \le C\varepsilon^{1/(2n)}$ *for all* t *with* $|t| \le \tau$;
(d) (uniqueness) for every $L \in [0, \tau]$ *and every* $y : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$ *with* $y(0) = (\theta_0, I_0)$ *and* `IsSolutionOn` $H\, B(0,R)\, [-L, L]\, y$: $y(t) = z(t)$ *for all* $t \in [-L, L]$.

The order of choice is: n; $R, s, M, m, M_{\mathbb{C}}, \omega_{\min}$; *then* C, T, c, ε_0, *depending only on these seven data; then* h, f, ε, *then* θ_0, I_0; *the solution* z *may depend on all of them. There is no dependence on* A *in Definition 1.11(iii). In (H4)–(H5) the set is* $\mathcal{B}_s(R) = B(0, R+s)$ *(Remark 1.9), in (H3) the smaller ball* $B(0,R)$.

Frozen Lean declaration: `KAM.Nekhoroshev.quasiConvex_nekhoroshev_complexBound`

Source: Pos93 Theorem 1, pp. 190–191 (with the conversions and anchors of `docs/tracks/N/sources.md`*); exponent* $1/(2n)$ *as BM11 p. 5 with* $\delta = (2n(n-1))^{-1}$; *ruling D18 Q2, Q4; deviation D20: the extra inputs* $M_{\mathbb{C}}$ *and* $\omega_{\min}$.

Source gap: The printed theorem is for a 2π*-periodic torus, Euclidean norms and exceptional initial frequencies* $\|\omega(I_0)\| \le mr_0/8$; *the conversion to the frozen statement (rescaling* $H \mapsto 2\pi H(\varphi/2\pi, I)$, *choice of* r_0, s_0 *from the seven data, exclusion of the exceptional branch by* $\omega_{\min}$, *existence and uniqueness of the solution) is given only in* `sources.md` *and is proved in Part 10 of* `PLAN.md`.

Theorem 1.20 (N-ZERO: exact integrable dynamics)**.** *Let* $n \in \mathbb{N}$, $R, s > 0$, h *with* `IsAnalyticIntegrable` $R\, s\, h$ *and* f *with* `IsAnalyticPerturbation` $R\, s\, 0\, f$. *For every* $\theta_0 \in \mathbb{R}^n$ *and* $I_0 \in B(0, R/2)$, *with* $\gamma = \gamma_{\theta_0, I_0}$ *(Definition 1.17):*

$$\gamma(0) = (\theta_0, I_0), \qquad \texttt{IsSolutionOn}\, H_{h,f}\, B(0,R)\, \mathbb{R}\, \gamma, \qquad \gamma(t)_2 = I_0 \quad (t \in \mathbb{R}).$$

Frozen Lean declaration: `KAM.Nekhoroshev.zero_perturbation`

Source: Ours (supplement to BM11 Theorem 2.1, p. 5, whose strict bound excludes $\varepsilon = 0$*); from* $H = h(I)$, *BM11 p. 1.*

Theorem 1.21 (N-QC: nonsingular isoenergetic derivative)**.** *Let* $n \in \mathbb{N}$, $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$, $m \in \mathbb{R}$ *with* `IsStrictlyQuasiConvex` $U\, h\, m$. *For* $I \in U$ *and* $\lambda > 0$, *the derivative at* (I, λ) *of*

$$\Psi : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}^n, \qquad \Psi(I', \lambda') = \bigl(h_{\mathbb{R}}(I'),\ \lambda'\omega_h(I')\bigr)$$

is a bijective linear map $\mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}^n$. *Explicitly it is* $D\Psi(I,\lambda)[v,u] = \bigl(\omega_h(I)\cdot v,\ u\,\omega_h(I) + \lambda D^2 h_{\mathbb{R}}(I)v\bigr)$. *No uniform radius of invertibility is asserted.*

Frozen Lean declaration: `KAM.Nekhoroshev.quasiConvex_isoenergetic_derivative`

Source: BM11 Lemma 3.3 and its proof, pp. 9–10 (only the derivative assertion).

2. Interfaces and conversion of hypotheses

This part connects the frozen hypotheses of track N (Definitions of Part 0) to the form in which the proofs use them. It proves: the real action domain and the inclusions of tubes (§2.2); that holomorphy gives a smooth real Hamiltonian with $X_H = X_{h+f}$ (§2.3); the translation of the derivative bounds $B(M)$, the complex Hessian bound and $QC(m)$ into the Euclidean, uniform form

of Pöschel's Theorem 1 (§2.4); the change of normalisation 2π-periodic$\leftrightarrow$ 1-periodic and the sup-to-weighted Fourier norm (§2.5); and the dictionary of constants (§2.6). *Convention ruling.* All of Parts 1–10 work in the frozen conventions (period 1, max norms, characters $e^{2\pi i k\cdot\theta}$). Pöschel's normalisation (period 2π, Euclidean norms on actions, weights $e^{|k|_1 s}$) enters only through the labelled conversions Lemmas 2.8, 2.9, 2.7, 2.10; quantities in his normalisation carry a tilde or the subscript P. This overrules the suggestion in `PLAN.md` to work in Pöschel's normalisation. The Euclidean norm is written $|\cdot|_2$, the max norm $|\cdot|$ (or $|\cdot|_\infty$), and $|k|_1 = \sum_i |k_i|$; the phase-space norm is $\max(|\theta|_\infty, |I|_\infty)$, Euclidean structure is used only for the dot product and where stated.

2.1. **External facts.**

(E1) *Finite-dimensional linear algebra.* An injective linear map $L : V \to W$ between finite-dimensional real vector spaces with $\dim V = \dim W$ is bijective (rank–nullity; Mathlib `LinearMap.injective_iff_surjective_of_finrank_eq_finrank`, in `LinearAlgebra/FiniteDimensional/Lemmas.lean`; the endomorphism form `LinearMap.injective_iff_surjective` does not apply directly to $V \neq W$). The finranks used below are $\dim(\mathbb{R}^n \times \mathbb{R}) = \dim(\mathbb{R} \times \mathbb{R}^n) = n + 1$ (`Module.finrank_prod`, `Module.finrank_fin_fun`). Exists.

(E2) *Norm comparison and Cauchy–Schwarz.* For $x \in \mathbb{R}^n$ or $\mathbb{C}^n$: $|x|_\infty \le |x|_2 \le \sqrt{n}|x|_\infty$ and $|x|_1 \le \sqrt{n}|x|_2$, $|x|_2 \le |x|_1$; $|x \cdot y| \le |x|_2|y|_2$ (for the bilinear, not Hermitian, product too). Exists.

(E3) *Restriction of scalars and chain rule.* A $\mathbb{C}$-differentiable (resp. C^k) map is $\mathbb{R}$-differentiable (resp. C^k) with the same derivative; the chain rule for the $\mathbb{R}$-linear inclusion $\iota : \mathbb{R}^N \to \mathbb{C}^N$ and for Re. (Mathlib `DifferentiableAt.restrictScalars`, `ContDiffOn.restrict_scalars`, `HasFDerivAt.comp`.) Exists.

(E4) *Product rule* for $(x, \lambda) \mapsto \lambda\, v(x)$ with v differentiable, and differentiability of a vector function coordinatewise. Exists.

Track-K results used: [K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:real-restriction`], [K:`bp:p1:hamiltonian-real`], [K:`bp:p1:ext-schwarz`], [K:`bp:p1:decay`], [K:`bp:p1:summation`], [K:`bp:p1:exp-sums`].

2.2. **Domains and norms.**

Lemma 2.1 (Norm and lattice comparisons)**.** *(a) For $x \in \mathbb{R}^n$ (or $\mathbb{C}^n$), every $n \ge 0$ (all norms are 0 when $n = 0$): $|x|_\infty \le |x|_2 \le \sqrt{n}\,|x|_\infty$, i.e. $|x|_2^2 \le n|x|_\infty^2$; for $n > 0$ this gives $|x|_\infty^2 \ge |x|_2^2/n$. (b) For $k \in \mathbb{Z}^n$: $|k|_2 \le |k|_1 \le \sqrt{n}|k|_2$. Consequently $\{|k|_1 \le K\} \subset \{|k|_2 \le K\}$ and $\{|k|_2 \le K\} \subset \{|k|_1 \le \sqrt{n}K\}$; so a geometric cutoff K_g in the Euclidean norm and an analytic cutoff K_a in $|\cdot|_1$ are compatible when $\sqrt{n}\,K_g \le K_a$. (c) The bilinear bound: if $|Q_{ij}| \le M_0$ for a matrix Q then $|x^\top Q y| \le n M_0 |x|_2 |y|_2$ for $x, y \in \mathbb{C}^n$.*
Source: Standard ((E2)); the lattice inclusions correct the direction stated in `PLAN.md` (review N06).

Proof. **Step S2.1: Vectors.** (a),(b) are (E2): $|x|_2^2 = \sum |x_i|^2 \le n \max |x_i|^2$, $|x|_1 \le \sqrt{n}|x|_2$ is Cauchy–Schwarz with $(1, \dots, 1)$.

Step S2.2: Matrices. $|x^\top Q y| \le M_0 |x|_1 |y|_1 \le M_0\, n |x|_2 |y|_2$ by (b). □

Lemma 2.2 (Domains)**.** *Let $R, s > 0$, $n \ge 0$.*

(a) $\mathcal{B}_s(R) = B(0, R + s)$; in particular it is open and $B(0, R) \subset \mathcal{B}_s(R)$ (Remark 1.9); $W_s := \mathbb{R}^n \times \mathcal{B}_s(R)$ is open in $\mathbb{R}^n \times \mathbb{R}^n$ and $D_s(R) \cap (\mathbb{R}^n \times \mathbb{R}^n) = W_s$ (real points of $D_s(R)$: $\theta \in \mathbb{R}^n \subset T_s$ since $s > 0$).

(b) $B(0, R)_s \cap \mathbb{R}^n = \mathcal{B}_s(R)$; $D_s(R)$ and $B(0, R)_s$ are open.

(c) *(Euclidean tubes) Let* $P = \{J \in \mathbb{R}^n : |J| \le R/2\}$ *and* $0 < r_0 \le \min(s/2, R/8)$. *If* $I \in \mathbb{C}^n$ *and* $J \in P$ *with* $|I - J|_2 < r_0$, *then* $I \in B(0,R)_s$ *and* $|I - J|_\infty < s$. *If moreover* $I \in \mathbb{R}^n$ *then* $|I|_\infty < 5R/8 \le 3R/4$.

(d) $T_\rho \subset T_s$ *for* $\rho \le s$; $T_s = \mathbb{C}^0$ *is the universal singleton for* $n = 0$ *(no* $n > 0$ *hypothesis is needed anywhere in N-ZERO, N-QC).*

Source: Ours (Remark 1.9); BM11 p. 4 for the notation.

Proof. **Step S2.3: Real tube.** ($\subset$) $|I| \le |J| + |I - J| < R + s$. ($\supset$) If $|I| < R + s$ put $J = \frac{R}{R+s} I$; then $|J| < R$ and $|I - J| = \frac{s}{R+s}|I| < s$ (for $I = 0$ take $J = 0$).

Step S2.4: Openness and real points. $B(0,R)_s = \bigcup_{J \in B(0,R)} \{|I - J| < s\}$ is a union of open balls; T_s is open as a preimage of an open set; products of open sets are open; the rest is the definition of ι and $\mathcal{B}_s(R) = B(0,R)_s \cap \mathbb{R}^n$ by Step S2.3.

Step S2.5: Euclidean tubes. By (E2) $|I - J|_\infty \le |I - J|_2 < r_0 \le s/2 < s$, and $|J| < R$ since $J \in P$; so $I \in B(0,R)_s$. For real I: $|I|_\infty < R/2 + R/8$. □

2.3. **Holomorphy implies real smoothness.** Throughout this subsection $R, s > 0$, h satisfies `IsAnalyticIntegrable` $R\, s\, h$, f satisfies `IsAnalyticPerturbation` $R\, s\, \varepsilon\, f$ ($\varepsilon \ge 0$), $G(\theta, I) = h(I) + f(\theta, I)$ on $D_s(R)$, $H = H_{h,f}$ and $\iota : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{C}^n \times \mathbb{C}^n$ the componentwise inclusion. Complex partials are $\partial_{\theta_i} G(w) = DG(w)[(e_i, 0)]$, $\partial_{I_i} G(w) = DG(w)[(0, e_i)]$.

Lemma 2.3 (Real smoothness and real Hamiltonian field)**.** (a) *G is holomorphic and C^∞ on $D_s(R)$; h is holomorphic and C^∞ on $B(0,R)_s$.*

(b) *H is C^∞ on the open set $W_s = \mathbb{R}^n \times B(0, R+s)$ (Lemma 2.2); on W_s it is real and $H = G \circ \iota$, and for all orders $D^k H(z)[\zeta_1, \dots, \zeta_k] = D^k G(\iota z)[\iota\zeta_1, \dots, \iota\zeta_k] \in \mathbb{R}$. Likewise $h_\mathbb{R} = h \circ \iota$ is C^∞ on $\mathcal{B}_s(R)$, with $D^k h_\mathbb{R}(I)[a_1, \dots, a_k] = D^k h(I)[\iota a_1, \dots]$.*

(c) *(Hamiltonian field) For $z = (\theta, I) \in W_s$: H is differentiable at z and $X_H(z) = \big((\partial_{I_i} G(\iota z))_i, (-\partial_{\theta_i} G(\iota z))_i\big) = X_{h+f}(\iota z)$, the complex Hamiltonian field of G at the real point ιz. The frequency map is $\omega_h(I)_i = \partial_i h(I) \in \mathbb{R}$ and $D^2 h_\mathbb{R}(I)[e_i, e_j] = \partial_i \partial_j h(I) = \partial_j \partial_i h(I) \in \mathbb{R}$ for $I \in \mathcal{B}_s(R)$.*

(d) *(Periodicity) $H(\theta + k, I) = H(\theta, I)$ for $(\theta, I) \in W_s$, $k \in \mathbb{Z}^n$; consequently DH, $D^k H$ and X_H are 1-periodic in θ on W_s.*

(e) *If $\varepsilon = 0$ then $f = 0$ on $D_s(R)$ and $H(\theta, I) = h_\mathbb{R}(I)$ on W_s.*

Source: BM11 p. 4 (regularity of h on B_s); [K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:hamiltonian-real`]*; ours for the packaging. Part 3 uses this as* [N:`bpN:p3:holo-input`].

Proof. **Step S2.6: Holomorphy.** h is complex differentiable on the open set $B(0,R)_s$ and f on $D_s(R)$ (Definitions 1.11(i), 1.12(i); openness Lemma 2.2(b)); the map $(\theta, I) \mapsto h(I)$ is holomorphic on $D_s(R)$ (composition with the linear projection; $B(0,R)_s$ is the second factor). So G is holomorphic on $D_s(R)$, hence C^∞ with holomorphic derivatives of all orders ([K:`bp:p1:holomorphic-regularity`]), which is (a); by (E3) G is C^∞ in the real sense.

Step S2.7: Reality. $\iota(W_s) \subset D_s(R)$ (Lemma 2.2(a)) and $W_s = \iota^{-1}(D_s(R))$ is open. For $z = (\theta, I) \in W_s$, $\operatorname{Im} h(I) = 0$ (Definition 1.11(iv), $I \in \mathcal{B}_s(R)$) and $\operatorname{Im} f(\theta, I) = 0$ (Definition 1.12(iii)); so $G(\iota z) \in \mathbb{R}$ and $H(z) = \operatorname{Re} G(\iota z) = G(\iota z)$. Likewise $h_\mathbb{R} = h \circ \iota$ on $\mathcal{B}_s(R)$.

Step S2.8: Derivatives of all orders. On the open set W_s, $H = G \circ \iota$ with ι $\mathbb{R}$-linear and G real-C^∞ (Step S2.6): chain rule (E3) gives $H \in C^\infty(W_s)$ and the displayed formula for $D^k H$, with values real because H is real-valued (the derivative of a real-valued function along real directions is a limit of real difference quotients). The same for $h_\mathbb{R}$.

Step S2.9: The field. By Definition 1.3, $X_H(z) = ((DH(z)[(0, e_i)])_i, (-DH(z)[(e_i, 0)])_i)$; by Step S2.8 these are $\partial_{I_i} G(\iota z)$ and $-\partial_{\theta_i} G(\iota z)$. This is [K:`bp:p1:hamiltonian-real`] (with $V =$

$D_s(R)$, H agreeing with $\operatorname{Re} G$ on the real neighbourhood W_s). Taking $k = 1, 2$ for $h_{\mathbb{R}}$ gives $\omega_h(I)_i = Dh_{\mathbb{R}}(I)[e_i] = \partial_i h(I)$ and $D^2 h_{\mathbb{R}}(I)[e_i, e_j] = \partial_i \partial_j h(I)$, symmetric by [K:bp:p1:holomorphic-regularity] ([K:bp:p1:ext-schwarz]).

Step S2.10: Periodicity. For $(\theta, I) \in W_s$, $k \in \mathbb{Z}^n$ and real θ: $(\theta, I), (\theta + k, I) \in D_s(R)$ and $f(\theta + k, I) = f(\theta, I)$ (Definition 1.12(ii)), $h(I)$ does not depend on θ. So $H \circ \tau_k = H$ on W_s, $\tau_k(\theta, I) = (\theta + k, I)$, which maps W_s onto itself with $D\tau_k = \mathrm{id}$; differentiating (chain rule) gives $D^j H(\tau_k z) = D^j H(z)$ and hence $X_H(\tau_k z) = X_H(z)$.

Step S2.11: Zero perturbation. $|f(w)| \le 0$ on $D_s(R)$ gives $f = 0$ there, so $H = \operatorname{Re} h(I) = h_{\mathbb{R}}(I)$ on W_s. □

Lemma 2.4 (Ordered partials and first-order bounds)**.** *Let $U \subset \mathbb{R}^n$ be open with* `HasDerivativeBounds` $U\, h\, M$ *(Definition 1.13). Then for $I \in U$, $a, b \in \mathbb{R}^n$: $|\omega_h(I)_i| \le M$, $|\omega_h(I)|_\infty \le M$, $|\omega_h(I)|_2 \le \sqrt{n}\, M$, and $|D^2 h_{\mathbb{R}}(I)[e_i, e_j]| \le M$. The ordered partial $D^k h_{\mathbb{R}}(I)[e_{i_1}, \dots, e_{i_k}]$ ($k \le 3$) is invariant under permutations of $(i_1, \dots, i_k)$, so it equals the multi-index partial of the source.*
Source: BM11 p. 4 ($B(M)$); [K:bp:p1:ext-schwarz].

Proof. **Step S2.12: First order.** $k = 1$ in the definition gives $|Dh_{\mathbb{R}}(I)[e_i]| = |\omega_h(I)_i| \le M$ (Definition 1.2); then $|\omega|_2 \le \sqrt{n}|\omega|_\infty$ (Lemma 2.1).

Step S2.13: Symmetry. $h_{\mathbb{R}} \in C^3$ on the open set U: $D^2 h_{\mathbb{R}}(I)$ and $D^3 h_{\mathbb{R}}(I)$ are symmetric ([K:bp:p1:ext-schwarz] applied to $h_{\mathbb{R}}$, $Dh_{\mathbb{R}}$, $D^2 h_{\mathbb{R}}$), which is the permutation invariance; the entry bound is $k = 2$. □

Lemma 2.5 (Complex Hessian bound in Euclidean form)**.** *Let* `HasComplexHessianBound` $R\, s\, h\, M_{\mathbb{C}}$. *For $I \in B(0, R)_s$ let $Q(I) = (\partial_i \partial_j h(I))_{ij}$ (complex symmetric). Then $|Q(I)_{ij}| \le M_{\mathbb{C}}$ and $|x^{\top} Q(I) y| \le n M_{\mathbb{C}} |x|_2 |y|_2$ for $x, y \in \mathbb{C}^n$; hence the operator norm of $Q(I) : (\mathbb{C}^n, |\cdot|_2) \to (\mathbb{C}^n, |\cdot|_2)$ is $\le n M_{\mathbb{C}}$. For real $I \in \mathcal{B}_s(R)$ this matrix is the real Hessian $D^2 h_{\mathbb{R}}(I)[e_i, e_j]$ (Lemma 2.3(c)), and for real x, y: $|D^2 h_{\mathbb{R}}(I)[x, y]| \le n M_{\mathbb{C}} |x|_2 |y|_2$.*
Source: Pos93 p. 190 (Hessian bound, anchor POS190-HESS); the conversion factor n is ours (`docs/tracks/N/sources.md`).
Source gap: Pos93 p. 190 says that for the symmetric Q the Euclidean operator norm "amounts to the spectral radius norm". For complex symmetric matrices this is false ($Q = \left(\begin{smallmatrix} 1 & i \\ i & -1 \end{smallmatrix}\right)$ has $Q^2 = 0$). The frozen definition uses the operator norm of $D^2 h$ for the max norm, and so does this blueprint; no spectral-radius claim is used.

Proof. **Step S2.14: Entries.** $D^2 h(I) = D(Dh)(I) \in \mathcal{L}(\mathbb{C}^n, \mathcal{L}(\mathbb{C}^n, \mathbb{C}))$ has norm $\le M_{\mathbb{C}}$ by hypothesis and $|e_i| = 1$ for the max norm, so $|\partial_i \partial_j h(I)| = |D^2 h(I)[e_i, e_j]| \le M_{\mathbb{C}}$ (the order of arguments is immaterial by symmetry, [K:bp:p1:holomorphic-regularity]).

Step S2.15: Euclidean bound. Lemma 2.1(c). The operator-norm statement: $|Qx|_2 = \sup_{|y|_2 = 1} |\sum_i \bar{y}_i (Qx)_i| \le n M_{\mathbb{C}} |x|_2$ by the same estimate applied to $\bar{y}$. □

2.4. **Quasi-convexity: the uniform Euclidean form.**

Lemma 2.6 (Transverse coercivity gives conic coercivity (linear algebra core))**.** *Let $n \ge 1$, $B, q > 0$, $Q : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ bilinear (not necessarily symmetric) with $|Q[x, y]| \le B|x|_2 |y|_2$, $\omega \in \mathbb{R}^n \setminus \{0\}$, and*

$$\omega \cdot v = 0 \implies Q[v, v] \ge q|v|_2^2 \qquad (v \in \mathbb{R}^n).$$

If $\ell' \ge 0$ and $\ell' \le |\omega|_2\, q/(8(B + q))$, then for every $\xi \in \mathbb{R}^n$:

$$|\omega \cdot \xi| \le \ell' |\xi|_2 \implies Q[\xi, \xi] \ge \tfrac{5}{8} q |\xi|_2^2 \quad (\ge \tfrac{q}{2} |\xi|_2^2).$$

Source: Ours; the computation is in `docs/tracks/N/sources.md` ("Uniform conversion"); all inequalities re-checked here.

Proof. **Step S2.16: Reduction to unit vectors.** Both hypothesis and conclusion are homogeneous (degrees 1 and 2) in ξ, and the case $\xi = 0$ is trivial; so let $|\xi|_2 = 1$.

Step S2.17: Decomposition. Put $e = \omega/|\omega|_2$, $t = \xi \cdot e$, $u = \xi - te$. Then $u \cdot \omega = 0$, $|u|_2^2 = 1 - t^2$, $|t| \le 1$, and $|t| = |\omega \cdot \xi|/|\omega|_2 \le \ell'/|\omega|_2 \le \kappa := q/(8(B+q)) < \frac{1}{8}$.

Step S2.18: Expansion. $Q[\xi,\xi] = Q[u,u] + tQ[u,e] + tQ[e,u] + t^2 Q[e,e]$. By hypothesis $Q[u,u] \ge q(1-t^2)$; by the bound on Q and $|u|_2, |e|_2 \le 1$: $|Q[u,e]|, |Q[e,u]|, |Q[e,e]| \le B$. So $Q[\xi,\xi] \ge q - (q+B)t^2 - 2B|t|$.

Step S2.19: Numerics. $(q+B)t^2 \le (q+B)\kappa \cdot \kappa = \frac{q}{8}\kappa \le \frac{q}{8}$ and $2B|t| \le 2B\kappa = \frac{2Bq}{8(B+q)} \le \frac{q}{4}$ (as $B \le B+q$). Hence $Q[\xi,\xi] \ge q - \frac{q}{8} - \frac{q}{4} = \frac{5}{8}q$. □

Lemma 2.7 (Quasi-convexity in Pöschel's quantified form)**.** *Let $n \ge 1$, $R, s, m, M_{\mathbb{C}}, \omega_{\min} > 0$ and put*

$$q = \frac{m}{n}, \qquad B_c = nM_{\mathbb{C}} + q, \qquad \mu = \frac{q}{2}, \qquad \ell = \frac{\omega_{\min}\, q}{8(B_c + q)}.$$

Let h satisfy `IsAnalyticIntegrable` $R\, s\, h$ *and* `HasComplexHessianBound` $R\, s\, h\, M_{\mathbb{C}}$, `IsStrictlyQuasiConvex` $\mathcal{B}_s(R)\, h\, m$, *and $\omega_{\min} \le |\omega_h(I)|_\infty$ for all $I \in B(0,R)$. Then for every $I \in B(0,R)$, writing $\omega = \omega_h(I)$, $Q = D^2 h_{\mathbb{R}}(I)$:*

- *(a) $|Q[x,y]| \le B_c|x|_2|y|_2$; $\omega \cdot v = 0 \Rightarrow Q[v,v] \ge q|v|_2^2$;*
- *(b) (Pöschel's alternative,* conditional*) for every $\xi \in \mathbb{R}^n$: $|\omega \cdot \xi| \le \ell|\xi|_2 \implies Q[\xi,\xi] \ge \mu|\xi|_2^2$; equivalently, either $|\omega \cdot \xi| > \ell|\xi|_2$ or $Q[\xi,\xi] \ge \mu|\xi|_2^2$ (Pos93 p. 190, "ℓ, m-quasi-convex" with $m := \mu$);*
- *(c) $|\omega|_2 \ge |\omega|_\infty \ge \omega_{\min}$ and $|\omega|_2 \le \sqrt{n}|\omega|_\infty$.*

The constants q, B_c, μ, ℓ depend only on $(n, m, M_{\mathbb{C}}, \omega_{\min})$. In Pöschel's normalisation ($\tilde{h} = 2\pi h$, Lemma 2.8) the same holds with $(\tilde{\ell}, \tilde{\mu}, \tilde{B}_c, \tilde{\omega}, \tilde{Q}) = 2\pi(\ell, \mu, B_c, \omega, Q)$. Warning. *Full positive definiteness $Q[\xi,\xi] \ge \mu|\xi|_2^2$ for all ξ is false in general and is not claimed: for $h(I) = I_1 + \frac{1}{2}(I_2^2 + I_3^2)$ one has $\omega(I) = (1, I_2, I_3) \ne 0$ (equal to e_1 on $I_2 = I_3 = 0$), $QC(m)$ holds on every bounded set for some $m > 0$ depending on the set (for $\omega \cdot v = 0$: $Q[v,v] = v_2^2 + v_3^2$ and $|v_1| \le (|I_2| + |I_3|)\max(|v_2|, |v_3|)$, so $m = (1+2K)^{-2}$ works on $|I|_\infty \le K$), but $Q[e_1, e_1] = 0$ at every point. The estimate (b) is only needed (and only true) in the cone $|\omega \cdot \xi| \le \ell|\xi|_2$.*

Source: Pos93 p. 190 (anchors POS190-QC, POS190-HESS, POS191-LOW; I read the scan, p. 190, ℓ, m-quasi-convexity, hypothesis $r_0 \le 4\ell/m$); conversion: `docs/tracks/N/sources.md`*, "Uniform conversion"; BM11 p. 4 for $QC(m)$. Review N05.*

Proof. **Step S2.20: Regularity and Hessian.** Since $B(0,R) \subset \mathcal{B}_s(R)$, $I \in \mathcal{B}_s(R)$; Q is the real Hessian of Lemma 2.5, so $|Q[x,y]| \le nM_{\mathbb{C}}|x|_2|y|_2 \le B_c|x|_2|y|_2$, the first half of (a).

Step S2.21: Transverse coercivity in the Euclidean norm. Definition 1.14 at $I \in \mathcal{B}_s(R)$ gives $\omega \cdot v = 0 \Rightarrow Q[v,v] \ge m|v|_\infty^2 \ge \frac{m}{n}|v|_2^2 = q|v|_2^2$ (Lemma 2.1(a)). Also $\omega \ne 0$.

Step S2.22: Frequency size. (c): $|\omega|_\infty \ge \omega_{\min}$ on $B(0,R)$ by hypothesis, $|\omega|_2 \ge |\omega|_\infty$; $|\omega|_2 \le \sqrt{n}|\omega|_\infty$ by Lemma 2.1.

Step S2.23: Conic coercivity. Apply Lemma 2.6 with $B = B_c$, q and $\ell' = \ell$: $\ell = \omega_{\min} q/(8(B_c + q)) \le |\omega|_2 q/(8(B_c+q))$ by (c). The conclusion is $Q[\xi,\xi] \ge \frac{5}{8}q|\xi|_2^2 \ge \mu|\xi|_2^2$.

Step S2.24: Scaling. Multiplying h by 2π multiplies ω, Q, B_c and the three of μ, q, ℓ by 2π, and the implications are homogeneous, so they persist. □

2.5. **Normalisations.**

Lemma 2.8 (Period conversion)**.** *Let $\Lambda(\theta, I) = (2\pi\theta, I)$ and, for $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ and arbitrary $h : \mathbb{C}^n \to \mathbb{C}$, $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$, define*

$$\tilde{H}(\varphi, I) = 2\pi H(\varphi/2\pi, I), \qquad \tilde{h} = 2\pi h, \qquad \tilde{f}(\varphi, I) = 2\pi f(\varphi/2\pi, I),$$

so $\tilde{H} = H_{\tilde{h},\tilde{f}}$ when $H = H_{h,f}$. Clauses (a), (b) are unconditional (no analyticity or periodicity is assumed). Clauses (c), (d) and the periodicity statement (e) are about f satisfying `IsAnalyticPerturbation` $R\, s\, \varepsilon\, f$ *(Definition 1.12), $R, s > 0$. Then:*

- *(a) (Dynamics) For $U \subset \mathbb{R}^n$, $J \subset \mathbb{R}$: z satisfies* `IsSolutionOn` $H\, U\, J\, z$ *iff $\Lambda \circ z$ satisfies* `IsSolutionOn` $\tilde{H}\, U\, J$ *(same time, same actions). In particular $|(\Lambda z)_2 - I_0| = |z_2 - I_0|$.*
- *(b) (Quantities) $\omega_{\tilde{h}} = 2\pi\omega_h$, $D^2\tilde{h}_{\mathbb{R}} = 2\pi D^2 h_{\mathbb{R}}$, and so:* `IsStrictlyQuasiConvex` $U\, h\, m \Rightarrow$ `IsStrictlyQuasiConvex` $U\, \tilde{h}\, 2\pi m$*; $B(M) \Rightarrow$ coordinate partials of $\tilde{h}_{\mathbb{R}}$ of order $1, 2, 3$ bounded by $2\pi M$; the complex Hessian bound $M_{\mathbb{C}} \to 2\pi M_{\mathbb{C}}$; $\omega_{\min} \to 2\pi\omega_{\min}$; Euclidean sizes of ω: $|\tilde{\omega}|_2 = 2\pi|\omega|_2$. Time is unchanged.*
- *(c) (Domains; needs holomorphy of f on $D_s(R)$ and the bound $|f| \le \varepsilon$) $\tilde{f}$ is holomorphic on $\{|\mathrm{Im}\,\varphi| < 2\pi s\} \times B(0, R)_s$ and $|\tilde{f}| \le 2\pi\varepsilon$ there; the strip T_ρ (ρ in θ) corresponds to the strip of width $2\pi\rho$ in φ.*
- *(d) (Periodicity; needs $f(\theta + k, I) = f(\theta, I)$ on $D_s(R)$) $\tilde{f}(\varphi + 2\pi k, I) = \tilde{f}(\varphi, I)$ for $k \in \mathbb{Z}^n$ and φ in that strip. (Without this premise the claim fails: $f = \theta_1$ gives $\tilde{f} = \varphi_1$.)*
- *(e) (Fourier; needs (c), (e)) $\tilde{f}(\varphi, I) = \sum_k \hat{\tilde{f}}_k(I) e^{ik\cdot\varphi}$ with $\hat{\tilde{f}}_k(I) = 2\pi\hat{f}_k(I)$, $\hat{f}_k(I) = \int_{[0,1]^n} f(\theta, I) e^{-2\pi ik\cdot\theta}\, d\theta$, whenever the Fourier series of $f(\cdot, I)$ converges to it (*[K:`bp:p1:summation`]*); the coefficient radius s' of Pöschel's weight $e^{|k|_1 s'}$ equals $2\pi\rho$.*

Source: `docs/tracks/N/sources.md` *("For the period conversion"); Pos93 p. 189 (set-up: $\theta \in \mathbb{R}^n/2\pi\mathbb{Z}^n$, strip W_{s_0} of width s_0 in the max norm of $\mathrm{Im}\,\theta$).*

Proof. **Step S2.25: Chain rule.** $\tilde{H} = 2\pi\, H \circ \Lambda^{-1}$; Λ is a linear isomorphism, so $\tilde{H}$ is differentiable at Λz iff H is at z, with $\partial_{I_i}\tilde{H}(\Lambda z) = 2\pi\partial_{I_i}H(z)$ and $\partial_{\varphi_i}\tilde{H}(\Lambda z) = \partial_{\theta_i}H(z)$.

Step S2.26: Vector field. Hence $X_{\tilde{H}}(\Lambda z) = (2\pi\partial_I H(z), -\partial_\theta H(z)) = (2\pi\dot{\theta}, \dot{I}) = \frac{d}{dt}\Lambda z$ when $\dot{z} = X_H(z)$, and conversely; the condition $z(t)_2 \in U$ is unchanged. This is (a). (Rescaling the angle without the factor 2π in H would change time.)

Step S2.27: Scalars. (b) is linearity of $h \mapsto \omega_h, D^2 h_{\mathbb{R}}, D^k h_{\mathbb{R}}$ and of the operator norm. (c)–(f) are, under the stated premises, the substitution $\varphi = 2\pi\theta$, $|\mathrm{Im}\,\varphi| = 2\pi|\mathrm{Im}\,\theta|$, and $e^{ik\cdot\varphi} = e^{2\pi ik\cdot\theta}$. □

Lemma 2.9 (From sup norm on a strip to the weighted Fourier norm)**.** *Let $u : T_\rho \to \mathbb{C}$ be holomorphic, bounded and 1-periodic, $\|u\|_\rho = \sup_{T_\rho}|u|$, and $0 < \delta < \rho$. Then*

$$\sum_{k\in\mathbb{Z}^n} |\hat{u}_k|\, e^{2\pi|k|_1(\rho-\delta)} \;\le\; \coth(\pi\delta)^n\, \|u\|_\rho,$$

the left side is Pöschel's weighted norm with weight $e^{|k|_1 s'}$, $s' = 2\pi(\rho - \delta)$, and the factor is $\coth^n\sigma$ with $\sigma = \pi\delta$ (Pos93 p. 190: $|u|_{D,r,s} \le \coth^n\sigma\, |u|_{D,r,s+2\sigma}$, where the right side is the sup norm on the strip of width $s + 2\sigma$ in φ, i.e. $\rho = (s' + 2\sigma)/2\pi$). Conversely $\|u\|_\rho \le \sum_k |\hat{u}_k| e^{2\pi|k|_1\rho}$. For I-dependent $u(\theta, I)$ the statement holds for each fixed I with the same constant, so the sup over I in the weighted norm is $\le \coth(\pi\delta)^n \sup_{\theta,I}|u|$.

Source: Pos93 p. 190 (the displayed relation between the weighted and sup norms); frozen-convention statement: [K:`bp:p1:summation`], [K:`bp:p1:decay`], [K:`bp:p1:exp-sums`]*(b).*

Proof. **Step S2.28: Decay.** For $\rho' \in (\rho-\delta,\rho)$, [K:bp:p1:decay] gives $|\hat{u}_k| \le e^{-2\pi|k|_1\rho'}\|u\|_\rho$; hence $|\hat{u}_k|e^{2\pi|k|_1(\rho-\delta)} \le e^{-2\pi|k|_1(\rho'-\rho+\delta)}\|u\|_\rho$, and letting $\rho' \uparrow \rho$ (termwise, then summing, or directly [K:bp:p1:summation], converse part) the bound is $e^{-2\pi\delta|k|_1}\|u\|_\rho$ per k.

Step S2.29: Sum. $\sum_{k\in\mathbb{Z}^n} e^{-2\pi\delta|k|_1} = \coth(\pi\delta)^n$ ([K:bp:p1:exp-sums](b) with $x = 2\pi\delta$). The case with parameters is pointwise in I. □

2.6. **The dictionary of constants.**

Lemma 2.10 (Constants dictionary for N-HEAD-C)**.** *Let $n \ge 3$ (only $n \ge 2$ is used by Pöschel) and $R, s, M, m, M_{\mathbb{C}}, \omega_{\min} > 0$, $a = \frac{1}{2n}$. Define successively*

$$q = \tfrac{m}{n},\ B_c = nM_{\mathbb{C}} + q,\ \mu = \tfrac{q}{2},\ \ell = \tfrac{\omega_{\min} q}{8(B_c+q)}, \qquad r_0 = \min\Big(\tfrac{s}{2}, \tfrac{R}{8}, \tfrac{2\ell}{\mu}, \tfrac{4\omega_{\min}}{\mu}\Big), \qquad A = \tfrac{11B_c}{\mu},$$

$$s_0 = \pi s, \quad \sigma = \tfrac{\pi s}{4}, \quad F = 2\pi \coth^n\big(\tfrac{\pi s}{4}\big), \quad \varepsilon_* = \frac{2\pi\mu\, r_0^2}{2^{10}A^{2n}},$$

$$C = \tfrac{r_0}{A}\Big(\tfrac{F}{\varepsilon_*}\Big)^a, \quad c = \tfrac{s_0}{6}\Big(\tfrac{\varepsilon_*}{F}\Big)^a, \quad T = \frac{A^2 s_0}{4\pi n M},$$

$$\varepsilon_0 = \min\Big(\tfrac{\varepsilon_*}{F},\ \big(\tfrac{R}{4C}\big)^{2n}\Big).$$

Pöschel's data (his normalisation, for $\tilde{h} = 2\pi h$): $m_P = 2\pi\mu$, $\ell_P = 2\pi\ell$, $M_P = 2\pi B_c$, $A_P = 11M_P/m_P = A$, $\varepsilon_{0,P} = m_P r_0^2/(2^{10}A_P^{2n}) = \varepsilon_*$, $R_0 = r_0/A$, and the set $P = \{J \in \mathbb{R}^n : |J| \le R/2\}$. Then:

- *(a) All constants are positive and finite and depend only on $(n, R, s, M, m, M_{\mathbb{C}}, \omega_{\min})$ — not on $h, f, \varepsilon, \theta_0, I_0$, nor on the bound A of Definition 1.11(iii). Order of choice: $q, B_c, \mu, \ell, r_0, A$; then s_0, σ, F; ε_*; C, c; T; ε_0. And $C\varepsilon_0^a \le R/4$, $0 < \varepsilon_0 \le \varepsilon_*/F$.*
- *(b) $r_0 \le 4\ell_P/m_P$ $(= 4\ell/\mu)$; $A \ge 22$; $R_0 = r_0/A \le r_0/22 \le R/176$.*
- *(c) (Tubes) With $V_{r_0}P = \{I \in \mathbb{C}^n : \exists J \in P,\ |I-J|_2 < r_0\}$ and $U_{r_0}P = V_{r_0}P \cap \mathbb{R}^n$: $V_{r_0}P \subset B(0,R)_s$ and $U_{r_0}P \subset B(0, 5R/8) \subset B(0,R)$.*
- *(d) (Hypotheses of Theorem 1) If h satisfies (H1)–(H5) of Theorem 1.19, then for $I \in U_{r_0}P$: $\tilde{h}$ is ℓ_P, μ_P-quasi-convex (Lemma 2.7) with $\mu_P = 2\pi\mu$; the complex Hessian operator norm (Euclidean) of $\tilde{h}$ on $V_{r_0}P$ is $\le M_P$ (Lemma 2.5); Pöschel's m, M are m_P, M_P and his hypothesis $r_0 \le 4\ell_P/m_P$ holds.*
- *(e) (Perturbation) If f satisfies (H6) with ε, then Pöschel's norm of $\tilde{f}$ on $V_{r_0}P \times$(strip of width s_0) is $\le F\varepsilon$; and $F\varepsilon \le \varepsilon_{0,P}$ if $\varepsilon \le \varepsilon_0$.*
- *(f) (Exceptional branch is empty) For $I_0 \in B(0, R/2)$: $|\tilde{\omega}(I_0)|_2 \ge 2\pi\omega_{\min} > \pi\omega_{\min} \ge m_P r_0/8$.*
- *(g) (Time prefactor) For $I_0 \in B(0,R/2)$, $0 < 2\pi\omega_{\min} \le \Omega_0 = \sup_{|I-I_0|_2 \le R_0}|\tilde{\omega}(I)|_2 \le 2\pi\sqrt{n}M \le 2\pi n M$, so Pöschel's $T_0 = A_P^2 s_0/\Omega_0 \ge 2T$.*
- *(h) (Conclusion shape, for $\varepsilon > 0$; $\varepsilon = 0$ is N-ZERO) Pöschel's $R_0(\varepsilon_P/\varepsilon_{0,P})^a$ and $T_0 \exp\big(\frac{s_0}{6}(\varepsilon_{0,P}/\varepsilon_P)^a\big)$ with $\varepsilon_P = F\varepsilon$ are $C\varepsilon^a$ and $T_0\exp(c\varepsilon^{-a})$, and $T\exp(c\varepsilon^{-a}) \le \frac{1}{2}T_0\exp(c\varepsilon^{-a})$.*

Source: `docs/tracks/N/sources.md` "Uniform conversion" (all constants), checked against Pos93 p. 190 (hypotheses, $A = 11M/m$, $\varepsilon_0 = mr_0^2/(2^{10}A^{2n})$, $r_0 \le 4\ell/m$), p. 191 (parameters a, R_0, T_0, exceptional branch) and p. 204 (assembly: $K = (\varepsilon_0/\varepsilon)^a$). Pos93 requires $\|f_\varepsilon\| \le \varepsilon$ (p. 190) with $\varepsilon \le \varepsilon_{0,P}$.
Source gap: Pos93 states no theorem about the $\omega_{\min}$-conversion and the frozen-to-source dictionary: it exists only in `sources.md` and here (Part 0 `bpN:p0:head`). The choice of T with the factor $1/2$ is slack, not needed by the source.

Proof. **Step S2.30: Positivity and dependence.** Each constant is a positive finite expression in earlier ones (denominators: $n, \mu, 8(B_c+q), A, F, \varepsilon_*, 4C, 4\pi nM$, all > 0); $\varepsilon_0 > 0$ as a minimum of

positive numbers, and $C\varepsilon_0^a \le C \cdot \frac{R}{4C} = R/4$ since $((R/4C)^{2n})^{1/(2n)} = R/(4C)$. The list of inputs in each formula contains only $n, R, s, M, m, M_{\mathbb{C}}, \omega_{\min}$. This is (a).

Step S2.31: Size relations. $B_c = nM_{\mathbb{C}} + q \ge q = 2\mu$ so $A = 11B_c/\mu \ge 22$. $4\ell_P/m_P = 4\ell/\mu \ge 2\ell/\mu \ge r_0$, and $R_0 = r_0/A \le r_0/22 \le R/176$. This is (b).

Step S2.32: Tubes. $r_0 \le \min(s/2, R/8)$: Lemma 2.2(c) gives (c).

Step S2.33: Quasi-convexity and Hessian. For (d): $U_{r_0}P \subset B(0,R)$ by (c), so Lemma 2.7(b) applies at every point of $U_{r_0}P$ and, after Lemma 2.8(b), gives the (ℓ_P, μ_P)-alternative for $\tilde{h}$. For $I \in V_{r_0}P \subset B(0,R)_s$ the Euclidean operator norm of $\tilde{Q}(I)$ is $\le 2\pi n M_{\mathbb{C}} \le 2\pi B_c = M_P$ (Lemma 2.5). Pöschel's $A = 11M/m$ with $M = M_P$ and $m = \mu_P = m_P$ is $11B_c/\mu = A$.

Step S2.34: Perturbation. By (H6) $|f| \le \varepsilon$ on $D_s(R) \supset T_{3s/4} \times V_{r_0}P$ ($V_{r_0}P \subset B(0,R)_s$, (c)). For each fixed $I \in V_{r_0}P$ apply Lemma 2.9 to $u = f(\cdot, I)$ with $\rho = 3s/4$, $\delta = s/4$: $\sum_k |\hat{f}_k(I)| e^{2\pi|k|_1 s/2} \le \coth(\pi s/4)^n \varepsilon$. Multiplying by 2π (Lemma 2.8(f); f satisfies (H6), so (c), (e) apply) and noting $2\pi \cdot s/2 = \pi s = s_0$ gives $\le F\varepsilon$. (In Pöschel's terms $s_0 + 2\sigma = 3\pi s/2 = 2\pi \cdot \frac{3s}{4}$.) $\varepsilon \le \varepsilon_*/F$ gives $F\varepsilon \le \varepsilon_* = \varepsilon_{0,P}$. This is (e).

Step S2.35: Exceptional branch. $I_0 \in B(0, R/2) \subset B(0,R)$ so $|\omega(I_0)|_2 \ge \omega_{\min}$ (Lemma 2.7(c)) and $|\tilde{\omega}|_2 = 2\pi|\omega|_2$. And $m_P r_0/8 = 2\pi\mu r_0/8 \le 2\pi\mu \cdot \frac{4\omega_{\min}}{8\mu} = \pi\omega_{\min}$ by $r_0 \le 4\omega_{\min}/\mu$. This is (f).

Step S2.36: Time. The Euclidean ball $\{|I - I_0|_2 \le R_0\}$ with $R_0 \le r_0$ lies in $\{|I|_\infty < R/2 + R_0\} \subset B(0, 5R/8)$ ($|I - I_0|_\infty \le |I - I_0|_2$), inside $\mathcal{B}_s(R)$ where Lemma 2.4 gives $|\omega(I)|_2 \le \sqrt{n}M$. So $\Omega_0 \le 2\pi\sqrt{n}M \le 2\pi nM$ (finite); also I_0 belongs to the closed ball, $I_0 \in B(0,R)$, so $\Omega_0 \ge |\tilde{\omega}(I_0)|_2 \ge 2\pi\omega_{\min} > 0$ (Lemma 2.7(c)). Inverting the positive quantities, $T_0 = A^2 s_0/\Omega_0 \ge A^2 s_0/(2\pi nM) = 2T$. (g).

Step S2.37: Exponent and constants. $R_0(F\varepsilon/\varepsilon_*)^a = \frac{r_0}{A}(F/\varepsilon_*)^a \varepsilon^a = C\varepsilon^a$ and $\frac{s_0}{6}(\varepsilon_*/(F\varepsilon))^a = c\varepsilon^{-a}$. (h) follows from $T \le T_0/2$. □

Remark 2.11 (Forward use and an unused constant). Part 10 chooses C, T, c, ε_0 as in Lemma 2.10, verifies Pos93's hypotheses by (d),(e) and excludes the exceptional branch by (f). The time prefactor T enters only through (g); the remaining slack $\varepsilon_0 \le (R/4C)^{2n}$ gives $C\varepsilon^a \le R/4$ so that the a-priori confinement keeps $I(t) \in B(0, R/2 + R/4) = B(0, 3R/4)$ (continuation, [N:`bpN:p3:continuation`]). The value $M_{\mathbb{C}}$ enters only through B_c; M only through T.

3. N-QC and N-ZERO

This part proves two frozen theorems completely: N-QC (Theorem 1.21, the nonsingular isoenergetic derivative) and N-ZERO (Theorem 1.20, the exact integrable dynamics). Neither uses the normal-form machinery. The linear-algebra core of N-QC is stated in a reusable form (Lemma 3.1, cited by track I as the isoenergetic gate, [N:`bpN:p2:qc-kernel`]). Frozen conventions throughout (max norms, period 1); Euclidean structure appears only as the dot product $\omega \cdot v = \sum \omega_i v_i$.

3.1. N-QC.

Lemma 3.1 (Kernel core: bordered Hessian is nonsingular). *Let $n \ge 1$, $\omega \in \mathbb{R}^n$, A a real $n \times n$ matrix (not necessarily symmetric, no norm bound), $\lambda \ne 0$, and suppose*

$$\omega \cdot v = 0,\ v \ne 0 \implies v^\top A v \ne 0 \qquad \text{and} \qquad \omega \ne 0.$$

Let $L : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}^n$, $L(v,u) = (\omega \cdot v,\ u\,\omega + \lambda A v)$. Then $\ker L = 0$, hence L is bijective. In particular this holds if $\lambda > 0$, $\omega \ne 0$ and $v^\top A v \ge m|v|^2$ for some norm and $m > 0$ whenever $\omega \cdot v = 0$. (For $n = 0$ no $\omega \ne 0$ exists; N-QC is then vacuous, Remark 3.2.)

Source: BM11 Lemma 3.3 and its proof, pp. 9–10 (kernel computation); linear algebra (E1).

Proof. **Step S3.1: Kernel equations.** Let $L(v,u)=0$: $\omega\cdot v=0$ and $u\omega+\lambda Av=0$.

Step S3.2: Pairing with v. Dot the second equation with v: $u(\omega\cdot v)+\lambda v^\top Av=0$, so $\lambda v^\top Av=0$, i.e. $v^\top Av=0$ as $\lambda\neq 0$. Since also $\omega\cdot v=0$, the hypothesis forces $v=0$.

Step S3.3: The scalar. With $v=0$: $u\omega=0$; as $\omega\neq 0$, $u=0$.

Step S3.4: Bijectivity. L is a linear map between spaces of dimension $n+1$ with trivial kernel, hence bijective by (E1) (`LinearMap.injective_iff_surjective_of_finrank_eq_finrank`, with $\dim(\mathbb{R}^n\times\mathbb{R})=\dim(\mathbb{R}\times\mathbb{R}^n)=n+1$). □

Remark 3.2 ($n=0$). For $n=0$ the space $\mathbb{R}^0$ is a point, so $\omega=0$ and Definition 1.14 requires $\omega_h(I)\neq 0$: no I satisfies it. Hence for $n=0$ the hypotheses of N-QC are never met with $I\in U$ and the theorem is vacuous; no restriction $n\geq 1$ is added to it (review N03).

Lemma 3.3 (The derivative of Ψ)**.** *Let $U\subset\mathbb{R}^n$ be open, h with $h_{\mathbb{R}}\in C^2(U)$, $I\in U$, $\lambda\in\mathbb{R}$, $\Psi(I',\lambda')=(h_{\mathbb{R}}(I'),\lambda'\omega_h(I'))$. Then Ψ is differentiable at (I,λ) (so `fderiv` is the genuine derivative; Remark 1.18(e)) and*

$$D\Psi(I,\lambda)[v,u]=\big(\omega\cdot v,\ u\,\omega+\lambda Qv\big),\qquad \omega=\omega_h(I),\ Q_{ij}=D^2h_{\mathbb{R}}(I)[e_i,e_j]=Q_{ji}.$$

Source: BM11 p. 10; (E4), [K:`bp:p1:ext-schwarz`].

Proof. **Step S3.5: First component.** $(I',\lambda')\mapsto h_{\mathbb{R}}(I')$ is differentiable at (I,λ) ($h_{\mathbb{R}}$ is C^2 on the open set $U\ni I$) with derivative $[v,u]\mapsto Dh_{\mathbb{R}}(I)[v]=\sum_i v_i Dh_{\mathbb{R}}(I)[e_i]=\omega\cdot v$.

Step S3.6: The frequency map is C^1. $\omega_h(I')_i=Dh_{\mathbb{R}}(I')[e_i]$ is the composition of $I'\mapsto Dh_{\mathbb{R}}(I')\in\mathcal{L}(\mathbb{R}^n,\mathbb{R})$ (which is C^1 on U since $h_{\mathbb{R}}\in C^2(U)$ and U is open) with the continuous linear evaluation $\ell\mapsto\ell(e_i)$. So ω_h is differentiable on U with $D\omega_h(I)[v]_i=D(Dh_{\mathbb{R}})(I)[v][e_i]=D^2h_{\mathbb{R}}(I)[v,e_i]$.

Step S3.7: Symmetry. $D^2h_{\mathbb{R}}(I)$ is symmetric ([K:`bp:p1:ext-schwarz`]), so $D\omega_h(I)[v]=Qv$ with Q the symmetric Hessian matrix.

Step S3.8: Product. By the product rule (E4), $(I',\lambda')\mapsto\lambda'\omega_h(I')$ is differentiable at (I,λ) with derivative $[v,u]\mapsto u\,\omega+\lambda Qv$. Both components are differentiable, so Ψ is, with the stated derivative. □

Theorem 3.4 (N-QC)**.** *Theorem 1.21 holds: for `IsStrictlyQuasiConvexU h m`, $I\in U$, $\lambda>0$, the map $D\Psi(I,\lambda)$ is bijective.*
Frozen Lean declaration: `KAM.Nekhoroshev.quasiConvex_isoenergetic_derivative`
Source: BM11 Lemma 3.3 and proof, pp. 9–10.

Proof. **Step S3.9: Unpack the hypothesis.** Definition 1.14: $m>0$, U open, $h_{\mathbb{R}}\in C^2(U)$, $\omega:=\omega_h(I)\neq 0$, and $\omega\cdot v=0\Rightarrow m|v|^2\leq D^2h_{\mathbb{R}}(I)[v,v]$ (max norm on v). Note $n\geq 1$ because $\omega\neq 0$ (Remark 3.2).

Step S3.10: The derivative. By Lemma 3.3, $D\Psi(I,\lambda)=L$ of Lemma 3.1 with $A=Q$ the Hessian matrix, $v^\top Qv=D^2h_{\mathbb{R}}(I)[v,v]$ (bilinearity).

Step S3.11: Positivity on $\omega^\perp$. For $\omega\cdot v=0$, $v\neq 0$: $v^\top Qv\geq m|v|_\infty^2>0$, since $m>0$ and $|v|_\infty>0$. So the hypothesis of Lemma 3.1 holds (with $\lambda>0$).

Step S3.12: Conclusion. Lemma 3.1 gives bijectivity of $L=D\Psi(I,\lambda)=\texttt{fderiv}\,\mathbb{R}\,\Psi\,(I,\lambda)$. □

Remark 3.5 (Reusable form for track I). Lemma 3.1 needs no norm, no symmetry of A and no uniform constant: any quasi-convexity-type hypothesis ("$v^\top Av>0$ on $\omega^\perp\setminus 0$") together with $\omega\neq 0$ and $\lambda\neq 0$ gives nonsingularity of the bordered matrix $\begin{pmatrix}\omega^\top & 0\\ \lambda A & \omega\end{pmatrix}$. A quantitative inverse bound is

not asserted by N-QC and is not provided here; it would follow, if needed, from the compactness of $\{v : |v|_2 = 1,\ \omega \cdot v = 0\}$ for a fixed (I, λ) but is not uniform in (I, λ).

3.2. N-ZERO.

Lemma 3.6 (The field at the integrable orbit)**.** *Let $R, s > 0$, `IsAnalyticIntegrable` $R\, s\, h$, `IsAnalyticPerturbation` $R\, s\, 0\, f$, $H = H_{h,f}$, and $W_s = \mathbb{R}^n \times B(0, R+s)$. Then $H(\theta, I) = h_{\mathbb{R}}(I)$ on W_s; H is differentiable at every $z = (\theta, I) \in W_s$ with $DH(z)[(a,b)] = Dh_{\mathbb{R}}(I)[b]$ and*

$$X_H(\theta, I) = (\omega_h(I), 0).$$

Source: BM11 p. 1 ($H = h(I)$ when $f = 0$); ours.

Proof. **Step S3.13: f vanishes.** Definition 1.12(iv) with $\varepsilon = 0$: $|f| \le 0$ on $D_s(R)$, so $f = 0$ there. For $z = (\theta, I) \in W_s$, $(\theta, I) \in D_s(R)$ (Lemma 2.2(a)), so $H(z) = \operatorname{Re} h(I) = h_{\mathbb{R}}(I)$ (Lemma 2.3(e)).

Step S3.14: Differentiability. W_s is open and on it $H = h_{\mathbb{R}} \circ \mathrm{pr}_2$ with $h_{\mathbb{R}} \in C^3(\mathcal{B}_s(R))$ (Definition 1.11(v)), $\mathcal{B}_s(R) = B(0, R+s)$ open. So H is differentiable at each point of W_s with $DH(z)[(a,b)] = Dh_{\mathbb{R}}(I)[b]$ (chain rule with the continuous linear projection). The point-wise hypotheses of the totality remark 1.18 are met.

Step S3.15: Field. By Definition 1.3, $\partial_{I_i} H = Dh_{\mathbb{R}}(I)[e_i] = \omega_h(I)_i$ (Definition 1.2) and $\partial_{\theta_i} H = DH(z)[(e_i, 0)] = 0$. □

Theorem 3.7 (N-ZERO)**.** *Theorem 1.20 holds.*
Frozen Lean declaration: `KAM.Nekhoroshev.zero_perturbation`
Source: Ours (supplement to BM11 Theorem 2.1, p. 5).

Proof. **Step S3.16: Initial value and action.** $\gamma(t) = (\theta_0 + t\omega_h(I_0), I_0)$ (Definition 1.17) gives $\gamma(0) = (\theta_0, I_0)$ and $\gamma(t)_2 = I_0$ for all t: the first and third conclusions.

Step S3.17: The action stays in $B(0,R)$. $I_0 \in B(0, R/2) \subset B(0, R)$ since $R > 0$ (max norm: $|I_0| < R/2 < R$).

Step S3.18: Differentiability of H along the orbit. $\gamma(t) \in \mathbb{R}^n \times B(0,R) \subset W_s$ (Lemma 2.2(a): $B(0,R) \subset B(0, R+s)$). By Lemma 3.6 H is differentiable at $\gamma(t)$ for each t.

Step S3.19: Derivative of the orbit. γ is affine in t, so differentiable at every $t \in \mathbb{R}$ (two-sided) with $\gamma'(t) = (\omega_h(I_0), 0)$.

Step S3.20: The equation. Lemma 3.6 gives $X_H(\gamma(t)) = (\omega_h(I_0), 0) = \gamma'(t)$ for every $t \in \mathbb{R}$. With Steps S3.17–S3.19, every clause of Definition 1.16 holds with $U = B(0,R)$ and $J = \mathbb{R}$. □

Remark 3.8 (Scope). N-ZERO uses neither the holomorphy of Part 1 (beyond the stated clauses of the definitions) nor the ODE layer of Part 3; Part 3 supplies uniqueness if ever required for $\varepsilon = 0$ (γ is the unique solution by [N:`bpN:p3:gen-uniqueness`]), which is not asserted by the frozen statement. The case $\varepsilon = 0$ is excluded from N-HEAD-C by (H7).

4. Hamiltonian ODE infrastructure (shared by tracks I, N, P)

This part is the single ODE layer of the three blueprints I, N, P. It has two levels. The *general layer* (§4.2–§4.6) is about an autonomous vector field F of class C^1 on an open subset W of a finite-dimensional real normed space E: local existence, uniqueness on *every* interval, maximal solutions, continuation (compact sets, and closed sets with uniform bounds), first exit from an open set. The *Hamiltonian layer* (§4.7) adds $F = X_H$: smoothness of X_H, energy conservation, translation in the angles. §4.8 shows that the frozen solution predicates of the three tracks (`IsSolutionOn`, `IsHamiltonianSolutionOn`, `IsRestrictedSolutionOn`, `IsNewtonSolution`, `IsFixedSolution`) are instances of the general predicate, and §4.9 gives the six track-N results of `PLAN.md`. Tracks I and P cite the general results as [N:`bpN:p3:gen-...`].

4.1. **Conventions and external facts.**

- E is a finite-dimensional real normed space (hence complete and proper: closed bounded sets are compact), $F : E \to E$ a *total* function, $W \subset E$ open. In the applications $E = \mathbb{R}^n \times \mathbb{R}^n$ with the max norm $|(\theta, I)| = \max(|\theta|, |I|)$ (the product norm of the Lean text, equal to the max norm on the $2n$ coordinates), or $E = (\mathbb{R}^2)^N \times (\mathbb{R}^2)^N$ for the Newtonian problems.
- F is C^1 *on* W means `ContDiffOn` $\mathbb{R}$ 1 F W; as W is open this gives $F \in C^1$ at each point of W, and no information off W. Off W the values of F never enter (Lean totality, below).
- An *interval* is an order-connected subset of $\mathbb{R}$ (Lean `OrdConnected`); open, closed, half-open, bounded or not, possibly a single point. $\overline{B}(x, \delta)$ is the closed ball.
- All derivatives $\dot{z}(t)$ in solution predicates are two-sided (Lean `HasDerivAt`) at every $t \in J$, including endpoints of J, exactly as in the frozen predicates ("z is defined on $\mathbb{R}$").

External facts. Mathlib-level statements, with the declarations that exist (checked in `Mathlib/Analysis/ODE/`); what is *not* in Mathlib is listed after them.

(E1) *Picard–Lindelöf, quantitative form.* Let $F : E \to E$, $x \in E$, $a, M, L \geq 0$ with $\|F(y)\| \leq M$ and F L-Lipschitz on $\overline{B}(x, a)$, $T \geq 0$ with $MT \leq a$. Then there is $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x$ and α has derivative $F(\alpha(t))$ within $[t_0 - T, t_0 + T]$ at each t of that interval. (Mathlib: the root-namespace structure `IsPicardLindelof` $f\, t_0\, x_0\, a\, r\, L\, K$ of `Analysis/ODE/PicardLindelof.lean`, with $r = 0$ and a time-independent field, and `IsPicardLindelof.exists_eq_forall_mem_Icc_hasDerivWithinAt`$_0$ (Unicode subscript zero) of `Analysis/ODE/ExistUnique.lean`; Mathlib's norm bound is the parameter L (our M) and its Lipschitz parameter is K (our L); the time constraint is the field `mul_max_le`: $M \max(t_{\max} - t_0, t_0 - t_{\min}) \leq a - r$. Parameters are `NNReal`. The contraction on iterates means no smallness of LT is needed.) Exists.

(E2) *Uniqueness with a Lipschitz constant on a set.* If $S \subset E$ and F is L-Lipschitz on S, $f, g : [a, b] \to E$ are continuous, have right derivative $F(f(t))$, $F(g(t))$ on $[a, b)$, take values in S and $f(a) = g(a)$, then $f = g$ on $[a, b]$; and the mirror statement with $f(b) = g(b)$ and left derivatives. (Mathlib: `ODE_solution_unique_of_mem_Icc_right`, `ODE_solution_unique_of_mem_Icc_left`.) Exists.

(E3) C^1 *implies locally Lipschitz.* If F is C^1 at p there are L and a neighbourhood on which F is L-Lipschitz (Mathlib: `ContDiffAt.exists_lipschitzOnWith`); and a function locally Lipschitz on a compact set is Lipschitz there (Mathlib: `LocallyLipschitzOn.exists_lipschitzOnWith_of_compact`, `Topology/Algebra/MetricSpace/Lipschitz.lean`), which packages Lemma 4.4 once local Lipschitzness on K is extracted from C^1 on W. Exists.

(E4) *Mean value inequality*: [K:`bp:p1:ext-mvt`]; for a differentiable map on a convex set with $\|DF\| \leq L$, F is L-Lipschitz there; for a function of one variable with derivative bounded by M on an interval, $\|\alpha(u) - \alpha(s)\| \leq M|u - s|$. Exists.

(E5) *Topology.* Heine–Borel in E (`ProperSpace`); continuous images of compact sets are compact; sequential compactness; for a *nonempty* closed set A, $x \mapsto \operatorname{dist}(x, A)$ is continuous and positive exactly on $E \setminus A$ (`Metric.infDist`, `IsClosed.notMem_iff_infDist_pos`, which requires `A.Nonempty`; for $A = \emptyset$ Mathlib has $\operatorname{infDist}(x, \emptyset) = 0$, `Metric.infDist_empty`, so no statement below uses infDist of an empty set: thickenings are defined by $K_\delta = \{y : \exists x \in K,\ \|y - x\| \leq \delta\}$, which is empty for $K = \emptyset$). Exists.

(E6) *Calculus.* Chain rule `HasFDerivAt.comp_hasDerivAt`; a function continuous on $[s, t]$ with right derivative 0 on $[s, t)$ is constant (`constant_of_has_deriv_right_zero`); two one-sided derivatives agree $\Rightarrow$ two-sided one (`HasDerivWithinAt.union` with $(-\infty, b] \cup [b, \infty) = \mathbb{R}$ and `hasDerivWithinAt_univ`); `ContDiffOn.fderiv_of_isOpen`: if H is C^{k+1} on an open set then DH is C^k there. Exists.

(E7) *Uniform existence time near a point.* If F is C^1 at x_0 and $t_0 \in \mathbb{R}$ there are $r, \varepsilon > 0$ such that for every $x \in \overline{B}(x_0, r)$ there is α with $\alpha(t_0) = x$ and derivative $F(\alpha(t))$ for all $t \in (t_0 - \varepsilon, t_0 + \varepsilon)$ (Mathlib: `ContDiffAt.exists_forall_mem_closedBall_exists_eq_forall_mem_Ioo_hasDerivAt`, `ExistUnique.lean:145`; note the solution is for the total F and is not asserted to stay in the neighbourhood of x_0). Exists. It covers Theorem 4.10 for a single point after the staying-in-W step; the explicit $\delta/\max(M, 1)$ time of Proposition 4.9, uniform over arbitrary sets Φ with uniform data, is derived from (E1).

Not in Mathlib (checked in `Analysis/ODE` and `Geometry/Manifold/IntegralCurve`: no maximal-solution, continuation or first-exit statement, and no uniqueness on arbitrary intervals with only local Lipschitz constants): the following must be proved from the blueprint below: uniform data on arbitrary sets and its compact and periodic instances (Lemmas 4.6, 4.7), Lemma 4.8, Proposition 4.9 (derived from (E1); the existing statement (E7) may be reused for the single-point case), Theorems 4.11, 4.13, 4.14, Lemma 4.19 and the Hamiltonian layer.

Remark 4.1 (Lean totality). `KAM.hamiltonianVectorField` H uses `fderiv`, which is 0 at a point of non-differentiability, and the Newtonian fields use $1/\ell^3$ with $\ell =$ `planeLength` or `radius`. In every use below these are guarded: X_H occurs only on an open set where H is C^2 (or inside the frozen predicates, which assert differentiability at every visited point), and $\ell > 0$ at every visited point because the frozen predicates require it. The general layer never uses F off W.

4.2. The general solution predicate; Lipschitz data.

Definition 4.2 (Solutions in W)**.** For $F : E \to E$, $W \subset E$, $J \subset \mathbb{R}$ and $z : \mathbb{R} \to E$: `IsSolIn` $F\, W\, J\, z$ holds if for every $t \in J$, $z(t) \in W$ and z has derivative $F(z(t))$ at t (two-sided). Suggested Lean name `KAM.ODE.IsSolutionIn`; it is *not* a frozen declaration.
Source: Standard; shape dictated by the frozen predicates of Definition 1.16 and §4.8.

Elementary consequences (each a one-line Lean lemma, (E6)): (i) restriction: `IsSolIn` $F\, W\, J\, z$ and $J' \subset J$ give `IsSolIn` $F\, W\, J'\, z$; (ii) z is continuous on J (differentiable at each point of J); (iii) *local nature*: if J' is a neighbourhood of each of its points and $z = y$ on it, then `IsSolIn` $F\, W\, J'\, y \Leftrightarrow$ `IsSolIn` $F\, W\, J'\, z$ for open J'.

Lemma 4.3 (Symmetries)**.** *Let* `IsSolIn` $F\, W\, J\, z$.

(a) (time shift) For $c \in \mathbb{R}$, $t \mapsto z(t + c)$ is a solution on $J - c$.
(b) (time reversal) $t \mapsto z(-t)$ is a solution of $-F$ on $-J$, and $-F$ is C^1 on W if F is.
(c) (space translation) If $v \in E$ satisfies $W + v = W$ and $F(x + v) = F(x)$ for $x \in W$, then $t \mapsto z(t) + v$ is a solution of F in W on J.

Source: Standard.

Proof. **Step S4.1: Chain rule.** (a) and (b) are the chain rule for $t \mapsto t + c$, $t \mapsto -t$ ((E6)); (c): $z + v$ has the same derivative $F(z(t)) = F(z(t) + v)$, and $z(t) + v \in W + v = W$. □

Lemma 4.4 (Bounds and Lipschitz constants on compact sets)**.** *Let F be C^1 on the open set W and $K \subset W$ compact. Then F is bounded on K and there is $L \geq 0$ with $\|F(x) - F(y)\| \leq L\|x - y\|$ for $x, y \in K$.*
Source: Standard.

Proof. **Step S4.2: Via Mathlib.** Alternatively: F is locally Lipschitz at every point of W ((E3)), hence locally Lipschitz on K, and `LocallyLipschitzOn.exists_lipschitzOnWith_of_compact` ((E3)) gives L. The remaining steps are the explanatory proof of that fact.

Step S4.3: Bounded. F is continuous on $W \supset K$, so $\|F\| \leq M$ on the compact K ((E5)).

Step S4.4: Contradiction hypothesis. If no L works, for every $k \in \mathbb{N}$, $k \geq 1$, there are $x_k \neq y_k$ in K with $\|F(x_k) - F(y_k)\| > k\|x_k - y_k\|$; then $\|x_k - y_k\| < 2M/k \to 0$.

Step S4.5: Compactness. A subsequence $x_{k_j} \to p \in K$, hence also $y_{k_j} \to p$. By (E3) there are L_p and a neighbourhood N_p of p on which F is L_p-Lipschitz. For large j both points lie in N_p and $k_j > L_p$, which contradicts Step S4.4. □

Definition 4.5 (Uniform data)**.** Let F be C^1 on the open set W. A set $\Phi \subset E$ has *uniform data* (δ, M, L), with $\delta > 0$, $M, L \geq 0$, if for every $x \in \Phi$: $\overline{B}(x, \delta) \subset W$, $\|F\| \leq M$ on $\overline{B}(x, \delta)$ and F is L-Lipschitz on $\overline{B}(x, \delta)$. (Balls are convex, so by (E4) a bound $\|DF\| \leq L$ on $\overline{B}(x, \delta) \subset W$ gives the Lipschitz clause.)

Lemma 4.6 (Compact sets have uniform data)**.** *Let F be C^1 on the open set W and $K \subset W$ compact. Then K has uniform data (δ, M, L) for some $\delta > 0$, and $K_\delta = \{y : \exists x \in K,\ \|y - x\| \leq \delta\} \subset W$ is compact.*

Proof. **Step S4.6: Choice of δ.** If $K = \emptyset$ put $\delta = 1$: then $K_\delta = \emptyset$ and all assertions are vacuous (no infDist of the empty set is used). If $W = E$ put $\delta = 1$. Otherwise $K \neq \emptyset$ and $A = E \setminus W$ is closed and nonempty, so $x \mapsto \operatorname{dist}(x, A)$ is continuous and positive on K ((E5)); it has a minimum $2\delta > 0$ on the compact nonempty K, and then $K_\delta \subset W$. K_δ is the image of the compact $K \times \overline{B}(0, \delta)$ under $(x, w) \mapsto x + w$, hence compact.

Step S4.7: Data. Apply Lemma 4.4 to K_δ: $\|F\| \leq M$ and F is L-Lipschitz on $K_\delta \supset \overline{B}(x, \delta)$ for $x \in K$. □

Lemma 4.7 (Uniform data for angle-periodic fields)**.** *Let $E = \mathbb{R}^n \times \mathbb{R}^n$ (max norm), $U \subset \mathbb{R}^n$ open, $W = \mathbb{R}^n \times U$, F C^1 on W with $F(\theta + k, I) = F(\theta, I)$ for $(\theta, I) \in W$, $k \in \mathbb{Z}^n$. If $K \subset U$ is compact, then $\Phi = \mathbb{R}^n \times K$ has uniform data (δ, M, L) for some $\delta \in (0, 1]$.*

Proof. **Step S4.8: A compact cell.** If $K = \emptyset$ then $\Phi = \emptyset$ and any δ, M, L work. Otherwise choose $\delta \in (0, 1]$ with $K_\delta = \{I : \exists J \in K,\ |I - J| \leq \delta\} \subset U$ (as in Step S4.6, using the nonempty closed set $\mathbb{R}^n \setminus U$, or $\delta = 1$ if $U = \mathbb{R}^n$). $C = [-1, 2]^n \times K_\delta$ is compact and contained in W; let M, L_C be the constants of Lemma 4.4 for C.

Step S4.9: Translating a ball into the cell. Let $x = (\theta, I) \in \Phi$, $k = (\lfloor \theta_i \rfloor)_i \in \mathbb{Z}^n$ and $x' = x - (k, 0) \in [0, 1)^n \times K$. Since the norm is the max norm and $\delta \leq 1$, $\overline{B}(x', \delta) \subset C$, and $\overline{B}(x, \delta) = \overline{B}(x', \delta) + (k, 0) \subset W$. By periodicity $F(y + (k, 0)) = F(y)$ on $\overline{B}(x', \delta)$, so $\|F\| \leq M$ and F is L_C-Lipschitz on $\overline{B}(x, \delta)$. □

4.3. **Local existence.**

Lemma 4.8 (A priori short-time confinement)**.** *Let J be an interval, $t_0 \in J$, $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x$ and $\alpha'(t) = F(\alpha(t))$ for all $t \in J$ (derivative two-sided; F is only assumed to be a function), and suppose $\|F\| \leq M$ on $\overline{B}(x, \delta)$ with $M > 0$. Then $\|\alpha(t) - x\| \leq M|t - t_0|$ for all $t \in J$ with $|t - t_0| < \delta/M$; in particular $\alpha(t) \in \overline{B}(x, \delta)$.*
Source: Standard.

Proof. **Step S4.10: Forward bootstrap.** Let $t \in J$, $t_0 < t < t_0 + \delta/M$ (the case $t < t_0$ follows by Lemma 4.3(b) and the case $t = t_0$ is trivial). Let $A = \{s \in [t_0, t] : \|\alpha(u) - x\| \leq M(u - t_0)\ \forall u \in [t_0, s]\}$ and $s^* = \sup A$. $t_0 \in A$ and A is closed by continuity of α on J, so $s^* \in A$.

Step S4.11: Extension. If $s^* < t$, then $\|\alpha(s^*) - x\| \leq M(s^* - t_0) < \delta$, so by continuity there is $\eta > 0$, $s^* + \eta \leq t$, with $\alpha([s^*, s^* + \eta]) \subset \overline{B}(x, \delta)$. On this interval $\|\alpha'\| = \|F(\alpha)\| \leq M$, so by (E4) $\|\alpha(u) - \alpha(s^*)\| \leq M(u - s^*)$ and hence $\|\alpha(u) - x\| \leq M(u - t_0)$ there, contradicting the definition of s^*. So $s^* = t$. □

Proposition 4.9 (Local existence with uniform time)**.** *Let F be C^1 on the open set W and Φ have uniform data (δ, M, L); put $M' = \max(M, 1)$ and $\varepsilon = \delta/M'$. Then for every $x \in \Phi$ and every $t_0 \in \mathbb{R}$ there is $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x$, $\|\alpha(t) - x\| < \delta$ for all $t \in (t_0 - \varepsilon, t_0 + \varepsilon)$ and* `IsSolIn` $F\, W\, (t_0 - \varepsilon, t_0 + \varepsilon)\, \alpha$.
Source: Standard (Picard–Lindelöf).

Proof. **Step S4.12: Picard–Lindelöf.** Apply (E1) with $a = \delta$, bound M', Lipschitz constant L and $T = \varepsilon$ ($M'T = \delta \le a$): there is α with $\alpha(t_0) = x$ and derivative $F(\alpha(t))$ within $[t_0 - \varepsilon, t_0 + \varepsilon]$ at each point of it. At points t of the open interval this is a two-sided derivative, since the closed interval is a neighbourhood of t.

Step S4.13: Staying in W. By Lemma 4.8 on $J = (t_0 - \varepsilon, t_0 + \varepsilon)$, $\|\alpha(t) - x\| \le M'|t - t_0| < \delta$, so $\alpha(t) \in \overline{B}(x, \delta) \subset W$. □

Theorem 4.10 (Local existence, both time directions)**.** *Let F be C^1 on the open set W, $x_0 \in W$, $t_0 \in \mathbb{R}$. There are $\varepsilon > 0$ and $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x_0$ and* `IsSolIn` $F\, W\, (t_0 - \varepsilon, t_0 + \varepsilon)\, \alpha$. *Moreover ε may be taken uniform on any compact $K \subset W$ (for all $x_0 \in K$).*
Source: Standard.

Proof. **Step S4.14: Reduction.** $\{x_0\}$ (resp. K) is compact; Lemma 4.6 gives uniform data and Proposition 4.9 the claim. □

4.4. Uniqueness on every interval.

Theorem 4.11 (Uniqueness on arbitrary intervals)**.** *Let F be C^1 on the open set W, $J \subset \mathbb{R}$ an interval (of any kind: open, closed, half-open, unbounded, or a point), $t_0 \in J$, and $y_1, y_2 : \mathbb{R} \to E$ with* `IsSolIn` $F\, W\, J\, y_i$ *and $y_1(t_0) = y_2(t_0)$. Then $y_1 = y_2$ on J. No confinement, boundedness or smallness of either solution is assumed. In particular, if y_1, y_2 are solutions on intervals J_1, J_2 and agree at one point of $J_1 \cap J_2$, they agree on $J_1 \cap J_2$ (an interval), and two solutions which agree at one time of J agree at all times of J.*
Source: Standard; used for BM11 Theorem 2.1 (uniqueness) and in the maximal-solution argument of BM11 p. 13.

Proof. **Step S4.15: A compact piece.** Let $t_1 \in J$, $t_1 > t_0$ ($t_1 < t_0$ is the mirror image, using the left-derivative form of (E2); $t_1 = t_0$ is the hypothesis). Then $[t_0, t_1] \subset J$, each y_i is continuous on it, so $K = y_1([t_0, t_1]) \cup y_2([t_0, t_1])$ is compact ((E5)) and $K \subset W$.

Step S4.16: Lipschitz on K. Lemma 4.4 gives L with F L-Lipschitz on K.

Step S4.17: Apply (E2). Both y_i are continuous on $[t_0, t_1]$, have right derivative $F(y_i(t))$ on $[t_0, t_1)$ (a two-sided derivative gives a right derivative) and take values in K; so $y_1(t_1) = y_2(t_1)$. □

4.5. Maximal solutions and continuation.

Lemma 4.12 (Extension to an open interval)**.** *Let F be C^1 on the open set W, $J \neq \emptyset$ an interval and* `IsSolIn` $F\, W\, J\, y$. *Then there are an open interval $J' \supset J$ and y' with $y' = y$ on J and* `IsSolIn` $F\, W\, J'\, y'$.

Proof. **Step S4.18: Right end.** If J has a largest element b, apply Theorem 4.10 at $x_0 = y(b) \in W$, $t_0 = b$: α on $(b - \varepsilon, b + \varepsilon)$. Put $y_1(s) = y(s)$ for $s \le b$ and $y_1(s) = \alpha(s)$ for $s > b$. Then y_1 is a solution on $J_1 = J \cup (b, b + \varepsilon)$: at $t \in J$, $t < b$, and at $t \in (b, b + \varepsilon)$ it agrees with y, resp. α, near t; at $t = b$ it has derivative $F(y(b))$ within $(-\infty, b]$ (it equals y there) and within $[b, \infty)$ (it equals α there, as $\alpha(b) = y(b)$), so by (E6) it has the two-sided derivative. If J has no largest element, $J_1 = J$, $y_1 = y$.

Step S4.19: Left end. Apply Step S4.18 to the reversed solution $y_1(-\cdot)$ (Lemma 4.3(b)). The result has no endpoint, so is an open interval. □

Theorem 4.13 (Maximal solution)**.** *Let F be C^1 on the open set W and $x_0 \in W$. There are $T_- \in [-\infty, 0)$, $T_+ \in (0, +\infty]$ and $z : \mathbb{R} \to E$ with $z(0) = x_0$ and* `IsSolIn` $F\, W\, (T_-, T_+)\, z$*, such that every solution y with $y(0) = x_0$ on an interval $J \ni 0$ satisfies $J \subset (T_-, T_+)$ and $y = z$ on J. Then (T_-, T_+) is the* maximal interval *and $z|_{(T_-,T_+)}$ the* maximal solution*; both are unique. For initial time t_0 instead of 0 use Lemma 4.3(a).*
Source: BM11 p. 13 ("maximal time of existence"); standard.

Proof. **Step S4.20: The union.** Let $\mathcal{J}$ be the set of open intervals $J \ni 0$ carrying a solution y_J with $y_J(0) = x_0$; $\mathcal{J} \neq \emptyset$ by Theorem 4.10. The union I of $\mathcal{J}$ is an open interval $(T_-, T_+) \ni 0$ (a union of intervals containing 0).

Step S4.21: Gluing. For $t \in I$ choose $J \in \mathcal{J}$ with $t \in J$ and set $z(t) = y_J(t)$ (and $z(t) = x_0$ outside I). This is independent of J: for $J, J' \in \mathcal{J}$, $y_J = y_{J'}$ on $J \cap J' \ni 0$ by Theorem 4.11. Each $t \in I$ has an open neighbourhood $J \in \mathcal{J}$ on which $z = y_J$, so z is a solution on I (local nature, (iii) above).

Step S4.22: Maximality. If y is a solution on an interval $J \ni 0$ with $y(0) = x_0$, Lemma 4.12 gives an open $J' \supset J$ and a solution y' extending y, so $J' \in \mathcal{J}$, $J \subset J' \subset I$, and $y = y' = z$ on J (Theorem 4.11). Uniqueness of the interval (T_-, T_+), and of z as a function on it (equality on this interval of any two representatives; values outside it are arbitrary), is immediate from the maximality property applied twice. □

Theorem 4.14 (Continuation criterion, uniform form)**.** *Let F be C^1 on the open set W, $x_0 \in W$, (T_-, T_+) and z the maximal interval and solution of Theorem 4.13. Let $\Phi \subset E$ have uniform data (δ, M, L) (Definition 4.5). If $T_+ < \infty$ there is no $t_* \in [0, T_+)$ with $z(t) \in \Phi$ for all $t \in [t_*, T_+)$. Symmetrically, if $T_- > -\infty$ there is no $t_* \in (T_-, 0]$ with $z(t) \in \Phi$ for all $t \in (T_-, t_*]$. Equivalently: if the solution stays in Φ near the right (left) end of its maximal interval, that end is $+\infty$ ($-\infty$).*
Source: Standard; BM11 p. 13 ("this solution is defined for all time"), Pöschel 1993 p. 198 ("$T = T_*$").

Proof. **Step S4.23: Set-up.** Suppose $T_+ < \infty$ and $z(t) \in \Phi$ on $[t_*, T_+)$. Let $\varepsilon = \delta / \max(M, 1)$ from Proposition 4.9. Choose $t_1 \in [t_*, T_+)$ with $t_1 > T_+ - \varepsilon$.

Step S4.24: A solution through $z(t_1)$. Proposition 4.9 with $x = z(t_1) \in \Phi$ and $t_0 = t_1$ gives α on $(t_1 - \varepsilon, t_1 + \varepsilon)$ with $\alpha(t_1) = z(t_1)$. By Theorem 4.11 on $(t_1 - \varepsilon, t_1 + \varepsilon) \cap (T_-, T_+)$, $\alpha = z$ there.

Step S4.25: Gluing past T_+. Put $y = z$ on (T_-, T_+) and $y = \alpha$ on $[T_+, t_1 + \varepsilon)$; then $y = \alpha$ on $(t_1 - \varepsilon, t_1 + \varepsilon)$ and $y = z$ on (T_-, T_+), two open sets covering $(T_-, t_1 + \varepsilon)$ on each of which y is a solution. So y is a solution on $(T_-, t_1 + \varepsilon) \ni 0$ with $y(0) = x_0$, and $t_1 + \varepsilon > T_+$ contradicts Theorem 4.13. The left end follows by Lemma 4.3(b). □

Corollary 4.15 (Compact continuation)**.** *In the setting of Theorem 4.14, if $K \subset W$ is compact and $z(t) \in K$ for all $t \in [t_*, T_+)$ (some $t_* \in [0, T_+)$), then $T_+ = +\infty$; likewise on the left.*

Proof. Lemma 4.6 gives uniform data for K; apply Theorem 4.14. □

Corollary 4.16 (Closed sets, periodic fields)**.** *Let $E = \mathbb{R}^n \times \mathbb{R}^n$, $U \subset \mathbb{R}^n$ open, $W = \mathbb{R}^n \times U$, F C^1 on W with $F(\theta + k, I) = F(\theta, I)$ for $k \in \mathbb{Z}^n$. If the maximal solution through $x_0 \in W$ satisfies $z(t)_2 \in K$ for $t \in [t_*, T_+)$ with $K \subset U$ compact, then $T_+ = +\infty$ (likewise on the left). The angle components $z(t)_1$ may be unbounded.*

Proof. Lemma 4.7 gives uniform data for $\Phi = \mathbb{R}^n \times K$; apply Theorem 4.14. □

Remark 4.17 (What the closed-set variant really needs). Staying in a closed non-compact set $\Phi \subset W$ does *not* suffice in general (the solution of $\dot{x} = x^2$ on $E = \mathbb{R}$, $W = \mathbb{R}$, $x_0 = 1$, $x(t) = 1/(1 - t)$, stays in the closed set $[1, \infty)$ for $0 \le t < 1$ and blows up at $T_+ = 1$). What suffices is the uniform data (δ, M, L) of Definition 4.5: a δ-neighbourhood of Φ inside W on which F is bounded and

Lipschitz, uniformly. This holds for compact sets, for angle-periodic fields over a compact action set (Lemma 4.7), and for the Newtonian regions bounded away from collisions with bounded velocities (Corollary 4.27).

Theorem 4.18 (Confinement glue)**.** *Let F be C^1 on the open set W, $x_0 \in W$, $\tau > 0$, and $\Phi \subset E$ a set with uniform data (e.g. compact $\Phi \subset W$, or as in Lemma 4.7). Assume the* a priori confinement*: every solution y on an open interval $J \ni 0$ with $y(0) = x_0$ satisfies $y(t) \in \Phi$ for all $t \in J$ with $|t| \le \tau$. Then the maximal interval contains $[-\tau, \tau]$ in its interior, so z is a solution on $[-\tau, \tau]$ with $z(t) \in \Phi$ for $|t| \le \tau$, and every solution on $[-L, L]$ $(0 \le L \le \tau)$ with initial value x_0 equals z there.*
Source: BM11 p. 13 (the two-case argument "Therefore this solution is defined for all time"); Pöschel 1993 p. 198.

Proof. **Step S4.26: No exit before τ.** Suppose $T_+ \le \tau$. The maximal solution z on (T_-, T_+) satisfies, for $t \in [0, T_+)$ $(|t| < \tau)$, $z(t) \in \Phi$ by confinement. Theorem 4.14 gives $T_+ = \infty$, a contradiction. So $T_+ > \tau$, and likewise $T_- > -\tau$.

Step S4.27: Conclusion. $[-\tau, \tau] \subset (T_-, T_+)$ so z is a solution on $[-\tau, \tau]$ and $z(t) \in \Phi$ for $|t| \le \tau$ by confinement applied to z itself. Uniqueness is Theorem 4.11. □

4.6. **First exit from an open set; barriers.**

Lemma 4.19 (First exit)**.** *Let* `IsSolIn` $F\, W\, J\, z$ *with J an interval, $0 \in J$, $V \subset E$ open and $z(0) \in V$. Put $A = \{t \in J : t \ge 0,\ z(t) \notin V\}$. Then exactly one of the following holds.*

- *(i) $A = \emptyset$: $z(t) \in V$ for all $t \in J$, $t \ge 0$.*
- *(ii) $A \neq \emptyset$. Then $t^* = \min A$ exists, $t^* > 0$, $t^* \in J$, $z(t) \in V$ for $t \in [0, t^*)$ and $z(t^*) \in \partial V = \overline{V} \setminus V$. The number t^* is the* first exit time *and equals $\sup\{t \in J : t \ge 0,\ z([0,t]) \subset V\}$; in particular t^* is finite and lies in J (t^* is "inside the solution's interval" and the solution is defined, continuous, at t^*).*

The same holds backwards in time (replace $z(t)$ by $z(-t)$, Lemma 4.3(b)). Only continuity of z on J is used; F and W play no role.
Source: Standard; used for BM11 p. 13 ("smallest time t_* such that $\omega(t_*) \in R_K$") and Pöschel 1993 p. 198–199 (stopping-time arguments).

Proof. **Step S4.28: Closedness.** z is continuous on J (consequence (ii)), $E \setminus V$ is closed, so A is closed in $J \cap [0, \infty)$.

Step S4.29: The minimum. If $A \neq \emptyset$, $t^* = \inf A \in [0, \infty)$; $t^* \in J$ since J is an interval containing 0 and a point of A that is $\ge t^*$. By closedness $t^* \in A$, i.e. $z(t^*) \notin V$. As $z(0) \in V$, $t^* > 0$.

Step S4.30: Boundary point. For $t \in [0, t^*)$, $t \notin A$, so $z(t) \in V$. By continuity $z(t^*) = \lim_{t \uparrow t^*} z(t) \in \overline{V}$, and $z(t^*) \notin V$; as V is open, $\overline{V} \setminus V = \partial V$.

Step S4.31: The supremum formula. $z([0,t]) \subset V$ iff $t < t^*$ (for $t \in J$, $t \ge 0$), so the supremum is t^*. □

Definition 4.20 (Invariant sets)**.** $S \subset E$ is *invariant* for (F, W) if for every interval J' and every y with `IsSolIn` $F\, W\, J'\, y$: if $y(t_1) \in S$ for one $t_1 \in J'$ then $y(t) \in S$ for all $t \in J'$. (Equilibria $\{x : F(x) = 0\}$ are invariant by Theorem 4.11; so is any union of such sets and any set $\{G = c\}$ for a first integral G.)

Corollary 4.21 (Barrier)**.** *Let F be C^1 on the open set W, V open,* `IsSolIn` $F\, W\, J\, z$ *with J an interval, $0 \in J$, $z(0) \in V$. Let $G \subset E$ with $z(J) \subset G$, and suppose $\partial V \cap G \subset S$ where S is invariant and $z(0) \notin S$ (e.g. G an energy level, Theorem 4.23, and $\partial V \cap G$ a union of invariant sets not meeting the initial point). Then $z(t) \in V$ for all $t \in J$ (both time directions).*

Proof. **Step S4.32: Forward.** If not, Lemma 4.19(ii) gives $t^* \in J$ with $z(t^*) \in \partial V \cap G \subset S$; invariance gives $z(0) \in S$, a contradiction.

Step S4.33: Backward. Apply Step S4.32 to $t \mapsto z(-t)$, a solution of $-F$ on $-J$; G, ∂V are unchanged and S is invariant for $-F$ as well (reversal maps solutions of F to solutions of $-F$ and back). □

4.7. **Hamiltonian layer.** Phase space $P = \mathbb{R}^n \times \mathbb{R}^n$, points $z = (\theta, I)$, max norm. X_H is the frozen `KAM.hamiltonianVectorField`, Definition 1.3 (convention of [K:`bp:p0:hamiltonian`]): $X_H(z) = \big((\partial_{I_i} H(z))_i, (-\partial_{\theta_i} H(z))_i\big)$, with $\partial_{I_i} H(z) = DH(z)(0, e_i)$ and $\partial_{\theta_i} H(z) = DH(z)(e_i, 0)$.

Lemma 4.22 (Regularity of X_H)**.** *Let $W \subset P$ be open and $H : P \to \mathbb{R}$ be C^{k+1} on W, $k \geq 0$. Then X_H is C^k on W; for $k \geq 1$ it is C^1 (hence locally Lipschitz) on W, and H is differentiable at every point of W.*
Source: Standard.

Proof. **Step S4.34: Derivative is C^k.** $DH : W \to \mathcal{L}(P, \mathbb{R})$ is C^k on the open W ((E6), `ContDiffOn.fderiv_of_isOpen`).

Step S4.35: Components. Each coordinate of X_H is $\pm$ the composition of DH with the continuous linear evaluation $\ell \mapsto \ell(0, e_i)$ or $\ell \mapsto \ell(e_i, 0)$; hence C^k, and a map into $\mathbb{R}^n \times \mathbb{R}^n$ with C^k coordinates is C^k. □

Theorem 4.23 (Energy conservation)**.** *Let $H : P \to \mathbb{R}$, $J \subset \mathbb{R}$ an interval and $z : \mathbb{R} \to P$ such that for every $t \in J$, H is differentiable at $z(t)$ and z has derivative $X_H(z(t))$ at t. Then $t \mapsto H(z(t))$ is constant on J. (No regularity of H beyond the stated pointwise differentiability, and no confinement.)*
Source: BM11 p. 13 and Pöschel 1993 p. 198 ("energy conservation", "$\Delta H = H(I(t), \theta(t))|_0^T$ vanishes"); standard.

Proof. **Step S4.36: Chain rule.** For $t \in J$, $\frac{d}{dt} H(z(t)) = DH(z(t))\big[X_H(z(t))\big]$ ((E6)).

Step S4.37: Cancellation. Write $z = z(t)$, $\ell = DH(z)$ and $X_H(z) = (a, b)$ with $a_i = \ell(0, e_i)$, $b_i = -\ell(e_i, 0)$. By linearity, $\ell(a, b) = \sum_i a_i\, \ell(e_i, 0) + \sum_i b_i\, \ell(0, e_i) = \sum_i \ell(0, e_i)\ell(e_i, 0) - \sum_i \ell(e_i, 0)\ell(0, e_i) = 0$.

Step S4.38: Constancy. Let $s < t$ in J. $H \circ z$ is continuous on $[s, t] \subset J$ and has derivative 0 at every point, so $H(z(t)) = H(z(s))$ ((E6)). □

Lemma 4.24 (Periodicity in the angles passes to solutions)**.** *Let $W \subset P$ be open with $W + (k, 0) = W$ for all $k \in \mathbb{Z}^n$, and $H : P \to \mathbb{R}$ with $H(\theta + k, I) = H(\theta, I)$ for $(\theta, I) \in W$, $k \in \mathbb{Z}^n$. Then $X_H(\theta + k, I) = X_H(\theta, I)$ on W. Consequently, if `IsSolIn` $X_H\, W\, J\, z$ then `IsSolIn` $X_H\, W\, J\, (z + (k, 0))$ for every $k \in \mathbb{Z}^n$, and $H \circ (z + (k, 0)) = H \circ z$.*
Source: Standard.

Proof. **Step S4.39: Field.** Let $\tau_k(x) = x + (k, 0)$. On the open set W, $H \circ \tau_k = H$, so $DH = D(H \circ \tau_k) = DH \circ \tau_k$ at each point of W (the derivative of a translate; both sides are 0 if H is not differentiable, by the definition of `fderiv`). Evaluating on $(0, e_i)$ and $(e_i, 0)$ gives $X_H \circ \tau_k = X_H$ on W.

Step S4.40: Solutions. Lemma 4.3(c) with $v = (k, 0)$ applies to $F = X_H$. □

4.8. **The frozen solution predicates are instances.**

Proposition 4.25 (Track N and track I Hamiltonian predicates)**.** *Let $H : P \to \mathbb{R}$, $U \subset \mathbb{R}^n$ open, $W \subset P$ open with $\mathbb{R}^n \times U \subset W$, and H be C^2 on W. Put $W_U = \mathbb{R}^n \times U$. For every $J \subset \mathbb{R}$ and $z : \mathbb{R} \to P$,*

$$\texttt{IsSolutionOn}\, H\, U\, J\, z \iff \texttt{IsSolIn}\, X_H\, W_U\, J\, z \iff \texttt{IsHamiltonianSolutionOn}\, H\, U\, J\, z,$$

and X_H is C^1 on W_U. (The two frozen Hamiltonian predicates have the same formula up to the letter naming the action set, and under the regularity hypotheses both are equivalent to `IsSolIn` *as shown below,* `KAM.Nekhoroshev.IsSolutionOn` *and* `KAM.Iso.IsHamiltonianSolutionOn`*.)*
Frozen Lean declaration: `KAM.Nekhoroshev.IsSolutionOn, KAM.Iso.IsHamiltonianSolutionOn`
Source: Ours (bookkeeping).

Proof. **Step S4.41: Membership.** $z(t)_2 \in U \iff z(t) \in W_U$ by definition of W_U.

Step S4.42: Differentiability is free. For $z(t) \in W_U \subset W$, H is C^2 on the open W, hence differentiable at $z(t)$ (Lemma 4.22); so the clause $\texttt{DifferentiableAt}\,\mathbb{R}\,H\,(z\,t)$ is implied by $z(t) \in W_U$, and the two predicates coincide. Smoothness of X_H: Lemma 4.22, $k = 1$, restricted to $W_U \subset W$. □

Proposition 4.26 (Newtonian predicates)**.** *Let $N \geq 2$, $G \in \mathbb{R}$, $\texttt{mass} : Fin\,N \to \mathbb{R}$. Let $E_N = (Fin\,N \to \mathbb{R}^2) \times (Fin\,N \to \mathbb{R}^2)$ (max norm), $\ell = \texttt{planeLength}$,*

$$W_N = \{(q,v) : \ell(q_j - q_i) > 0\ \forall i \neq j\}, \qquad F_N(q,v) = \Big(v,\ \big(\textstyle\sum_{j\neq i} \frac{G\,\mathrm{mass}_j}{\ell(q_j-q_i)^3}(q_j - q_i)\big)_i\Big).$$

Then W_N is open, F_N is C^∞ on W_N, and for $Q, V : \mathbb{R} \to (Fin\,N \to \mathbb{R}^2)$ with $x(t) = (Q(t), V(t))$: $\texttt{IsNewtonSolution}\,G\,\mathrm{mass}\,Q\,V \iff \texttt{IsSolIn}\,F_N\,W_N\,\mathbb{R}\,x$. Likewise for $\texttt{IsFixedSolution}\,G\,m_0\,m\,\varepsilon$ with $W_N' = W_N \cap \{\ell(q_i) > 0\ \forall i\}$ and the field $F'(q,v) = (v, (-\frac{Gm_0}{\ell(q_i)^3}q_i + \varepsilon\sum_{j\neq i}\frac{Gm_j}{\ell(q_j-q_i)^3}(q_j - q_i))_i)$, and for $\texttt{IsRestrictedSolutionOn}\,\mu\,J$ with $W_\mu = \{(q,p) : \texttt{CollisionFree}\,\mu\,(q,p)\}$ and the total extension F_μ of $\texttt{restrictedVectorField}\,\mu$ by the same formula (with the values off W_μ irrelevant): $\texttt{IsRestrictedSolutionOn}\,\mu\,J\,z \iff \texttt{IsSolIn}\,F_\mu\,W_\mu\,J\,z$. In all three cases F is C^∞ on the open set W and the general layer applies with $J = \mathbb{R}$ (resp. the given J).
Frozen Lean declaration: `KAM.Planetary.IsNewtonSolution, KAM.Planetary.IsFixedSolution, KAM.Iso.IsRestrictedSolutionOn`
Source: Ours (bookkeeping); Fejoz Section 6.1, Newton equations as in the frozen text.

Proof. **Step S4.43: Open domain and smoothness.** $\ell(w)^2 = \texttt{planeSq}(w) = w_0^2 + w_1^2$ is a polynomial, so $\{\ell > 0\} = \{\texttt{planeSq} > 0\}$ is open, and $\ell = \sqrt{\cdot}$ is C^∞ where $\texttt{planeSq} > 0$. Hence $\ell(q_j - q_i)^{-3}$ is C^∞ on W_N and F_N is a finite sum of products of C^∞ maps. Same for F' and F_μ ($\texttt{radius} = \sqrt{\texttt{radiusSq}}$ with the shifts by $\texttt{primary}_{1,2}\mu$).

Step S4.44: Derivatives of pairs. A map into a finite product has derivative (a,b) at t iff each component has the corresponding derivative (`hasDerivAt_pi`, `HasDerivAt.prodMk` and converse). The first block of the frozen definition (distance positivity) is membership in W_N; the second block says Q has derivative $V(t)$; the third says V has derivative the force. Together they say exactly that x has derivative $F_N(x(t))$ at t. □

Corollary 4.27 (Newtonian continuation)**.** *In the setting of Proposition 4.26, let x be the maximal solution through $x_0 \in W_N$ (Theorem 4.13), $\rho > 0$, $\bar{V} \geq 0$ and*

$$\Phi_{\rho,\bar{V}} = \{(q,v) :\ \ell(q_j - q_i) \geq \rho\ (i \neq j),\ |v| \leq \bar{V}\}.$$

If $T_+ < \infty$ then for every $\rho, \bar{V}$ and every $t_* \in [0, T_+)$ there is $t \in [t_*, T_+)$ with $x(t) \notin \Phi_{\rho,\bar{V}}$ (a finite-time singularity needs near-collisions or unbounded velocities; positions may be unbounded).

The same for F' with the additional condition $\ell(q_i) \geq \rho$, and for F_μ with $\ell \geq \rho$ from both primaries and $|q|, |p| \leq \bar{V}$.

Proof. **Step S4.45: Uniform data.** Let $\delta = \rho/8$. If $|y - x| \leq \delta$ (max norm) with $x \in \Phi_{\rho,\bar{V}}$ then each $\ell(q'_j - q'_i) \geq \rho - 2\sqrt{2}\,\delta \geq \rho/2$, so $y \in W_N$ and $\overline{B}(x, \delta) \subset W_N$. On this ball $\ell \geq \rho/2$ and $|v'| \leq \bar{V} + \delta$; the entries of F_N and DF_N are polynomials in v', the differences $q'_j - q'_i$ (Euclidean length $\leq \ell(q'_j - q'_i)$) and the inverse powers ℓ^{-3}, ℓ^{-5} ($D(w/\ell^3) = \ell^{-3}I - 3\ell^{-5}ww^\top$), hence bounded by a constant $C(N, G, \mathrm{mass}, \rho, \bar{V})$ *independent of the position of* x. By (E4) on the convex ball, F_N is C-Lipschitz there. So $\Phi_{\rho,\bar{V}}$ has uniform data.

Step S4.46: Apply. Theorem 4.14 gives the claim; the other two fields are identical in structure (for F_μ the extra terms are linear in (q, p) and bounded on $|q|, |p| \leq \bar{V}$). □

4.9. **The track-N statements.** Throughout, $R, s > 0$, $h : \mathbb{C}^n \to \mathbb{C}$ with `IsAnalyticIntegrable` $R\,s\,h$, $\varepsilon \geq 0$ and $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ with `IsAnalyticPerturbation` $R\,s\,\varepsilon\,f$ (Definitions 1.11, 1.12), $H = H_{h,f}$ (Definition 1.4), $W_s = \mathbb{R}^n \times B(0, R + s) \subset P$ (open; $B(0, R + s) = \mathcal{B}_s(R)$ by Remark 1.9), and $U \subset B(0, R + s)$ open (in N-HEAD-C, $U = B(0, R)$).

Lemma 4.28 (Input from Part 1: H is smooth on W_s)**.** *$H_{h,f}$ is C^∞ (in particular C^2) on the open set W_s, $X_H = X_{h+f}$ on $W_s \cap (\mathbb{R}^n \times \mathbb{R}^n)$, and $H(\theta + k, I) = H(\theta, I)$ for $(\theta, I) \in W_s$, $k \in \mathbb{Z}^n$.*
Source: Part 1, Lemma 2.3(b)–(d) (real smoothness on W_s, field $X_H = X_{h+f} \circ \iota$, periodicity); ingredients [K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:hamiltonian-real`].

Proof. **Step S4.47: Citation.** (b), (c), (d) of Lemma 2.3 are exactly the three claims. For orientation, the argument is as follows.

Step S4.48: Regularity. The function $G(\theta, I) = h(I) + f(\theta, I)$ is holomorphic on the open set $D_s(R) \supset W_s$ (real points (θ, I) with $I \in \mathcal{B}_s(R) = B(0, R)_s \cap \mathbb{R}^n$ lie in $T_s \times B(0, R)_s$), hence C^∞ in the real sense ([K:`bp:p1:holomorphic-regularity`]); it is real on W_s by Definition 1.11(iv) and 1.12(iii). So $H = \mathrm{Re}\,G = G$ on W_s is the restriction of C^∞ map to a real subspace: C^∞ on W_s; $X_H = X_G$ at real points is [K:`bp:p1:hamiltonian-real`]. (Packaged as Lemma 2.3.)

Step S4.49: Periodicity. $f(\theta + k, I) = f(\theta, I)$ for real θ, $I \in \mathcal{B}_s(R)$, and h depends on I only. □

Theorem 4.29 (X_H is C^1, hence locally Lipschitz)**.** *$X_{H_{h,f}}$ is C^1 on $W_s \supset \mathbb{R}^n \times B(0, R)$; consequently it is Lipschitz on every compact subset of W_s, and on every set $\mathbb{R}^n \times K$ with $K \subset B(0, R+s)$ compact there are δ, M, L as in Lemma 4.7.*
Source: BM11 p. 13 (implicit in "the solution"); standard.

Proof. **Step S4.50: Smoothness.** Lemma 4.28 and Lemma 4.22 ($k = 1$).

Step S4.51: Lipschitz. Lemma 4.4 on compact sets, Lemma 4.7 (with $U = B(0, R+s)$, using the periodicity of X_H from Lemma 4.24 and Lemma 4.28) for the cylinders. □

Theorem 4.30 (Local existence and uniqueness for `IsSolutionOn`)**.** *For every $t_0 \in \mathbb{R}$ and $z_0 \in \mathbb{R}^n \times U$ there are $\epsilon > 0$ and $y : \mathbb{R} \to P$ with $y(t_0) = z_0$ and* `IsSolutionOn` $H\,U\,(t_0 - \epsilon, t_0 + \epsilon)\,y$ *(both time directions). If J is any interval, $t_0 \in J$ and y_1, y_2 satisfy* `IsSolutionOn` $H\,U\,J\,y_i$ *with $y_1(t_0) = y_2(t_0)$, then $y_1 = y_2$ on J.*
Source: BM11 p. 13 and Theorem 2.1 p. 5; Picard–Lindelöf (external fact (E1)).

Proof. **Step S4.52: Reduction.** Proposition 4.25 (with $W = W_s$, H C^2 there by Lemma 4.28) turns `IsSolutionOn` $H\,U$ into `IsSolIn` $X_H\,(\mathbb{R}^n \times U)$ with X_H C^1 on $\mathbb{R}^n \times U$.

Step S4.53: Existence, uniqueness. Theorem 4.10 and Theorem 4.11. □

Theorem 4.31 (Energy conservation for `IsSolutionOn`). *If `IsSolutionOn` $H\,U\,J\,z$ for an interval J (here H may be any function $P \to \mathbb{R}$ and U any set), then $H(z(t)) = H(z(t'))$ for $t, t' \in J$. In particular, for $H = H_{h,f}$ and $J = [0,T]$ the quantity $\Delta H = H(z(t))|_0^T$ of Pöschel's Lemma 1 vanishes.*
Source: Pöschel 1993 p. 198 ("by energy conservation ... ΔH vanishes"; page read); BM11 p. 13.

Proof. **Step S4.54: Direct.** The definition gives differentiability of H at every $z(t)$ and $z'(t) = X_H(z(t))$; Theorem 4.23. □

Theorem 4.32 (Maximal solutions and continuation). *Let $z_0 = (\theta_0, I_0)$ with $I_0 \in U$. There is a unique maximal interval $(T_-, T_+) \ni 0$ and a solution z with $z(0) = z_0$ and `IsSolutionOn` $H\,U\,(T_-,T_+)\,z$, such that every y with $y(0) = z_0$ and `IsSolutionOn` $H\,U\,J\,y$ on an interval $J \ni 0$ has $J \subset (T_-,T_+)$ and $y = z$ on J (Theorem 4.13 through Proposition 4.25). If $K \subset U$ is compact and $z(t)_2 \in K$ for all $t \in [t_*, T_+)$ for some $t_* \in [0, T_+)$, then $T_+ = +\infty$; likewise on the left. The angle components $z(t)_1$ may be unbounded: no obstruction arises because X_H is 1-periodic in θ.*
Source: BM11 p. 13 ("From this estimate we can deduce that the solution $I(t)$ belongs to some compact ball ... this solution is defined for all time"); Pöschel 1993 p. 198 ("$T = T_*$").

Proof. **Step S4.55: General statements.** By Proposition 4.25, `IsSolutionOn` $H\,U\,J\,y \iff$ `IsSolIn` $X_H\,(\mathbb{R}^n \times U)\,J\,y$ and X_H is C^1 on $W = \mathbb{R}^n \times U$; maximality is Theorem 4.13.

Step S4.56: Periodic cylinder. X_H is 1-periodic in θ on W (Lemmas 4.24, 4.28). Corollary 4.16 yields $T_+ = \infty$. □

Theorem 4.33 (Uniqueness on closed intervals). *For $L \geq 0$ and y_1, y_2 with $y_1(0) = y_2(0)$ and `IsSolutionOn` $H\,U\,[-L,L]\,y_i$ one has $y_1 = y_2$ on $[-L,L]$. In particular a solution on $[-L,L]$ agrees with every solution on a larger interval containing $[-L,L]$ and the same initial value (clause (d) of Theorem 1.19, where $U = B(0,R)$, $L \in [0,\tau]$, y arbitrary and z the exhibited solution). No confinement of y is assumed: Gronwall is replaced by Theorem 4.11, whose Lipschitz constant comes from the compact set $y_1([-L,L]) \cup y_2([-L,L])$.*
Source: Standard (Gronwall); clause (d) of the frozen N-HEAD-C.

Proof. **Step S4.57: Apply.** $J = [-L,L]$, $t_0 = 0$ in Theorem 4.30 (uniqueness part). □

Theorem 4.34 (Confinement glue). *Let $\theta_0 \in \mathbb{R}^n$, $I_0 \in U$, $\tau > 0$ and $\rho > 0$ with $\overline{B}(I_0, \rho) \subset U$ (e.g. $|I_0| < R/2$, $\rho \leq R/4$, $U = B(0,R)$). Assume that for every open interval $J \ni 0$ and every y with $y(0) = (\theta_0, I_0)$ and `IsSolutionOn` $H\,U\,J\,y$ one has $|y(t)_2 - I_0| \leq \rho$ for all $t \in J$ with $|t| \leq \tau$. Then there is z with $z(0) = (\theta_0, I_0)$, `IsSolutionOn` $H\,U\,[-\tau,\tau]\,z$ and $|z(t)_2 - I_0| \leq \rho$ for $|t| \leq \tau$; and for every $L \in [0,\tau]$ and every y with $y(0) = z(0)$ and `IsSolutionOn` $H\,U\,[-L,L]\,y$, $y = z$ on $[-L,L]$. This is the assembly of clauses (a)–(d) of Theorem 1.19 from an a priori estimate (Part 10); $\rho = C\varepsilon^{1/(2n)}$ there.*
Source: BM11 p. 13 (two cases, "Therefore this solution is defined for all time"); Pöschel 1993 p. 198.

Proof. **Step S4.58: Compact confinement set.** $K = \overline{B}(I_0, \rho)$ is compact and $\subset U$; $\Phi = \mathbb{R}^n \times K$ has uniform data by Lemma 4.7 (with $W = \mathbb{R}^n \times U$, $F = X_H$ periodic by Lemma 4.24).

Step S4.59: General glue. By Proposition 4.25 the hypothesis reads: every general solution on an open $J \ni 0$ with this initial value stays in Φ for $|t| \leq \tau$. Theorem 4.18 gives z on $[-\tau,\tau]$ in Φ, i.e. $|z(t)_2 - I_0| \leq \rho$, and uniqueness on $[-L,L]$ (it is also Theorem 4.33). □

4.10. **Label index for tracks I and P.**

`bpN:p3:gen-sol`, `gen-symmetry`	predicate; time shift, reversal, space translation
`gen-lip`, `gen-uniform`, `gen-uniform-compact`, `gen-uniform-periodic`	Lipschitz and uniform data
`gen-short`, `gen-uniform-exist`, `gen-local`	local existence, uniform time
`gen-uniqueness`	uniqueness on all intervals, no confinement
`gen-extend`, `gen-maximal`	maximal solutions
`gen-continuation`, `-compact`, `-periodic`, `gen-confinement`	continuation, closed sets with uniform bounds, confinement glue
`gen-exit`, `gen-invariant`, `gen-barrier`	first exit, barrier
`ham-smooth`, `ham-energy`, `ham-periodic`	Hamiltonian layer
`ham-reduction`, `newton-reduction`, `newton-continuation`	frozen predicates of I, N, P
`holo-input`, `lipschitz`, `local-exist`, `energy`, `continuation`, `uniqueness-closed`, `confinement-glue`	track N specializations

5. Weighted Fourier norms and Cauchy estimates (shared by tracks N, P, K)

This part is Appendix B and Appendix C of Pöschel 1993 (pp. 213–215) in the frozen conventions. It is shared infrastructure: track P (averaging, two-scale KAM) cites its results as [N:`bpN:p4:...`]. The statements are therefore made for functions $u(\theta, I)$ with $\theta \in \mathbb{T}^n$ and I in a neighbourhood of an *arbitrary* set $D \subset \mathbb{C}^m$, with m independent of n; only the Poisson bracket (Lemma 5.7) needs $m = n$. All norms on $\mathbb{R}^N$, $\mathbb{C}^N$ are max norms; no Euclidean norm occurs in this part.
Normalisation. Angles have period 1, characters are $\chi_k(\theta) = e^{2\pi i k\cdot\theta}$, and the strip width s is the *geometric* width, $T_s = \{\theta \in \mathbb{C}^n : |\operatorname{Im}\theta|_\infty < s\}$ ([K:`bp:p0:domains`]); the weight is $e^{2\pi|k|_1 s}$, that is, the coefficient radius is $2\pi s$ as in `docs/fourier-conventions.md`. With this choice every formula of Pöschel's Appendix B holds *verbatim*, with no factor 2π (Remark 5.11); for the θ-derivative the factor 2π of $\partial_{\theta_j}\chi_k = 2\pi i k_j \chi_k$ is absorbed by the weight. Widths are r (actions) and s (angles); the *losses* are ρ (actions) and σ (angles), as in Pöschel's (ρ, σ). (The brief's $|u|_{D,r,\sigma}$ is $|u|_{D,r,s}$ here.)

5.1. **External facts.** Besides [K:`bp:p1:ext-cauchy`], [K:`bp:p1:ext-integral`], [K:`bp:p1:ext-fourier`], [K:`bp:p1:ext-mvt`], the proofs use:

(E1) *Holomorphic dependence on a parameter.* Let $Q \subset \mathbb{R}^p$ be a compact box, $U \subset \mathbb{C}^q$ open, $G : Q \times U \to \mathbb{C}^d$ continuous, holomorphic in $I \in U$ for each $x \in Q$, with $D_I G$ continuous on $Q \times U$. Then $g(I) = \int_Q G(x, I)\,dx$ is holomorphic on U and $Dg(I) = \int_Q D_I G(x, I)\,dx$. (Mathlib: `hasFDerivAt_integral_of_dominated_of_fderiv_le` with a bound on a compact polydisc.) Exists.

(E2) *Uniform limits.* If K_j are holomorphic on an open $U \subset \mathbb{C}^q$ and $K_j \to K$ uniformly on U, then K is holomorphic on U, and $DK_j \to DK$ uniformly on $\{z : \bar{P}(z, \eta) \subset U\}$ for each $\eta > 0$ (the derivatives by Lemma 5.8 applied to $K_j - K_l$; holomorphy by [K:`bp:p1:uniform-limits`]). Exists in this form modulo that lemma.

(E3) *Series and integrals.* Tonelli for nonnegative double series and for integrals of nonnegative functions; the Cauchy product of two absolutely convergent series; termwise integration of a uniformly convergent series over a box ([K:`bp:p1:ext-integral`]). Exists.

5.2. **Domains, function class, norm.**

Definition 5.1 (Domains)**.** Let $m, n \geq 0$, $D \subset \mathbb{C}^m$ any subset, $r, s > 0$. Put $V_r D = \{I \in \mathbb{C}^m : \exists I' \in D,\ |I - I'|_\infty < r\}$ and $\mathcal{D}_{r,s} = T_s \times V_r D \subset \mathbb{C}^n \times \mathbb{C}^m$ (Pöschel's $V_{r,s}$). For $z = (\theta, I)$ let

$\bar P(z;a,b)=\{(\theta',I'):|\theta'-\theta|_\infty\le a,|I'-I|_\infty\le b\}$. Then: (a) V_rD and $\mathcal D_{r,s}$ are open, and $\mathcal D_{r',s'}\subset\mathcal D_{r,s}$ if $r'\le r$, $s'\le s$. (b) If $0<r'<r$, $0<s'<s$ and $z\in\mathcal D_{r',s'}$, then $\bar P(z;s-s',r-r')\subset\mathcal D_{r,s}$.
Source: Pos93 p. 213 ("analytic on complex neighbourhoods of a fixed domain $D\times\mathbb T^n$"); ours (max norms).

Proof. **Step S5.1: Openness and monotonicity.** $V_rD=\bigcup_{I'\in D}P(I',r)$ is a union of open polydiscs, and T_s is open; monotonicity is clear.

Step S5.2: Polydisc. Let $|\theta'-\theta|_\infty\le s-s'$ and $|I'-I|_\infty\le r-r'$. Then $|\operatorname{Im}\theta_j'|\le|\operatorname{Im}\theta_j|+s-s'<s$; and with $I_0\in D$, $|I-I_0|<r'$: $|I'-I_0|<r'+(r-r')=r$. □

Definition 5.2 (The class $\mathcal O(r,s)$ and the weighted norm)**.** $\mathcal O(r,s)$ is the set of $u:\mathcal D_{r,s}\to\mathbb C$ that are holomorphic (complex differentiable on the open set $\mathcal D_{r,s}\subset\mathbb C^{n+m}$) and 1-periodic in each θ_j. For $u\in\mathcal O(r,s)$ and $k\in\mathbb Z^n$, $I\in V_rD$, put

$$\hat u_k(I)=\int_{[0,1]^n}u(x,I)\,e^{-2\pi ik\cdot x}\,dx,\qquad |u|_{I,s}=\sum_{k\in\mathbb Z^n}|\hat u_k(I)|\,e^{2\pi|k|_1 s},\qquad |u|_{r,s}=\sup_{I\in V_rD}|u|_{I,s}\in[0,\infty].$$

(This is $|u|_{D,r,s}$; the integrand is continuous because $\mathbb R^n\subset T_s$.) For a vector- or matrix-valued $u=(u_a)_a$ (components in $\mathcal O(r,s)$) the coefficients are taken componentwise and $|\hat u_k|_\infty=\max_a|\hat u_{a,k}|$ replaces $|\hat u_k|$ in $|u|_{I,s}$. If $|u|_{r,s}<\infty$ we say u has *finite norm*.
Source: Pos93 p. 213 (definition of $|u|_{I,s}$, $|u|_{r,s}$), in the frozen convention of `docs/fourier-conventions.md` ("Bridge to strip sup norms": weight $e^{2\pi|k|_1 s}$).

Monotonicity: $|u|_{r',s'}\le|u|_{r,s}$ for $r'\le r$, $s'\le s$ (restriction to $\mathcal D_{r',s'}$; the coefficients at $I\in V_{r'}D$ are unchanged). For a vector, $\max_a|u_a|_{r,s}\le|u|_{r,s}\le\sum_a|u_a|_{r,s}$.

Lemma 5.3 (Coefficients, derivatives, series)**.** *Let $u\in\mathcal O(r,s)$.*

- *(a) Each $\hat u_k$ is holomorphic on V_rD.*
- *(b) $\partial_{\theta_j}u$, $\partial_{I_i}u\in\mathcal O(r,s)$, $\widehat{(\partial_{\theta_j}u)}_k=2\pi ik_j\hat u_k$, and $\widehat{(\partial_{I_i}u)}_k=\partial_{I_i}\hat u_k$.*
- *(c) (Series) Let $a_k:V_rD\to\mathbb C$ ($k\in\mathbb Z^n$) be holomorphic with $S:=\sup_{I\in V_rD}\sum_k|a_k(I)|e^{2\pi|k|_1 s}<\infty$. Then $v=\sum_k a_k\chi_k$ converges absolutely and uniformly on $\mathcal D_{r,s'}$ for each $s'<s$, $v\in\mathcal O(r,s)$, $\hat v_k=a_k$ and $|v|_{r,s}=S$.*
- *(d) (Representation) If $|u|_{r,s}<\infty$ then $u(\theta,I)=\sum_k\hat u_k(I)\chi_k(\theta)$ on $\mathcal D_{r,s}$, and $|\hat u_k(I)|e^{2\pi|k|_1 s}\le|u|_{r,s}$.*

Source: Standard; the one-variable ingredients are [K:`bp:p1:fourier-basic`], [K:`bp:p1:summation`].

Proof. **Step S5.3: (a).** Fix $I_0\in V_rD$ and $\eta>0$ with $\bar P(I_0,\eta)\subset V_rD$. The integrand $G(x,I)=u(x,I)e^{-2\pi ik\cdot x}$ is continuous on $[0,1]^n\times P(I_0,\eta)$, holomorphic in I, and D_IG is continuous there, because Du is holomorphic ([K:`bp:p1:holomorphic-regularity`]) hence continuous. (E1) gives holomorphy of $\hat u_k$ near I_0.

Step S5.4: (b). The derivatives are holomorphic ([K:`bp:p1:holomorphic-regularity`]) and periodic (differentiate the identity $u(\theta+e_j,I)=u(\theta,I)$). The formula for θ_j is [K:`bp:p1:fourier-basic`](a) at fixed I (the restriction of $u(\cdot,I)$ to $\mathbb R^n$ is continuous, periodic and C^1). The formula for I_i is the derivative formula of (E1) as in Step S5.3.

Step S5.5: (c). Each hypothesis $|a_k(I)|e^{2\pi|k|_1 s}\le S$ gives, on $\mathcal D_{r,s'}$, $|a_k(I)\chi_k(\theta)|\le|a_k(I)|e^{2\pi|k|_1 s'}\le Se^{-2\pi|k|_1(s-s')}$, which is summable ([K:`bp:p1:exp-sums`](b)). By the Weierstrass test the series converges uniformly on $\mathcal D_{r,s'}$, and each term is holomorphic, so v is holomorphic there ((E2)); as $s'<s$ is arbitrary, v is holomorphic on $\mathcal D_{r,s}$. It is periodic since each χ_k is. Integrating termwise ((E3)) against χ_{-k} over $[0,1]^n$ gives $\hat v_k=a_k$ (as in the coefficient step of [K:`bp:p1:summation`]); the norm identity is then the definition.

Step S5.6: (d). The bound is a single term of the sum $|u|_{I,s} \le |u|_{r,s}$. Fix I. The function $v = \sum_k \hat{u}_k(I)\chi_k$ is holomorphic and 1-periodic on T_s with coefficients $\hat{u}_k(I)$ ([K:bp:p1:summation], first part), and $u(\cdot, I) - v$ is holomorphic, 1-periodic, with all coefficients zero, so $u(\cdot, I) = v$ by [K:bp:p1:fourier-basic](c). □

5.3. **Lemma B.1: sup norm versus weighted norm.**

Lemma 5.4 (B.1)**.** *Let $u \in \mathcal{O}(r,s)$ (scalar, vector or matrix valued; $\sup|u|$ is the sup of the max norm of the values).*

(a) $\sup_{\mathcal{D}_{r,s}} |u| \le |u|_{r,s}$.
(b) If $\sigma > 0$ and $|u| \le M$ on $\mathcal{D}_{r,s+\sigma}$ (so $u \in \mathcal{O}(r, s+\sigma)$), then $|u|_{r,s} \le \coth(\pi\sigma)^n M$.

In Pöschel's form: $|u|^{\sup}_{r,s} \le |u|_{r,s} \le \coth^n \sigma_P \, |u|^{\sup}_{r,s+2\sigma_P}$ with $\sigma_P = \pi\sigma$ (the coefficient radius grows by $2\sigma_P = 2\pi\sigma$); see Remark 5.11.
Source: Pos93 p. 213, Lemma B.1; frozen-convention form: [K:bp:p1:summation], [K:bp:p1:decay], [K:bp:p1:exp-sums]*(b); this is Lemma 2.9 with the I-dependence made explicit.*

Proof. **Step S5.7: (a).** If $|u|_{r,s} = \infty$ there is nothing to prove. Otherwise Lemma 5.3(d) gives, for $(\theta, I) \in \mathcal{D}_{r,s}$, $|u(\theta, I)| \le \sum_k |\hat{u}_k(I)||\chi_k(\theta)| \le \sum_k |\hat{u}_k(I)| e^{2\pi|k|_1 s}$, using $|\chi_k(\theta)| \le e^{2\pi|k|_1 |\mathrm{Im}\,\theta|_\infty}$ ([K:bp:p0:domains]). For vector values apply this to each component and note $|\hat{u}_{a,k}| \le |\hat{u}_k|_\infty$.

Step S5.8: (b). Fix $I \in V_r D$. The function $u(\cdot, I)$ is holomorphic, periodic and bounded by M on $T_{s+\sigma}$, so [K:bp:p1:decay] gives $|\hat{u}_k(I)|_\infty \le e^{-2\pi|k|_1(s+\sigma)} M$ (limit $\rho' \uparrow s+\sigma$), and

$$|u|_{I,s} \le M \sum_k e^{-2\pi|k|_1\sigma} = \coth(\pi\sigma)^n M \qquad ([\text{K:bp:p1:exp-sums}](b),\ x = 2\pi\sigma).$$

Take the supremum over I. □

5.4. **Lemma B.2: algebra property.**

Lemma 5.5 (B.2)**.** *Let $u, v \in \mathcal{O}(r,s)$ be scalar with finite norms. Then $uv \in \mathcal{O}(r,s)$, $\widehat{(uv)}_k(I) = \sum_l \hat{u}_{k-l}(I)\hat{v}_l(I)$ (absolutely convergent), and $|uv|_{I,s} \le |u|_{I,s}\,|v|_{I,s}$ for every $I \in V_r D$; hence $|uv|_{r,s} \le |u|_{r,s}|v|_{r,s}$, and $|u^j|_{r,s} \le |u|^j_{r,s}$.*
Source: Pos93 p. 213, Lemma B.2. (The coefficient-level inequality is proved in Lean: `docs/fourier-conventions.md`, "Verified coefficient multiplication", `KAM/Fourier/Convolution.lean`; its identification with the product of functions is the remaining bridge.)

Proof. **Step S5.9: Product is in the class.** uv is holomorphic and periodic.

Step S5.10: Coefficients. Fix I. By Lemma 5.3(d), at real x, $u(x,I) = \sum_k \hat{u}_k \chi_k(x)$ and $v(x,I) = \sum_l \hat{v}_l \chi_l(x)$, both absolutely convergent ((E3)). The Cauchy product gives $u(x,I)v(x,I) = \sum_k c_k \chi_k(x)$ with $c_k = \sum_l \hat{u}_{k-l}\hat{v}_l$, and $\sum_k |c_k| \le |u|_{I,0}|v|_{I,0} < \infty$, so the convergence is uniform in x and termwise integration gives $\widehat{(uv)}_k(I) = c_k$.

Step S5.11: Norm. Since $|k|_1 \le |k-l|_1 + |l|_1$, $|uv|_{I,s} = \sum_k |c_k| e^{2\pi|k|_1 s} \le \sum_k \sum_l |\hat{u}_{k-l}| e^{2\pi|k-l|_1 s} |\hat{v}_l| e^{2\pi|l|_1 s} = |u|_{I,s}|v|_{I,s}$ (Tonelli, with the bijection $(k,l) \mapsto (k-l, l)$). Take the supremum over I; induct for u^j. □

5.5. **Lemma B.3: Cauchy estimates in the weighted norm.**

Lemma 5.6 (B.3)**.** *Let $u \in \mathcal{O}(r_0, s_0)$ have finite norm, $0 < r < r_0$, $0 < s < s_0$. Then*

(a) (pointwise in I) for every $I \in V_{r_0}D$: $\sum_{j=1}^{n}|\partial_{\theta_j}u|_{I,s} \le \frac{1}{e(s_0-s)}|u|_{I,s_0}$; *consequently* $\sup_{I\in V_{r_0}D}\sum_j|\partial_{\theta_j}u|_{I,s} \le \frac{|u|_{r_0,s_0}}{e(s_0-s)}$ *and, for each single j (global norm),* $|\partial_{\theta_j}u|_{r_0,s} \le \frac{1}{e(s_0-s)}|u|_{r_0,s_0}$. The sum over j of the global norms $\sum_j \sup_I|\partial_{\theta_j}u|_{I,s}$ is not bounded by this constant *(for D disconnected it can be larger by a factor up to n; with the norm $\sharp$ of Lemma 5.10 it is bounded);*

(b) $\max_{1\le i\le m}|\partial_{I_i}u|_{r,s_0} \le \frac{1}{r_0-r}|u|_{r_0,s_0}$.

In particular (taking $r_0 = r+\rho$, $s_0 = s+\sigma$): $\sup_{I\in V_rD}\sum_j|\partial_{\theta_j}u|_{I,s-\sigma} \le (e\sigma)^{-1}|u|_{r,s}$, $|\partial_{\theta_j}u|_{r,s-\sigma} \le (e\sigma)^{-1}|u|_{r,s}$ *for each j, and* $\max_i|\partial_{I_i}u|_{r-\rho,s} \le \rho^{-1}|u|_{r,s}$ *for $0<\sigma<s$, $0<\rho<r$. The same holds for vector- and matrix-valued u, componentwise (and with the pointwise sum over the θ-index outside, never the sum of global norms).*

Source: Pos93 pp. 213–214, Lemma B.3 (with s geometric; Pöschel's $1/(e\sigma)$ is the same number, Remark 5.11).

Proof. **Step S5.12: (a): coefficients.** Fix $I \in V_{r_0}D$. By Lemma 5.3(b), $\sum_j|\partial_{\theta_j}u|_{I,s} = \sum_k\sum_j 2\pi|k_j||\hat u_k|e^{2\pi|k|_1 s} = \sum_k 2\pi|k|_1 e^{-2\pi|k|_1(s_0-s)}|\hat u_k|e^{2\pi|k|_1 s_0}$. By [K:bp:p1:exp-sums](a) with $\tau = 1$, $a = 2\pi(s_0-s)$: $2\pi|k|_1e^{-2\pi|k|_1(s_0-s)} \le 2\pi/(e\cdot 2\pi(s_0-s))$. Hence the sum is $\le (e(s_0-s))^{-1}|u|_{I,s_0}$ for this I. The two global consequences follow by taking the supremum over I of the left side (resp. of a single summand, which is at most the sum) and using $|u|_{I,s_0} \le |u|_{r_0,s_0}$.

Step S5.13: (b): Cauchy integral. Fix $I \in V_rD$, i, and put $\rho' = r_0 - r$. For $|\zeta| \le \rho'$ the point $I+\zeta e_i$ lies in $V_{r_0}D$: if $I_0 \in D$ with $|I-I_0| < r$ then $|I+\zeta e_i - I_0| < r+\rho' = r_0$. By Lemma 5.3(a),(b) and [K:bp:p1:ext-cauchy], $\partial_{I_i}\hat u_k(I) = \frac{1}{2\pi i}\oint_{|\zeta|=\rho'}\hat u_k(I+\zeta e_i)\zeta^{-2}\,d\zeta$, so $|\partial_{I_i}\hat u_k(I)| \le \frac{1}{2\pi}\oint|\hat u_k(I+\zeta e_i)|\rho'^{-2}|d\zeta|$. Multiply by $e^{2\pi|k|_1 s_0}$, sum over k and use Tonelli ((E3)): $|\partial_{I_i}u|_{I,s_0} \le \frac{1}{2\pi}\oint\rho'^{-2}|u|_{I+\zeta e_i,s_0}|d\zeta| \le \rho'^{-1}|u|_{r_0,s_0}$. The vector case is componentwise, the matrix case likewise. □

5.6. **Lemma B.4: the Poisson bracket.** For $m=n$ and u,v holomorphic on an open subset of $\mathbb{C}^n\times\mathbb{C}^n$, $\{u,v\} = \sum_{i=1}^n\big(\partial_{\theta_i}u\,\partial_{I_i}v - \partial_{I_i}u\,\partial_{\theta_i}v\big)$ (Definition 6.2; $\{u,v\} = Du\cdot X_v$ for the frozen field $X_v = (\partial_I v, -\partial_\theta v)$).

Lemma 5.7 (B.4)**.** *Let $m=n$, $u\in\mathcal{O}(r_0,s_0)$, $v\in\mathcal{O}(r,s)$ with finite norms, $0<\rho<r$, $0<\sigma<s$, $r-\rho<r_0$, $s-\sigma<s_0$. Then $\{u,v\}\in\mathcal{O}(r-\rho,s-\sigma)$ and*

$$|\{u,v\}|_{r-\rho,s-\sigma} \le \frac{1}{e}\Big(\frac{1}{(r_0-r+\rho)\,\sigma} + \frac{1}{(s_0-s+\sigma)\,\rho}\Big)|u|_{r_0,s_0}|v|_{r,s}.$$

Source: Pos93 p. 214, Lemma B.4 (same formula, with Pöschel's $(r_0,s_0,r,s,\rho,\sigma)$ read as our geometric quantities).

Proof. Write $\{u,v\} = \sum_i \partial_{\theta_i}u\,\partial_{I_i}v - \sum_i\partial_{I_i}u\,\partial_{\theta_i}v$, all terms restricted to $\mathcal{D}_{r-\rho,s-\sigma}\subset\mathcal{D}_{r_0,s_0}\cap\mathcal{D}_{r,s}$ (monotonicity).

Step S5.14: Pointwise product estimate. Fix $I\in V_{r-\rho}D$ and write $|\cdot|_{I,\cdot} = |\cdot|_{I,s-\sigma}$, $t = s-\sigma$. By Lemma 5.5 (pointwise in I) and the triangle inequality for $|\cdot|_{I,t}$ (coefficients of a sum are sums), $|\{u,v\}|_{I,t} \le \sum_i|\partial_{\theta_i}u|_{I,t}|\partial_{I_i}v|_{I,t} + \sum_i|\partial_{I_i}u|_{I,t}|\partial_{\theta_i}v|_{I,t} \le \Big(\sum_i|\partial_{\theta_i}u|_{I,t}\Big)\max_i|\partial_{I_i}v|_{I,t} + \max_i|\partial_{I_i}u|_{I,t}\Big(\sum_i|\partial_{\theta_i}v|_{I,t}\Big)$. The angular sums are kept *pointwise in I*.

Step S5.15: Four derivative bounds. By Lemma 5.6, with (a) pointwise and (b) global, and monotonicity of the norm in both widths: $\sum_i |\partial_{\theta_i} u|_{I,t} \le \frac{|u|_{I,s_0}}{e(s_0-s+\sigma)} \le \frac{|u|_{r_0,s_0}}{e(s_0-s+\sigma)}$ ((a) with $s \to s-\sigma$; $I \in V_{r_0}D$ since $r-\rho<r_0$); $\max_i |\partial_{I_i} u|_{I,t} \le \frac{|u|_{r_0,s_0}}{r_0-r+\rho}$ ((b) with $r \to r-\rho$); $\sum_i |\partial_{\theta_i} v|_{I,t} \le \frac{|v|_{I,s}}{e\sigma} \le \frac{|v|_{r,s}}{e\sigma}$ ((a), pointwise) and $\max_i |\partial_{I_i} v|_{I,t} \le \frac{|v|_{r,s}}{\rho}$ ((b)). Inserting the four bounds in the previous display and taking the supremum over $I \in V_{r-\rho}D$ *last* gives the claim; the constant is unchanged and no factor n appears. □

5.7. **Lemma C.1: the general Cauchy inequality.**

Lemma 5.8 (C.1)**.** *Let A, B be finite-dimensional complex normed spaces, $v \in A$, $R>0$, and $G : P_A(v,R) = \{a : |a-v|_A < R\} \to B$ holomorphic with $|G|_B \le M$ on this ball. Then the operator norm satisfies $|DG(v)|_{B,A} \le M/R$.*
Source: Pos93 p. 215, Lemma C.1 ("analytic maps of Banach spaces"; only the finite-dimensional case is needed).

Proof. **Step S5.16: Reduction to one variable.** Let $u \in A$, $u \neq 0$. $f(\zeta) = G(v+\zeta u)$ is holomorphic on $|\zeta| < R/|u|_A$ with $|f| \le M$ and $f'(0) = DG(v)u$.

Step S5.17: Cauchy. For every $\varrho < R/|u|_A$, [K:`bp:p1:ext-cauchy`] gives $|f'(0)| \le M/\varrho$; let $\varrho \uparrow R/|u|_A$: $|DG(v)u|_B \le M|u|_A/R$. □

This also gives (E2)'s derivative statement: apply it to $K_j - K_l$ on the ball of radius η.

5.8. **Projections and truncation.**

Lemma 5.9 (P_S, T_K, averages)**.** *Let $u \in \mathcal{O}(r,s)$ have finite norm and $S \subset \mathbb{Z}^n$. Put $P_S u = \sum_{k\in S} \hat{u}_k \chi_k$. Then $P_S u \in \mathcal{O}(r,s)$, $\widehat{(P_S u)}_k = \hat{u}_k$ for $k \in S$ and 0 otherwise, and $|P_S u|_{r,s} \le |u|_{r,s}$. In particular, for $K \ge 0$ the truncation $T_K u = P_{\{|k|_1 \le K\}} u$ (a trigonometric polynomial in θ) and the average $\langle u\rangle(I) = \hat{u}_0(I) = P_{\{0\}} u$ (independent of θ) satisfy $|T_K u|_{r,s} \le |u|_{r,s}$, $|\langle u\rangle|_{r,s} \le |u|_{r,s}$, and*

$$u - T_K u = P_{\{|k|_1 > K\}} u, \qquad |u - T_K u|_{r,s-\sigma} \le e^{-2\pi K\sigma} |u|_{r,s} \qquad (0<\sigma<s).$$

If $\Lambda \subset \mathbb{Z}^n$ is any set (e.g. a lattice, Pos93 p. 192) and u is said to be in Λ-normal form when $u = P_\Lambda u$, then $P_\Lambda u$, $u - P_\Lambda u$ are in the class with norms $\le |u|_{r,s}$. For vector-valued u everything is componentwise.
Source: Pos93 p. 192 (definition of T_K, P_Λ) and p. 212 ("$|f - T_K f|_{r,s-\sigma} \le e^{-K\sigma}|f|_{r,s}$"), geometric form.

Proof. **Step S5.18: Class and coefficients.** The coefficient family $a_k = \hat{u}_k \mathbf{1}_S(k)$ consists of holomorphic functions (Lemma 5.3(a)) with $\sup_I \sum_k |a_k| e^{2\pi|k|_1 s} \le |u|_{r,s} < \infty$; Lemma 5.3(c) gives all claims including $|P_S u|_{r,s} \le |u|_{r,s}$.

Step S5.19: Tail. For $|k|_1 > K$: $e^{2\pi|k|_1(s-\sigma)} = e^{-2\pi|k|_1\sigma} e^{2\pi|k|_1 s} \le e^{-2\pi K\sigma} e^{2\pi|k|_1 s}$. Sum over k with $|k|_1 > K$ and take the supremum over I. Also $u = \sum_k \hat{u}_k \chi_k$ (Lemma 5.3(d)) gives $u - T_K u = P_{\{|k|_1>K\}} u$. □

5.9. **The stronger norm ♯ (sum of coefficient suprema).** Track P's averaging uses $\|f\|_{r,\Sigma} = \sum_k \sup_I |f_k(I)| e^{\Sigma|k|_1}$. With $s = \Sigma/(2\pi)$ this is, in our notation,

$$|u|^\sharp_{r,s} = \sum_{k\in\mathbb{Z}^n} \sup_{I \in V_r D} |\hat{u}_k(I)|_\infty \, e^{2\pi|k|_1 s} \ \in [0,\infty].$$

Always $|u|_{r,s} \le |u|^\sharp_{r,s}$; the reverse fails by an unbounded factor on disconnected D (on M disjoint action components carry one distinct mode each: $|\cdot|$, $= 1$, $|\cdot|^\sharp = M$), so the two norms are *not* interchangeable and a bound for one does not give a bound for the other. *Width conversion for track P:* every width and loss of P (including a non-loss width $\bar{s}$) is converted by $s = \Sigma/(2\pi)$,

$\sigma = \Delta\Sigma/(2\pi)$; the action losses are unchanged. Then the angular Cauchy constant $1/(e\sigma)$ reads $2\pi/(e\,\Delta\Sigma)$, the tail factor $e^{-2\pi K\sigma}$ reads $e^{-K\Delta\Sigma}$, and the composition parameter $\eta = 2|\phi|/(\rho\sigma)$ reads $4\pi|\phi|^\sharp/(\rho\,\Delta\Sigma)$.

Lemma 5.10 (Variants for $\sharp$)**.** *All of Lemma 5.4(a),(b), Lemma 5.5, Lemma 5.6, Lemma 5.7 and Lemma 5.9 hold with $|\cdot|$, replaced by $|\cdot|^\sharp$, with the same constants. In Lemma 5.6(a) the stronger* global *form holds: $\sum_j |\partial_{\theta_j} u|^\sharp_{r_0,s} \le |u|^\sharp_{r_0,s_0}/(e(s_0-s))$. For finite-norm u the series $\sum_k \hat u_k \chi_k$ of Lemma 5.3(c) may be taken with $S^\sharp := \sum_k \sup_I |a_k| e^{2\pi|k|_1 s} < \infty$ (the hypothesis is then stronger, and implies the one used there).*

Source: Ours; the blueprint's reading of Pos93 App. B for the norm of track P's two-scale iteration.

Proof. In every proof above the supremum over I is taken of a sum over k; for $\sharp$ it is taken inside, coefficient by coefficient, and each step survives.

Step S5.20: B.1. (a) $|u(\theta,I)| \le \sum_k |\hat u_k(I)| e^{2\pi|k|_1 s} \le |u|^\sharp_{r,s}$. (b) the decay $\sup_I |\hat u_k(I)|_\infty \le e^{-2\pi|k|_1(s+\sigma)} M$ is uniform in I, so $|u|^\sharp_{r,s} \le \coth(\pi\sigma)^n M$.

Step S5.21: B.2. $\sup_I |\sum_l \hat u_{k-l}(I)\hat v_l(I)| \le \sum_l \sup_I |\hat u_{k-l}| \sup_I |\hat v_l|$, and then the weights as in Step S5.11.

Step S5.22: B.3. (a): $\sum_j \sup_I |2\pi k_j \hat u_k(I)| = 2\pi|k|_1 \sup_I |\hat u_k|$ (the sum over j is of the coefficient factors $|k_j|$ only), then Step S5.12 coefficientwise. (b): $\sup_{I\in V_r D} |\partial_{I_i}\hat u_k(I)| \le (r_0 - r)^{-1} \sup_{I'\in V_{r_0}D} |\hat u_k(I')|$ by the Cauchy integral of Step S5.13; then sum over k.

Step S5.23: B.4. The convolution bound of the previous steps gives, coefficientwise in the suprema, $|\{u,v\}|^\sharp_{r,s} \le \sum_i |\partial_{\theta_i} u|^\sharp_{r,s} |\partial_{I_i} v|^\sharp_{r,s} + \dots$, and the global forms of (a),(b) bound the four factors as in Step S5.15.

Step S5.24: Projections. $|P_S u|^\sharp_{r,s} = \sum_{k\in S} \sup_I |\hat u_k| e^{2\pi|k|_1 s} \le |u|^\sharp_{r,s}$ and the tail estimate is the same. □

Remark 5.11 (Dictionary with Pöschel's normalisation and with Part 1). Pöschel works with 2π-periodic angles φ, characters $e^{ik\cdot\varphi}$, weight $e^{|k|_1 s_P}$ and losses (ρ_P, σ_P). Under the symplectic scaling $\varphi = 2\pi\theta$, $I_P = I/2\pi$ (so $d\varphi \wedge dI_P = d\theta \wedge dI$; time and the values of Hamiltonians are unchanged) his objects become ours with $s_P = 2\pi s$, $\sigma_P = 2\pi\sigma$, $r_P = r/2\pi$, $\rho_P = \rho/2\pi$. Then $e^{|k|_1 s_P} = e^{2\pi|k|_1 s}$ and $\rho_P\sigma_P = \rho\sigma$; the quantities in which Lemmas B.3–B.5 and the Iterative Lemma are written (for instance $|\phi|/(\rho\sigma)$ and the norm $|W\cdot|$) are therefore the same numbers. The conversion of Part 1, $\tilde H = 2\pi H(\varphi/2\pi, I)$ with I unchanged (Lemma 2.8), gives the same smallness ratio: $\tilde\phi/(\rho\sigma_P) = 2\pi\phi/(\rho\cdot 2\pi\sigma) = \phi/(\rho\sigma)$. So a condition "$|\phi| < \rho\sigma/2$" means the same in all three conventions, and the 2π-conversion does not affect the constants of the Iterative Lemma beyond the scalings of Lemma 2.8(b) (that bookkeeping is Part 6's). Where `docs/fourier-conventions.md` writes the derivative loss as $2\pi/(e\delta)$ for coefficient radius s, Lemma 5.6(a) is the same statement for the geometric width: a coefficient-radius loss $\delta = 2\pi\sigma$ gives $2\pi/(e\cdot 2\pi\sigma) = 1/(e\sigma)$.

6. Canonical transformations: time-one maps and the Lie transform (shared by tracks N, P)

This part is the canonical-transformation toolkit of Pöschel 1993, Appendix A (pp. 211–213: Taylor's formula, the flow X^t_ϕ of a small Hamiltonian field, its displacement and derivative bounds) and Lemma B.5 (pp. 214–215), in the frozen conventions of Part 5 (period 1, geometric widths, max norms). Track P cites it as [N:`bpN:p5:...`]. Generating functions ϕ live on general domains $\mathcal{D}_{r,s} = T_s \times V_r D$ with $D \subset \mathbb{C}^n$ arbitrary (Definition 5.1); $m = n$ because a Hamiltonian structure is needed. The ODE existence, uniqueness and continuation used are those of Part 4 (general layer); the weighted-norm estimates are those of Part 5.

Remark 6.1 (Decision for Lemma B.5: the Lie series, not the sup-norm fallback). **Route.** `PLAN.md` left open whether $|u \circ X^1_\phi| \le (1-2|\phi|/(\rho\sigma))^{-1}|u|$ is proved by Pöschel's convergent series $u \circ X^1_\phi = \sum_h \frac{1}{h!}\operatorname{ad}^h_\phi u$ or by a sup-norm estimate of the displaced function and a conversion back with Lemma 5.4(b). *We take the series*, for three reasons. (1) The Normal Form Lemma iterates $N \approx Ks/(12\ln 2)$ times and every step feeds the *weighted* norm of f_+ into the next step. The fallback re-enters the weighted norm through $\coth(\pi\sigma)^n$ in every step; this factor exceeds 1 by a fixed amount, so the contraction $\varepsilon_{i+1} \le \frac14\varepsilon_i$ of the Iterative Lemma would be destroyed and not merely perturbed in constants. (2) The fallback needs $u \circ X^1_\phi$ to be holomorphic and periodic *before* one can speak of its Fourier coefficients, that is, holomorphic dependence of the flow on the initial point (Lemma 6.4(b)), a statement not in Mathlib. The series route gets holomorphy of $u \circ X^1_\phi$ for free (it is a norm-convergent series of holomorphic functions); the dependence on initial data is then needed only for the claims on the map $\Phi = X^1_\phi$ itself (Proposition 6.13). (3) The ingredients of the series route are already labelled results: the bracket estimate (Lemma 5.7), a one-line induction (Lemma 6.9), a Taylor remainder along a solution of $\dot z = X_\phi(z)$ (Part 4 and Lemma 6.8) and a geometric series. **What is lost.** The equality $u \circ X^1_\phi = \sum_h \frac{1}{h!}\operatorname{ad}^h_\phi u$ is proved here for $\eta := 2|\phi|/(\rho\sigma) \le \frac12$ instead of Pöschel's $\eta < 1$ (Theorem 6.10); the bound itself is $(1-\eta)^{-1}$ as in the source. For $\eta < 1$ the equality also holds, by real-analyticity of $t \mapsto u(X^t_\phi z)$ and the identity theorem, but real-analytic dependence on time is another external fact; since the frozen headline only asserts the existence of constants, Part 6 simply uses $|\phi| \le \rho\sigma/4$ in place of $|\phi| < \rho\sigma/2$.

6.1. **Setting.**

Definition 6.2 (Hamiltonian field, bracket, scaled norm)**.** Points of $\mathbb{C}^n \times \mathbb{C}^n$ are $z = (\theta, I)$ (angles first, as in the frozen field); $J = \begin{pmatrix} 0 & \mathbf{1} \\ -\mathbf{1} & 0 \end{pmatrix}$, so $J^2 = -\mathbf{1}$, $J^\top = -J$. For u holomorphic on an open set let $\nabla u = (\partial_\theta u, \partial_I u)^\top$,

$$X_u = J\nabla u = (\partial_I u, -\partial_\theta u), \qquad \{u,v\} = \sum_{i=1}^n \big(\partial_{\theta_i} u\,\partial_{I_i} v - \partial_{I_i} u\,\partial_{\theta_i} v\big) = Du\cdot X_v = \nabla u^\top J \nabla v = -\{v,u\}.$$

X_u is the frozen Hamiltonian field (`KAM.hamiltonianVectorField`, Definition 1.3; for a real-valued function on real points see [K:`bp:p1:hamiltonian-real`]); the bracket is normalised so that $\frac{d}{dt}u(X^t_\phi(z)) = \{u,\phi\}(X^t_\phi(z))$ (Lemma 6.7). For $\rho,\sigma > 0$ the *scaled norm* on $\mathbb{C}^n \times \mathbb{C}^n$ is $\nu_{\rho,\sigma}(a,b) = \max(|a|_\infty/\sigma, |b|_\infty/\rho)$ (Pöschel's $|W\cdot|$, $W = \operatorname{diag}(\sigma^{-1},\rho^{-1})$ in angle-first order); it is a complex norm, and its closed ball of radius c about $z = (\theta, I)$ is $\bar P(z; c\sigma, c\rho)$. Iterated brackets: $\operatorname{ad}^0_\phi u = u$, $\operatorname{ad}^j_\phi u = \{\operatorname{ad}^{j-1}_\phi u, \phi\}$.

Source: Pos93 pp. 211–212 (flow X^t_ϕ, W), p. 214 (ad).

Source gap: We did not locate the printed definition of the bracket's sign in pp. 211–215; only the two-term structure (Lemma B.4) and $\{\phi,h\} = \omega\cdot\partial_\theta\phi$ for $h = h(I)$ (p. 211, homological equation, divisors $ik\cdot\omega$) are used. With the convention above $\{\phi,h\} = \partial_\theta\phi\cdot\partial_I h$, which is that equation.

The bracket of holomorphic periodic functions is holomorphic and periodic ([K:`bp:p1:holomorphic-regularity`]); $\operatorname{ad}^j_\phi u$ is holomorphic on the common domain of u and ϕ; $\{u,v\} = 0$ if u, v depend only on I.

Lemma 6.3 (Size of the field)**.** *Let $\phi \in \mathcal{O}(r,s)$ have finite norm, $0 < \rho < r$, $0 < \sigma < s$. On $\mathcal{D}_{r-\rho,s-\sigma}$: $|\partial_{I_i}\phi| \le |\phi|_{r,s}/\rho$, $|\partial_{\theta_i}\phi| \le |\phi|_{r,s}/(e\sigma)$, and*

$$\nu_{\rho,\sigma}(X_\phi) = \max\Big(\frac{\max_i|\partial_{I_i}\phi|}{\sigma}, \frac{\max_i|\partial_{\theta_i}\phi|}{\rho}\Big) \le \frac{|\phi|_{r,s}}{\rho\sigma}.$$

Source: Pos93 p. 212 ("$|\phi_I|_\infty \le q\delta/(\alpha\rho)$, $|WX_\phi| \le q\delta/(\alpha\rho\sigma)$").

Proof. **Step S6.1: Derivatives.** Lemma 5.6 (with $r_0 = r$, $s_0 = s$ and the target widths $r-\rho$, $s-\sigma$) gives, for each single index, $|\partial_{I_i}\phi|_{r-\rho,s} \le \rho^{-1}|\phi|_{r,s}$ and $|\partial_{\theta_j}\phi|_{r,s-\sigma} \le (e\sigma)^{-1}|\phi|_{r,s}$ (the individual global angular estimate, not a sum of global norms). Restrict to $\mathcal{D}_{r-\rho,s-\sigma}$ (monotonicity), and apply Lemma 5.4(a) to each derivative.

Step S6.2: The field. Insert into the definition of $\nu_{\rho,\sigma}$: $\frac{1}{\sigma}\cdot\frac{|\phi|}{\rho} = \frac{|\phi|}{\rho\sigma}$ and $\frac{1}{\rho}\cdot\frac{|\phi|}{e\sigma} \le \frac{|\phi|}{\rho\sigma}$. □

6.2. The flow of a small holomorphic vector field.

Lemma 6.4 (Flow of a small holomorphic field)**.** *Let ν be a complex norm on $\mathbb{C}^N$, $\Omega \subset \mathbb{C}^N$ open, $F : \Omega \to \mathbb{C}^N$ holomorphic with $\nu(F) \le \varepsilon$ on Ω and $0 \le \varepsilon \le \frac{1}{2}$. For $c > 0$ let $\Omega^c = \{z : \bar{B}_\nu(z, c) \subset \Omega\}$ (an open set).*

- *(a) (Existence, uniqueness, displacement) For $z \in \Omega^{1/2}$ there is $\gamma_z : \mathbb{R} \to \mathbb{C}^N$ with* `IsSolIn` $F\,\Omega\,[-1,1]\,\gamma_z$ *and $\gamma_z(0) = z$; any solution in Ω through z on an interval $J \ni 0$ equals γ_z on $J \cap [-1,1]$. Put $X^t(z) = \gamma_z(t)$ ($|t| \le 1$). Then $\nu(X^t(z) - z) \le |t|\,\varepsilon$.*
- *(b) (Regularity) $(t,z) \mapsto X^t(z)$ is continuous on $[-1,1]\times\Omega^{1/2}$; for each t the map $X^t : \Omega^{1/2} \to \Omega$ is holomorphic; $(t,z) \mapsto D_z X^t(z)$ is continuous, and $Y(t) = D_z X^t(z)$ satisfies $Y(t) = \mathbf{1} + \int_0^t DF(X^\tau z)Y(\tau)\,d\tau$.*
- *(c) (Inverse, derivative) For $z \in \Omega^1$: $X^1(z) \in \Omega^{1/2}$ and $X^{-1}(X^1 z) = z$; and the ν-operator norm satisfies $|DX^t(z) - \mathbf{1}|_\nu \le 2\varepsilon$ for $|t| \le 1$. In the same way $X^1(X^{-1}z) = z$.*
- *(d) (Translations) If $\tau \in \mathbb{C}^N$, $\Omega + \tau = \Omega$ and $F(w+\tau) = F(w)$ on Ω, then $\Omega^{1/2} + \tau = \Omega^{1/2}$ and $X^t(z+\tau) = X^t(z) + \tau$.*
- *(e) (Reality) Let $N = 2n$ and suppose $F(x) \in \mathbb{R}^N$ for $x \in \Omega \cap \mathbb{R}^N$. Then $X^t(x) \in \mathbb{R}^N$ for $x \in \Omega^{1/2} \cap \mathbb{R}^N$.*

Source: Pos93 p. 212 ("X^t_ϕ maps $V_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$ into $V_{r-\rho,s-\sigma}$ for $0 \le t \le 1$ and satisfies $|W(X^t_\phi - \mathrm{id})| \le \ldots$"; the derivative estimate by Lemma C.1). The general-ν formulation and the holomorphy in (b) are ours (Pöschel asserts "real analytic").

Proof. Regard $\mathbb{C}^N$ as a real normed space of dimension $2N$ with norm ν, extend F by 0 outside Ω (Remark 4.1); F is C^1 on Ω ([K:`bp:p1:holomorphic-regularity`], restriction of scalars).

Step S6.3: (a) The maximal solution stays in a compact set. Let $z \in \Omega^{1/2}$, (T_-, T_+) and γ the maximal interval and solution (Theorem 4.13). For $t \in [0, T_+)$, $\nu(\gamma'(t)) \le \varepsilon$, so $\nu(\gamma(t) - z) \le \varepsilon t$ ([K:`bp:p1:ext-mvt`] in one variable). If $T_+ \le 1$, then $\gamma(t) \in K = \bar{B}_\nu(z, \varepsilon) \subset \bar{B}_\nu(z, \frac{1}{2}) \subset \Omega$ for $t \in [0, T_+)$, with K compact, so $T_+ = \infty$ by Corollary 4.15, a contradiction. Hence $T_+ > 1$; symmetrically (time reversal, Lemma 4.3(b)) $T_- > -1$. Uniqueness on an interval $J \ni 0$ is Theorem 4.11; the displacement bound was shown.

Step S6.4: (b) Picard iterates. Fix $z_0 \in \Omega^{1/2}$. As $\bar{B}_\nu(z_0, \frac{1}{2})$ is compact in the open Ω, there is $\eta > 0$ with $\bar{B}_\nu(z_0, \frac{1}{2} + \eta) \subset \Omega$. Let $\Lambda = \sup\{|DF(w)|_\nu : w \in \bar{B}_\nu(z_0, \frac{1}{2} + \eta)\} < \infty$; F is Λ-Lipschitz on this convex ball ([K:`bp:p1:ext-mvt`]). For $(t,z) \in [-1,1] \times B_\nu(z_0, \eta)$ put $y_0(t,z) = z$, $y_{k+1}(t,z) = z + \int_0^t F(y_k(\tau,z))\,d\tau$. By induction on k: $\nu(y_k(t,z) - z) \le |t|\varepsilon$ (so $y_k(t,z) \in \bar{B}_\nu(z,\varepsilon) \subset \bar{B}_\nu(z_0, \frac{1}{2} + \eta) \subset \Omega$, and the integral is defined); y_k is continuous; $y_k(t,\cdot)$ is holomorphic; $D_z y_k$ is continuous in (t,z). The last two follow from (E1) applied to $t\int_0^1 F(y_k(t\tau', z))\,d\tau'$, whose z-derivative is $t\int_0^1 DF(y_k)D_z y_k\,d\tau'$.

Step S6.5: (b) Convergence and identification. $\nu(y_{k+1} - y_k)(t,z) \le \varepsilon\Lambda^k|t|^{k+1}/(k+1)!$ by induction, so $y_k \to y_\infty$ uniformly on $[-1,1] \times B_\nu(z_0, \eta)$. The limit is continuous and satisfies $y_\infty(t,z) = z + \int_0^t F(y_\infty)\,d\tau$ (F is Λ-Lipschitz), so $y_\infty(\cdot, z)$ is C^1 with derivative $F(y_\infty)$, i.e. a solution in Ω on $[-1,1]$ through z, and $y_\infty(t,z) = X^t(z)$ by (a). Hence X^t is continuous in (t,z) and, by (E2), holomorphic on $B_\nu(z_0,\eta)$ with $D_z y_k \to D_z X^t$ uniformly on $[-1,1] \times B_\nu(z_0, \eta/2)$, so

$D_z X^t(z)$ is jointly continuous. As z_0 was arbitrary, (b) holds except for the integral equation. Differentiating $X^t(w) = w + \int_0^t F(X^\tau w)\,d\tau$ in w under the integral ((E1); the integrand's w-derivative $DF(X^\tau w) D_w X^\tau(w)$ is jointly continuous) gives the equation for Y.

Step S6.6: (c). Let $z \in \Omega^1$, $w = X^1(z)$. Then $\nu(w - z) \le \varepsilon \le \frac{1}{2}$, so $\bar{B}_\nu(w, \frac{1}{2}) \subset \bar{B}_\nu(z, 1) \subset \Omega$, i.e. $w \in \Omega^{1/2}$. The map $t \mapsto \gamma_z(1 + t)$ is a solution on $[-1, 0]$ through w, so it equals γ_w there (Theorem 4.11) and $X^{-1}(w) = \gamma_z(0) = z$. The statement for $X^1 \circ X^{-1}$ is this one for $-F$ (Lemma 4.3(b)). For the derivative: $B_\nu(z, \frac{1}{2}) \subset \Omega^{1/2}$ (as above), $G = X^t - \mathrm{id}$ is holomorphic there by (b) with $\nu(G) \le \varepsilon$ by (a); Lemma 5.8 with $R = \frac{1}{2}$ gives $|DG(z)|_\nu \le 2\varepsilon$.

Step S6.7: (d),(e). (d): $\Omega^{1/2} + \tau = \Omega^{1/2}$ since the ν-balls translate; $t \mapsto \gamma_z(t) + \tau$ is a solution through $z + \tau$ (Lemma 4.3(c)), so equals $\gamma_{z+\tau}$ by uniqueness. (e): apply (a) to $E = \mathbb{R}^N$, $W = \Omega \cap \mathbb{R}^N$, the real field $F|_W$ (it is C^1 and real by hypothesis, [K:`bp:p1:real-restriction`]) and the compact set $\bar{B}_\nu(x, \varepsilon) \cap \mathbb{R}^N \subset W$: this gives a real solution on $[-1, 1]$ through x, which is also a solution in Ω, hence equals γ_x. □

Lemma 6.5 (Symplectic flow)**.** *In Lemma 6.4 let $N = 2n$, $\Omega \subset \mathbb{C}^n \times \mathbb{C}^n$, and $F = X_\phi$ with ϕ holomorphic on Ω. Then for $z \in \Omega^{1/2}$, $|t| \le 1$, $Y = DX^t(z)$ satisfies*

$$Y^\top J Y = J, \qquad Y J Y^\top = J, \qquad \det Y = \pm 1$$

($Y^\top$ the plain transpose of the complex matrix).
Source: Standard (Hamiltonian flows are symplectic; used implicitly by Pos93 pp. 211–212, "canonical transformation Φ").

Proof. **Step S6.8: Variational equation.** By Lemma 6.4(b), $Y \in C^1$ with $Y' = DF(X^t z) Y$, $Y(0) = \mathbf{1}$.

Step S6.9: The field. $F = J\nabla\phi$, so $DF = JS$ with $S = D\nabla\phi$ the Hessian, symmetric ([K:`bp:p1:ext-schwarz`]). Then $DF^\top J = S^\top J^\top J = S$ and $J\,DF = J^2 S = -S$, so $DF^\top J + J\,DF = 0$.

Step S6.10: Conservation. $\frac{d}{dt}(Y^\top J Y) = Y^\top (DF^\top J + J\,DF) Y = 0$, so $Y^\top J Y = J$ for all t. Taking determinants, $\det(Y)^2 = 1$; Y is invertible and inverting $Y^\top J Y = J$ gives $Y^{-1} J^{-1} Y^{-\top} = J^{-1}$, that is, $J = Y J Y^\top$ ($J^{-1} = -J$). □

Corollary 6.6 (Canonical maps: bracket invariance and conjugation)**.** *In the setting of Lemma 6.5, let $z \in \Omega^{1/2}$, $\Phi = X^t$ ($|t| \le 1$), and u, v, H holomorphic on a neighbourhood of $\Phi(z)$. Then $u \circ \Phi$ is holomorphic near z and*

- *(a) $\{u \circ \Phi, v \circ \Phi\}(z) = \{u, v\}(\Phi(z))$;*
- *(b) $D\Phi(z)\, X_{H \circ \Phi}(z) = X_H(\Phi(z))$;*
- *(c) if $y : J \to \Omega^{1/2}$ satisfies $\dot{y} = X_{H \circ \Phi}(y)$ with H holomorphic on a neighbourhood of $\Phi(y(J))$, then $x = \Phi \circ y$ satisfies $\dot{x} = X_H(x)$.*

Source: Standard.

Proof. **Step S6.11: Chain rule.** Φ is holomorphic (Lemma 6.4(b)) and $\nabla(u \circ \Phi) = D\Phi^\top (\nabla u) \circ \Phi$.

Step S6.12: (a). $\{u \circ \Phi, v \circ \Phi\} = \nabla u^\top\, Y J Y^\top\, \nabla v = \nabla u^\top J \nabla v$ at $\Phi(z)$ by Lemma 6.5.

Step S6.13: (b),(c). $D\Phi\, X_{H \circ \Phi} = Y\, J\, Y^\top (\nabla H) \circ \Phi = J(\nabla H) \circ \Phi = X_H \circ \Phi$; (c) is the chain rule $\dot{x} = D\Phi(y)\dot{y} = D\Phi(y) X_{H \circ \Phi}(y)$. □

For real data: if ϕ is real on real points then so is $F = X_\phi$ (Step S6.7), Φ maps real points to real points, its derivative at a real point is a real matrix ([K:`bp:p1:real-restriction`]), and (a)–(c) hold for real u, v, H with the real restriction of X ([K:`bp:p1:hamiltonian-real`]).

6.3. **Taylor's formula along the flow.**

Lemma 6.7 (Derivative along the flow)**.** *Let $\Omega \subset \mathbb{C}^n \times \mathbb{C}^n$, ϕ, $F = X_\phi$, $\varepsilon \le \frac{1}{2}$ as in Lemma 6.5; let $\Omega_1 \subset \Omega$ be open and u holomorphic on Ω_1, and $z \in \Omega^{1/2}$ with $X^t(z) \in \Omega_1$ for $|t| \le 1$. Then $g(t) = u(X^t(z))$ is C^∞ on $[-1,1]$ and $g^{(j)}(t) = (\mathrm{ad}_\phi^j u)(X^t(z))$ for all $j \ge 0$ (each $\mathrm{ad}_\phi^j u$ is holomorphic on Ω_1).*

Source: Pos93 p. 211 ("Taylor's formula" for $h \circ X_\phi^t$); standard.

Proof. **Step S6.14: One derivative.** For v holomorphic on Ω_1, v is real-differentiable with derivative Dv, and $\gamma = X^\cdot(z)$ has derivative $X_\phi(\gamma)$, so $\frac{d}{dt} v(\gamma(t)) = Dv(\gamma) \cdot X_\phi(\gamma) = \{v, \phi\}(\gamma(t))$ (Definition 6.2), which is continuous in t.

Step S6.15: Induction. Apply the previous step to $v = \mathrm{ad}_\phi^j u$, holomorphic on Ω_1 (derivatives of holomorphic functions, [K:`bp:p1:holomorphic-regularity`]). □

Lemma 6.8 (Taylor with integral remainder along the flow)**.** *In the setting of Lemma 6.7, for every $N \ge 1$,*

$$u(X^1 z) = \sum_{j=0}^{N-1} \frac{1}{j!} (\mathrm{ad}_\phi^j u)(z) + R_N(z),$$

$$R_N(z) = \int_0^1 \frac{(1-t)^{N-1}}{(N-1)!} (\mathrm{ad}_\phi^N u)(X^t z)\, dt, \qquad |R_N(z)| \le \frac{1}{N!} \max_{0 \le t \le 1} |(\mathrm{ad}_\phi^N u)(X^t z)|.$$

In particular ($N = 1$ for f_0, $N = 2$ for h), if h, f_0 are holomorphic on Ω_1 and the trajectory of z stays in Ω_1,

$$(h + f_0)(X^1 z) = h(z) + \{h, \phi\}(z) + f_0(z) + \int_0^1 \big\{(1-t)\{h, \phi\} + f_0,\ \phi\big\}(X^t z)\, dt. \tag{1}$$

Source: Pos93 p. 211, displayed formula for $H_0 \circ \Phi$.

Proof. **Step S6.16: Taylor.** Apply the Taylor formula with integral remainder for C^N functions of one real variable (external fact: induction on N with one integration by parts per step, [K:`bp:p1:ext-integral`]) to $g(t) = u(X^t z)$ at 0, evaluated at 1, using $g^{(j)}(0) = (\mathrm{ad}_\phi^j u)(z)$ and $g^{(N)}(t) = (\mathrm{ad}_\phi^N u)(X^t z)$ (Lemma 6.7). The bound uses $\int_0^1 (1-t)^{N-1}/(N-1)!\, dt = 1/N!$.

Step S6.17: The displayed formula. Apply the case $N = 2$ to h ($\mathrm{ad}^2 h = \{\{h, \phi\}, \phi\}$) and the case $N = 1$ to f_0 and add: the remainders are $\int_0^1 (1-t)\{\{h, \phi\}, \phi\}(X^t z)\, dt + \int_0^1 \{f_0, \phi\}(X^t z)\, dt$, and brackets are bilinear. □

6.4. **Lemma B.5: composition in the weighted norm.**

Lemma 6.9 (Iterated brackets)**.** *Let $\phi \in \mathcal{O}(r', s')$, $u \in \mathcal{O}(r, s)$ have finite norms, $0 < \rho < r$, $0 < \sigma < s$, $r + \rho \le r'$, $s + \sigma \le s'$, and $\eta = 2|\phi|_{r',s'}/(\rho\sigma)$. Then for every $h \ge 1$,*

$$|\mathrm{ad}_\phi^h u|_{r-\rho, s-\sigma} \le \Big(\frac{2h^2}{e\rho\sigma}\Big)^h \frac{h!}{(2h)!} |\phi|_{r',s'}^h |u|_{r,s}, \qquad \text{hence} \qquad \frac{1}{h!} |\mathrm{ad}_\phi^h u|_{r-\rho, s-\sigma} \le \Big(\frac{e\eta}{4}\Big)^h |u|_{r,s}.$$

Source: Pos93 p. 215 (the displayed estimates for $|\mathrm{ad}_\phi^h u|$ with $\tilde\rho = \rho/h$, $\tilde\sigma = \sigma/h$).

Proof. **Step S6.18: Setup.** Fix $h \ge 1$, $\kappa = \rho/h$, $\lambda = \sigma/h$, $r_i = r - i\kappa$, $s_i = s - i\lambda$ ($0 \le i \le h$), so $r_h = r - \rho$, $s_h = s - \sigma$. By induction each $\mathrm{ad}_\phi^i u \in \mathcal{O}(r_i, s_i)$.

Step S6.19: One step. Let $1 \le i \le h$. Since $\mathrm{ad}_\phi^i u = \{\mathrm{ad}_\phi^{i-1} u, \phi\} = -\{\phi, \mathrm{ad}_\phi^{i-1} u\}$, Lemma 5.7 with $(u, v; r_0, s_0; r, s; \rho, \sigma) \to (\phi, \mathrm{ad}_\phi^{i-1} u; r', s'; r_{i-1}, s_{i-1}; \kappa, \lambda)$ applies (note $\kappa < r_{i-1}$, $r_i < r'$, and

likewise for s) and gives $|\mathrm{ad}_\phi^i u|_{r_i,s_i} \le \frac{1}{e}\Big(\frac{1}{(r'-r_{i-1}+\kappa)\lambda}+\frac{1}{(s'-s_{i-1}+\lambda)\kappa}\Big)|\phi|_{r',s'}|\mathrm{ad}_\phi^{i-1}u|_{r_{i-1},s_{i-1}}$. Now $r'-r_{i-1}+\kappa=(r'-r)+i\kappa\ge\rho+i\kappa=(h+i)\kappa$ and likewise $s'-s_{i-1}+\lambda\ge(h+i)\lambda$; so the bracket is $\le\frac{2}{e(h+i)\kappa\lambda}$.

Step S6.20: Product. Multiplying over $i=1,\dots,h$ and using $\prod_{i=1}^h(h+i)=(2h)!/h!$ and $\kappa\lambda=\rho\sigma/h^2$ gives the first estimate. For the second, $(2h)!\ge(2h/e)^{2h}$ (from $e^N\ge N^N/N!$), so $\frac{1}{h!}|\mathrm{ad}_\phi^h u|\le\frac{(\eta h^2/e)^h}{(2h)!}|u|\le\big(\eta\frac{h^2}{e}\cdot\frac{e^2}{4h^2}\big)^h|u|=(\frac{e\eta}{4})^h|u|$. □

Theorem 6.10 (Lemma B.5: $|u\circ X_\phi^1|\le(1-\eta)^{-1}|u|$)**.** *Let $D\subset\mathbb{C}^n$, $0<\rho<r\le r'-\rho$, $0<\sigma<s\le s'-\sigma$, $\phi\in\mathcal{O}(r',s')$ with finite norm and $\eta=2|\phi|_{r',s'}/(\rho\sigma)\le\frac12$. Put $\Omega=\mathcal{D}_{r'-\rho,s'-\sigma}\supset\mathcal{D}_{r,s}$ and $\nu=\nu_{\rho,\sigma}$.*

(i) $\nu(X_\phi)\le\eta/2\le\frac14$ on Ω; the flow $X^t=X_\phi^t$, $|t|\le1$, of Lemma 6.4 is defined on $\Omega^{1/2}\supset\mathcal{D}_{r-\rho,s-\sigma}$, with $X^t(\mathcal{D}_{r-\rho,s-\sigma})\subset\mathcal{D}_{r,s}$.

(ii) For every $u\in\mathcal{O}(r,s)$ with finite norm, $\sum_{h\ge0}\frac{1}{h!}\mathrm{ad}_\phi^h u$ converges in $|\cdot|_{r-\rho,s-\sigma}$ to a function $S\in\mathcal{O}(r-\rho,s-\sigma)$ and $S=u\circ X_\phi^1$ on $\mathcal{D}_{r-\rho,s-\sigma}$. In particular $u\circ X_\phi^1\in\mathcal{O}(r-\rho,s-\sigma)$ and

$$|u\circ X_\phi^1|_{r-\rho,s-\sigma}\le\Big(1-\frac{e\eta}{4}\Big)^{-1}|u|_{r,s}\le\Big(1-\frac{2}{\rho\sigma}|\phi|_{r',s'}\Big)^{-1}|u|_{r,s}.$$

(iii) The same holds for X_ϕ^t, $0\le t\le1$ (and $-1\le t\le0$), with η replaced by $|t|\eta$.

(iv) If ϕ and u are real on real points, $u\circ X_\phi^1$ is real on real points of $\mathcal{D}_{r-\rho,s-\sigma}$.

Source: Pos93 pp. 214–215, Lemma B.5 (with $|\phi|_{r_0,s_0}<\rho\sigma/2$ there).
Source gap: Pöschel writes "Consider the Lie series expansion $u\circ X_\phi^1=\sum\frac{1}{h!}\mathrm{ad}_\phi^h u$" and estimates the series; the equality with the composition is not proved on p. 214. It is proved here for $\eta\le\frac12$; see Remark 6.1. The remark that the smallness condition implies $X_\phi^t:V_{r-\rho,s-\sigma}\to V_{r,s}$ is (i).

Proof. **Step S6.21: The field and the domain.** Since $r\le r'-\rho$, $s\le s'-\sigma$: $\mathcal{D}_{r,s}\subset\Omega$. Lemma 6.3 (for $\phi\in\mathcal{O}(r',s')$ and losses ρ,σ) gives $\nu(X_\phi)\le|\phi|/(\rho\sigma)=\eta/2=:\varepsilon\le\frac14$ on Ω. If $z\in\mathcal{D}_{r-\rho,s-\sigma}$, then $\bar B_\nu(z,\frac12)=\bar P(z;\sigma/2,\rho/2)\subset\mathcal{D}_{r-\rho/2,s-\sigma/2}\subset\mathcal{D}_{r,s}\subset\Omega$ (Definition 5.1(b)), so $z\in\Omega^{1/2}$. By Lemma 6.4(a), $X^t(z)\in\bar P(z;\varepsilon\sigma,\varepsilon\rho)\subset\mathcal{D}_{r-(1-\varepsilon)\rho,s-(1-\varepsilon)\sigma}\subset\mathcal{D}_{r,s}$. This is (i).

Step S6.22: The series. By Lemma 6.9 (applicable: $r+\rho\le r'$, $s+\sigma\le s'$), $\frac{1}{h!}\mathrm{ad}_\phi^h u\in\mathcal{O}(r-\rho,s-\sigma)$ has norm $\le(e\eta/4)^h|u|_{r,s}$, and $e\eta/4<1$. The sum of norms is $\le(1-e\eta/4)^{-1}|u|_{r,s}$. By Lemma 5.4(a) the partial sums converge uniformly on $\mathcal{D}_{r-\rho,s-\sigma}$, so S is holomorphic ((E2)) and periodic, and termwise integration ((E3)) shows that its Fourier coefficients are the sums of the coefficients; the triangle inequality in $\sum_k|\hat S_k(I)|e^{2\pi|k|_1(s-\sigma)}$ gives $|S|_{r-\rho,s-\sigma}\le(1-e\eta/4)^{-1}|u|_{r,s}$; and $1-e\eta/4\ge1-\eta$.

Step S6.23: The remainder tends to 0. Let $z\in\mathcal{D}_{r-\rho,s-\sigma}$, $t\in[0,1]$, $w=X^t(z)$. By Step S6.21, $w\in\mathcal{D}_{r-\hat\rho,s-\hat\sigma}$ with $\hat\rho=(1-\varepsilon)\rho$, $\hat\sigma=(1-\varepsilon)\sigma$, $0<\hat\rho<r$, $r+\hat\rho\le r'$, and likewise for s. Lemma 6.9 with losses $(\hat\rho,\hat\sigma)$ and $\hat\eta=2|\phi|/(\hat\rho\hat\sigma)=\eta/(1-\varepsilon)^2\le\frac{16}{9}\eta\le\frac89$ gives $|\mathrm{ad}_\phi^N u|_{r-\hat\rho,s-\hat\sigma}\le N!(e\hat\eta/4)^N|u|_{r,s}$, and $e\hat\eta/4\le\frac{2e}{9}<1$. By Lemma 5.4(a), $|(\mathrm{ad}_\phi^N u)(X^tz)|\le N!(e\hat\eta/4)^N|u|_{r,s}$. Lemma 6.8 (with $\Omega_1=\mathcal{D}_{r,s}$, Ω_1-trajectory by Step S6.21) gives $|R_N(z)|\le(e\hat\eta/4)^N|u|_{r,s}\to0$. Hence $u(X^1z)=\lim_N\sum_{j<N}\frac{1}{j!}(\mathrm{ad}_\phi^j u)(z)=S(z)$. This proves (ii).

Step S6.24: (iii), (iv). $X_\phi^t=X_{t\phi}^1$ (the field $X_{t\phi}=tX_\phi$ and $s\mapsto X_\phi^{ts}(z)$ solves it), and $|t\phi|=|t||\phi|$, so (iii) is (ii) for $t\phi$. (iv): Lemma 6.4(e) with $F=X_\phi$ real on real points (a real-valued ϕ on real points has real partials there). □

Proposition 6.11 (Lie transform with first-order remainder, in the weighted norm)**.** *In the setting of Theorem 6.10, let $h\in\mathcal{O}(r,s)$ (any norm, possibly ∞) with $w:=\{h,\phi\}\in\mathcal{O}(r,s)$ of finite norm,*

and $f_0 \in \mathcal{O}(r,s)$ with finite norm. Then on $\mathcal{D}_{r-\rho,s-\sigma}$

$$(h+f_0)\circ X_\phi^1 = h+w+f_0+\mathcal{R}, \qquad \mathcal{R} = \sum_{j\ge 1}\Big(\frac{1}{(j+1)!}\operatorname{ad}_\phi^j w + \frac{1}{j!}\operatorname{ad}_\phi^j f_0\Big) = \int_0^1 \{(1-t)w+f_0,\phi\}\circ X_\phi^t\,dt,$$

the series converges in $|\cdot|_{r-\rho,s-\sigma}$, $\mathcal{R}\in\mathcal{O}(r-\rho,s-\sigma)$, and

$$|\mathcal{R}|_{r-\rho,s-\sigma} \le \sum_{j\ge 1}\Big(\frac{e\eta}{4}\Big)^j\Big(\frac{|w|_{r,s}}{j+1}+|f_0|_{r,s}\Big) \le \frac{\eta}{1-\eta}\Big(\tfrac12|w|_{r,s}+|f_0|_{r,s}\Big).$$

The integral form is Lemma 6.8, (1); the pointwise identity requires no bound on h itself.
Source: Pos93 p. 211 (the decomposition $H_0\circ\Phi = h+\{h,\phi\}+f_0+\int\dots$) and p. 212–213 (estimate of f_+); the series form of the remainder is ours.

Proof. **Step S6.25: Derivatives of h.** Since $\operatorname{ad}_\phi^1 h = w$, $\operatorname{ad}_\phi^j h = \operatorname{ad}_\phi^{j-1} w$ $(j\ge 1)$. In Lemma 6.7 (with $\Omega_1 = \mathcal{D}_{r,s}$) $g(t) = h(X^t z)$ has $g^{(j)}(t) = (\operatorname{ad}_\phi^{j-1} w)(X^t z)$; Taylor's formula to order N gives $h(X^1 z) = h(z) + \sum_{i=0}^{N-2}\frac{1}{(i+1)!}(\operatorname{ad}_\phi^i w)(z) + R'_N$, with $|R'_N| \le \frac{1}{N!}\max_t|(\operatorname{ad}_\phi^{N-1} w)(X^t z)|$. By Step S6.23 (Lemma 6.9 with $u=w$ and the losses $\hat\rho,\hat\sigma$) this is $\le \frac{(N-1)!}{N!}(e\hat\eta/4)^{N-1}|w|_{r,s}\to 0$. So $h\circ X^1 = h + \sum_{i\ge 0}\frac{1}{(i+1)!}\operatorname{ad}_\phi^i w$ pointwise, the term $i=0$ being w.

Step S6.26: Derivatives of f_0. By Theorem 6.10(ii), $f_0\circ X^1 = \sum_{j\ge 0}\frac{1}{j!}\operatorname{ad}_\phi^j f_0$.

Step S6.27: Sum and bound. Adding gives the first identity with $\mathcal{R}$ as stated (terms $j=0$ are w, f_0). Lemma 6.9 bounds $\frac{1}{j!}|\operatorname{ad}_\phi^j w| \le (e\eta/4)^j|w|$, hence $\frac{1}{(j+1)!}|\operatorname{ad}_\phi^j w| \le \frac{1}{j+1}(e\eta/4)^j|w|$, and similarly for f_0. The geometric sum with $\frac{1}{j+1}\le\frac12$ and $e\eta/4\le\eta$ gives the last bound. Convergence in the norm and membership of $\mathcal{O}(r-\rho,s-\sigma)$ are as in Step S6.22.

Step S6.28: Integral form. Equality with the integral is (1), because $\{h,\phi\} = w$; the trajectory stays in $\mathcal{D}_{r,s}$ by Step S6.21. □

Corollary 6.12 (Integral remainders in the weighted norm)**.** *In the setting of Theorem 6.10 let $v_t\in\mathcal{O}(r,s)$ $(0\le t\le 1)$ have finite norms with $V = \sup_t|v_t|_{r,s}<\infty$, and suppose $(t,z)\mapsto v_t(z)$ and $(t,z)\mapsto Dv_t(z)$ are continuous on $[0,1]\times\mathcal{D}_{r,s}$. Then $G(z) = \int_0^1 v_t(X_\phi^t z)\,dt$ lies in $\mathcal{O}(r-\rho,s-\sigma)$ and $|G|_{r-\rho,s-\sigma}\le(1-\eta)^{-1}V$.*
Source: Pos93 p. 212–213 ("$|u\circ X_\phi^t|\le(1-2|\phi|/\rho\sigma)^{-1}|u|$ uniformly for $0\le t\le 1$", applied inside the integral).

Proof. **Step S6.29: Integrand.** $g_t = v_t\circ X^t \in \mathcal{O}(r-\rho,s-\sigma)$ with $|g_t|\le(1-t\eta)^{-1}V \le (1-\eta)^{-1}V$ (Theorem 6.10(iii)). The map $(t,z)\mapsto g_t(z)$ is continuous and $D_z g_t = Dv_t(X^t z)D_z X^t(z)$ is continuous (Lemma 6.4(b)).

Step S6.30: Holomorphy and coefficients. By (E1), G is holomorphic; it is periodic. Fubini on $[0,1]\times[0,1]^n$ gives $\hat G_k(I) = \int_0^1 \hat{(g_t)}_k(I)\,dt$, so by Tonelli $|G|_{I,s-\sigma}\le\int_0^1|g_t|_{I,s-\sigma}\,dt\le(1-\eta)^{-1}V$. □

6.5. The canonical map $\Phi = X_\phi^1$ on weighted domains.

Proposition 6.13 (Iterative-Lemma package for Φ)**.** *Let $D\subset\mathbb{C}^n$, $\phi\in\mathcal{O}(r,s)$ with finite norm, $0<\rho<r/2$, $0<\sigma<s/2$, and $\varepsilon := |\phi|_{r,s}/(\rho\sigma)\le\frac12$. Put $\Omega = \mathcal{D}_{r-\rho,s-\sigma}$, $\nu=\nu_{\rho,\sigma}$, $\Phi = X_\phi^1$, $\Psi = X_\phi^{-1}$ (Lemma 6.4). Then:*

- *(a) $\nu(X_\phi)\le\varepsilon$ on Ω; Φ,Ψ are defined and holomorphic on $\mathcal{D}_{r-\frac32\rho,s-\frac32\sigma}\subset\Omega^{1/2}$ and map it into Ω. For every $t\in[-1,1]$, $\nu(X^t-\mathrm{id})\le|t|\varepsilon$; explicitly $|\Phi_\theta-\theta|_\infty\le|\phi|_{r,s}/\rho$ and $|\Phi_I-I|_\infty\le|\phi|_{r,s}/(e\sigma)$, and the same for Ψ.*

(b) $\Phi(\theta+k,I)=\Phi(\theta,I)+(k,0)$ *for* $k\in\mathbb{Z}^n$*; if* ϕ *is real on real points,* Φ *maps real points to real points.*
(c) (Derivative) on $\mathcal{D}_{r-2\rho,s-2\sigma}$*:* $|D\Phi-\mathbf{1}|_\nu\le 2\varepsilon$*, i.e.* $|W(D\Phi-\mathbf{1})W^{-1}|\le 2|\phi|_{r,s}/(\rho\sigma)$ *for* $W=\operatorname{diag}(\sigma^{-1},\rho^{-1})$*.*
(d) (Canonical) $(D\Phi)^\top J\,D\Phi=J$*, and Corollary 6.6 holds for* Φ*.*
(e) (Inverse) $\Psi\circ\Phi=\mathrm{id}=\Phi\circ\Psi$ *on* $\mathcal{D}_{r-2\rho,s-2\sigma}$*, and* $\Phi(\mathcal{D}_{r-2\rho,s-2\sigma})\subset\mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$*.*

Source: Pos93 p. 212 ("Equivalently, X_ϕ *satisfies* $|WX_\phi|_{\mathcal{P}}\le\frac{q\delta}{\alpha\rho\sigma}\le\frac12$ *…* X_ϕ^t *maps* $V_{r-\frac32\rho,s-\frac32\sigma}$ *into* $V_{r-\rho,s-\sigma}$ *…* $|W(DX_\phi^t-I_{2n})W^{-1}|\le 2\frac{q\delta}{\alpha\rho\sigma}$ *uniformly on* $V_{r-2\rho,s-2\sigma}$*"), with* $|\phi|_{r,s}$ *in place of* $q\delta/\alpha$*.*

Proof. **Step S6.31: Field.** Lemma 6.3 gives $\nu(X_\phi)\le\varepsilon\le\frac12$ on Ω, and Ω is open. For $z\in\mathcal{D}_{r-c\rho,s-c\sigma}$ $(c\ge\frac32)$ the ball $\bar B_\nu(z,c-1)=\bar P(z;(c-1)\sigma,(c-1)\rho)$ lies in Ω (Definition 5.1(b)): so $\mathcal{D}_{r-\frac32\rho,s-\frac32\sigma}\subset\Omega^{1/2}$ and $\mathcal{D}_{r-2\rho,s-2\sigma}\subset\Omega^1$.

Step S6.32: (a),(b). Lemma 6.4(a),(b),(d),(e) (with $\tau=(k,0)$: $\Omega+\tau=\Omega$ and X_ϕ is periodic, as ϕ is). $X^t(z)\in\bar P(z;\varepsilon\sigma,\varepsilon\rho)\subset\mathcal{D}_{r-\rho,s-\sigma}$ for $z\in\mathcal{D}_{r-\frac32\rho,s-\frac32\sigma}$ because $\varepsilon\le\frac12$. The explicit component bounds are $|X_\theta^t-\theta|\le|t|\sup|\partial_I\phi|\le|t||\phi|/\rho$ and $|X_I^t-I|\le|t|\sup|\partial_\theta\phi|\le|t||\phi|/(e\sigma)$ (mean value inequality along the trajectory and Lemma 6.3). For $\Psi=X_\phi^{-1}=X^1_{-\phi}$ the same with $-\phi$.

Step S6.33: (c)–(e). (c) is Lemma 6.4(c) with Ω^1 and Step S6.31. (d) is Lemma 6.5 and Corollary 6.6. (e): Lemma 6.4(c) for $z\in\Omega^1$ gives $\Psi(\Phi z)=z$ and $\Phi(\Psi z)=z$; and $\Phi(z)\in\bar B_\nu(z,\varepsilon)\subset\mathcal{D}_{r-\frac32\rho,s-\frac32\sigma}$ for $z\in\mathcal{D}_{r-2\rho,s-2\sigma}$ $(\varepsilon\le\frac12)$. □

6.6. **Variants for the norm ♯.**

Proposition 6.14 (Composition in the norm ♯)**.** *Lemma 6.3, Lemma 6.9, Theorem 6.10(i)–(iv), Proposition 6.11, Corollary 6.12 and Proposition 6.13 hold with every weighted norm* $|\cdot|$ *replaced by* $|\cdot|^\sharp$ *(Lemma 5.10), with the same constants and the same hypothesis* $\eta=2|\phi|^\sharp_{r',s'}/(\rho\sigma)\le\frac12$*. For the consumer with coefficient radius* $\Sigma=2\pi s$ *and loss* $\Delta\Sigma=2\pi\sigma$ *the parameter is* $\eta=4\pi|\phi|^\sharp/(\rho\,\Delta\Sigma)$ *(Section 5.9).*

Proof. **Step S6.34: Where the norm enters.** The geometry (flow, symplecticity, Taylor, displacement) uses the norm only through sup bounds $|u(z)|\le|u|^\sharp$ (Lemma 5.10, B.1(a)), which hold for ♯ as for the pointwise norm; the vector-field bound Lemma 6.3 uses the individual derivative estimates, valid for ♯.

Step S6.35: Series. Lemma 6.9 uses only the bracket estimate; Theorem 6.10 Step S6.22 uses the triangle inequality, valid for ♯ since $\sup_I|\sum_h c_k^h(I)|\le\sum_h\sup_I|c_k^h|$. The integral corollary uses $\sup_I\int_0^1|(\hat g_t)_k|\,dt\le\int_0^1\sup_I|(\hat g_t)_k|\,dt$ and Tonelli. □

Remark 6.15 (Use by Part 6 and track P; what is not provided). Part 6 applies Proposition 6.11 with h the integrable part, $w=\{h,\phi\}=g_+-f_0$ by the homological equation (so $|w|$ is controlled by $|\chi|$ and $|T_Kf|$, not by $|h|$), $f_0=g+T_Kf$, and Theorem 6.10 for $(f-T_Kf)\circ X^1_\phi$, with the losses (ρ,σ) from (r,s) to $(r-\rho,s-\sigma)$ (so ϕ lives on $(r+\rho,s+\sigma)$, or one re-indexes as Pöschel: $\phi\in\mathcal{O}(r,s)$, $u\in\mathcal{O}(r-\rho,s-\sigma)$, result on $\mathcal{D}_{r-2\rho,s-2\sigma}$). The smallness used is $\eta=2|\phi|/(\rho\sigma)\le\frac12$; the displacement and derivative bounds of Proposition 6.13 need only $\varepsilon\le\frac12$. The sup-norm *fallback* of `PLAN.md` is not used. Not proved here: the stability transfer (Part 7) and the estimate of $|W(\Phi_1\circ\cdots\circ\Phi_N-\mathrm{id})|$ for compositions of the maps of successive steps, which is Part 6's (the displacement bounds of (a) add, since each Φ_i maps into the domain of Φ_{i+1} by the choice of losses).

7. The homological equation, the Iterative Lemma and the Normal Form Lemma (shared by tracks N, P)

This part is Section 2 of Pöschel 1993 (the Normal Form Lemma, pp. 192–196, with the Iterative Lemma, p. 193) and Appendix A (pp. 211–213), in the frozen conventions of Part 5: angles of period 1, weight $e^{2\pi|k|_1 s}$, geometric widths, max norms on $\mathbb{R}^n, \mathbb{C}^n$, and the Lie transform of Part 6. It is shared infrastructure: track P derives CP10's averaging lemma ($\alpha_1 = \alpha_2$, $\Lambda = \{0\}$) from Theorem 7.6, which is therefore stated for an arbitrary set $\Lambda \subset \mathbb{Z}^n$ (including $\{0\}$), arbitrary n, and an arbitrary domain $D \subset \mathbb{C}^n$, with all constants explicit. Nothing but Parts 5 and 6 is used.

Notation. $Z_K = \{k \in \mathbb{Z}^n : |k|_1 \le K\}$ for real $K \ge 0$ (the ℓ^1 ball: it is the cut-off of T_K, Lemma 5.9); $\omega = \partial_I h$ is the frequency; $p > 1$ and $q = p/(p-1)$, so $p^{-1} + q^{-1} = 1$ (Pöschel takes $q = 9$, $p = \frac{9}{8}$; nothing below depends on that choice); for $\alpha > 0$ and $s > 0$

$$\bar\alpha := 2\pi\alpha, \qquad \kappa := 2\pi K s.$$

By Remark 5.11, $\bar\alpha$ and κ are Pöschel's α and Ks (his ω is 2π times ours, his weight $e^{|k|_1 s_P}$ is $e^{2\pi|k|_1 s}$, his $\rho_P\sigma_P$ is our $\rho\sigma$), so every formula of pp. 192–196, 211–213 holds with $\alpha \to \bar\alpha$, $Ks \to \kappa$ and r unchanged in the combination αr; the arithmetic below is nevertheless redone in our variables.

Lattice cut-offs (finding N06). Only the ℓ^1 ball Z_K occurs here, and $|k \cdot v| \le |k|_1 |v|_\infty$ is all that is used about it. The Euclidean ball $\{\|k\|_2 \le K_g\}$ of the geometry of resonances (Part 8) is contained in $Z_{\sqrt{n}K_g}$ since $|k|_1 \le \sqrt{n}\|k\|_2$; so wherever the two meet, the analytic cut-off is $K = K_a \ge \sqrt{n}K_g$ (the factor $\sqrt{n}$ is carried by the caller). The results below are insensitive to the choice of K.

7.1. External facts.

(E1) *Numerics.* $2/e < 1$; $e^{-3\ln 2} = \frac{1}{8}$; $e^2 < 8$, hence $\ln 2 > \frac{2}{3}$ and $12\ln 2 > 8$; $\sum_{i\ge1} 4^{1-i} = \frac{4}{3}$; $(1 - \frac{1}{2^7})^{-1} = \frac{128}{127}$, $(1 - \frac{1}{8})^{-1} = \frac{8}{7}$. Exists.

(E2) *Chain rule for holomorphic maps.* Compositions of holomorphic maps are holomorphic and $D(F \circ G) = DF \circ G \cdot DG$. Exists.

Besides these, [K:`bp:p1:ext-mvt`] (mean value inequality) and the external facts of Parts 5 and 6 are used.

7.2. Setting and the divisor bound.

Definition 7.1 (Resonant setting)**.** Let $D \subset \mathbb{C}^n$, $r > 0$, $K \ge 0$, $\alpha > 0$, $L \ge 0$, $\Lambda \subset \mathbb{Z}^n$, and let h be holomorphic on $V_r D$ (Definition 5.1), regarded as a θ-independent function on $\mathcal{D}_{r,s}$ for every $s > 0$; $\omega = \partial_I h : V_r D \to \mathbb{C}^n$. We say that $(D, r, K, \alpha, L, \Lambda, h)$ satisfies

(N1) ((α, K)-*nonresonant modulo* Λ) $|k \cdot \omega(I)| \ge \alpha$ for all $I \in D$ and $k \in Z_K \setminus \Lambda$ (Pöschel p. 192);

(N2) (*frequency Lipschitz bound*) $|\omega(I) - \omega(I_0)|_\infty \le L|I - I_0|_\infty$ whenever $I_0 \in D$, $I \in V_r D$, $|I - I_0|_\infty < r$.

For $\Lambda = \{0\}$ and $\Lambda = \mathbb{Z}^n$ these are the completely nonresonant and the trivial cases. If $0 \notin \Lambda$ then (N1) forces $D = \emptyset$ (take $k = 0$), which is harmless. For $g = \sum_{j\in J} g_j$ (J finite, $g_j \in \mathcal{O}(r_j, s_j)$ with finite norm, $r_j \ge r$, $s_j \ge s$) and $0 < \rho < r$, $0 < \sigma < s$ put

$$|g|_* := \sum_{j\in J} \frac{1}{2}\Big(\frac{\rho}{r_j - r + \rho} + \frac{\sigma}{s_j - s + \sigma}\Big)|g_j|_{r_j,s_j} \qquad \text{(depends on the decomposition).}$$

If $J = \{1\}$, $g_1 = g \in \mathcal{O}(r, s)$ then $|g|_* = |g|_{r,s}$.

Source: Pos93 p. 192 (nonresonance), p. 193 ($|g|_*$), p. 211–212 ("on the domain D we have $|k\cdot\omega(I)| \ge \alpha \ldots$ by the mean value theorem and our assumptions on Q and r"): the Lipschitz bound is stated as hypothesis (N2) instead of via $\sup|Q|$, in max norms.

Lemma 7.2 (Lipschitz bound from the Hessian)**.** *If $\omega = \partial_I h$ is holomorphic on $V_r D$ and the operator norm of $D\omega = \partial_I^2 h$ (max norm on both sides) is $\le L$ on $V_r D$, then (N2) holds. In particular $L = nM_C$ if every entry of the Hessian has modulus $\le M_C$ (Lemma 2.5).*

Source: Pos93 p. 212 (mean value theorem); standard.

Proof. **Step S7.1: Segment.** For $I_0 \in D$, $I \in V_r D$ with $|I - I_0|_\infty < r$ the segment $[I_0, I]$ lies in the polydisc $P(I_0, r) \subset V_r D$ (convex, Definition 5.1). [K:bp:p1:ext-mvt] gives $|\omega(I) - \omega(I_0)|_\infty \le L|I - I_0|_\infty$. □

Lemma 7.3 (Divisor bound)**.** *Assume (N1), (N2) and $KLr \le \alpha/p$. Then $|k \cdot \omega(I)| \ge \alpha - KLr \ge \alpha/q$ for all $I \in V_r D$ and $k \in Z_K \setminus \Lambda$.*
Source: Pos93 p. 211–212 ("$|k \cdot \omega(I)| \ge \alpha - MKr \ge (1 - \frac{1}{p})\alpha = \alpha/q$").

Proof. **Step S7.2: Move to D.** Let $I \in V_r D$ and choose $I_0 \in D$ with $|I - I_0|_\infty < r$. Then $|k \cdot \omega(I)| \ge |k \cdot \omega(I_0)| - |k|_1 |\omega(I) - \omega(I_0)|_\infty \ge \alpha - KLr$ by (N1), (N2). Finally $\alpha - \alpha/p = \alpha/q$. □

7.3. The homological equation.

Proposition 7.4 (Homological equation)**.** *Assume (N1), (N2) and $KLr \le \alpha/p$. Let $s > 0$, $f \in \mathcal{O}(r, s)$ with $\delta = |f|_{r,s} < \infty$, and define*

$$\xi = P_\Lambda T_K f, \qquad \phi = \sum_{k \in Z_K \setminus \Lambda} \frac{\hat{f}_k(I)}{2\pi i\, k \cdot \omega(I)} \chi_k(\theta), \qquad \chi_k(\theta) = e^{2\pi i k \cdot \theta}.$$

Then $\xi, \phi \in \mathcal{O}(r, s)$ are finite sums of characters (trigonometric polynomials of degree $\le K$ in θ), and

- *(a) $\{\phi, h\} + \xi = T_K f$ on $\mathcal{D}_{r,s}$, where $\{\phi, h\} = \partial_\theta \phi \cdot \omega$; $P_\Lambda \phi = 0$, $P_\Lambda \xi = \xi$, $\xi = P_{\Lambda \cap Z_K} f$;*
- *(b) $|\xi|_{r,s} \le \delta$, $|T_K f|_{r,s} \le \delta$, and $|\phi|_{r,s} \le q\delta/\bar{\alpha} = q\delta/(2\pi\alpha)$;*
- *(c) if $\Lambda = -\Lambda$, f is real on real points and $\omega(I) \in \mathbb{R}^n$ for real $I \in V_r D$, then ξ and ϕ are real on real points.*

Source: Pos93 p. 211–212 ("$\xi = P_\Lambda T_K f$, $\phi = L^{-1}(T_K f - P_\Lambda T_K f)$, $\phi_k = f_k/(ik \cdot \omega(I))$ ... $|\xi|_{r,s} \le \delta$, $|\phi|_{r,s} \le q\delta/\alpha$"); the 2π in the divisor is the period-1 convention, so that the bound reads $q\delta/\bar{\alpha}$ as in the source.

Proof. **Step S7.3: Well defined and holomorphic.** For $k \in Z_K \setminus \Lambda$, $k \cdot \omega$ is holomorphic on $V_r D$ and has modulus $\ge \alpha/q > 0$ (Lemma 7.3), so $\hat{\phi}_k := \hat{f}_k/(2\pi i k \cdot \omega)$ is holomorphic (Lemma 5.3(a)). The sets $Z_K \cap \Lambda$ and $Z_K \setminus \Lambda$ are finite. Hence ϕ, ξ are finite sums of terms $a_k(I)\chi_k(\theta)$ with a_k holomorphic and $\sup_I \sum |a_k| e^{2\pi |k|_1 s} < \infty$, so they lie in $\mathcal{O}(r, s)$ with the stated coefficients (Lemma 5.3(c)); $\xi = P_\Lambda T_K f = P_{\Lambda \cap Z_K} f$ by Lemma 5.9.

Step S7.4: Norms. $|\hat{\phi}_k(I)| \le \frac{q}{2\pi\alpha} |\hat{f}_k(I)|$ by Lemma 7.3; multiplying by $e^{2\pi |k|_1 s}$, summing over $k \in Z_K \setminus \Lambda$ and taking the supremum over I gives $|\phi|_{r,s} \le q\delta/\bar{\alpha}$. The bounds for ξ and $T_K f$ are Lemma 5.9.

Step S7.5: The equation. By Lemma 5.3(b), $\partial_{\theta_j} \phi$ has coefficients $2\pi i k_j \hat{\phi}_k$; since h is independent of θ, $\{\phi, h\} = \sum_j \partial_{\theta_j} \phi\, \omega_j$, and multiplication by the θ-independent $\omega_j(I)$ multiplies each coefficient by $\omega_j(I)$ (integrate against χ_{-k}). So $\widehat{\{\phi, h\}}_k = 2\pi i (k \cdot \omega) \hat{\phi}_k = \hat{f}_k$ for $k \in Z_K \setminus \Lambda$ and 0 otherwise. Adding $\hat{\xi}_k = \hat{f}_k \mathbf{1}_{Z_K \cap \Lambda}(k)$ gives the coefficients of $T_K f$; coefficients determine the function (Lemma 5.3(d)).

Step S7.6: Reality. If f is real on real points, $\overline{\hat{f}_k(I)} = \hat{f}_{-k}(I)$ for real I (conjugate the integral defining $\hat{f}_k$). If $\omega(I)$ is real and $k \notin \Lambda$ with $-k \notin \Lambda$, then $\overline{\hat{\phi}_k} = \hat{f}_{-k}/\overline{2\pi i k \cdot \omega} = -\hat{f}_{-k}/(2\pi i k \cdot \omega) = \hat{\phi}_{-k}$; the set $Z_K \setminus \Lambda$ is symmetric. A finite sum $\sum a_k \chi_k$ with $\bar{a}_k = a_{-k}$ is real at real θ. The same for ξ. □

Lemma 7.5 (Bracket with a decomposed g). *Let $\phi \in \mathcal{O}(r,s)$ and $g = \sum_{j\in J} g_j$ as in Definition 7.1, $0 < \rho < r$, $0 < \sigma < s$. Then $\{g,\phi\} \in \mathcal{O}(r-\rho, s-\sigma)$ and $|\{g,\phi\}|_{r-\rho,s-\sigma} \le \dfrac{2}{e\rho\sigma}|g|_*|\phi|_{r,s} \le \dfrac{2}{\rho\sigma}|g|_*|\phi|_{r,s}$.*

Source: Pos93 p. 212 ("$|\{g,\phi\}|_{r-\rho,s-\sigma} \le \frac{q}{\alpha\rho\sigma}|g|_*|f|_{r,s}$"), with the bracket estimate of Lemma 5.7.

Proof. **Step S7.7: One term.** Lemma 5.7 with $(u,v;r_0,s_0) \to (g_j,\phi;r_j,s_j)$ (admissible: $r-\rho<r_j$, $s-\sigma<s_j$) gives $|\{g_j,\phi\}|_{r-\rho,s-\sigma} \le \frac{1}{e}\big(\frac{1}{(r_j-r+\rho)\sigma}+\frac{1}{(s_j-s+\sigma)\rho}\big)|g_j||\phi| = \frac{2}{e\rho\sigma}\cdot\frac{1}{2}\big(\frac{\rho}{r_j-r+\rho}+\frac{\sigma}{s_j-s+\sigma}\big)|g_j||\phi|$.

Step S7.8: Sum. The bracket is bilinear and the norm satisfies the triangle inequality; sum over j and use (E1) ($e>2$). □

7.4. The Iterative Lemma.

Theorem 7.6 (Iterative Lemma). *Let $(D,r,K,\alpha,L,\Lambda,h)$ satisfy (N1), (N2) with $KLr \le \alpha/p$ ($p>1$, $q=p/(p-1)$), let $s>0$, $0<\rho$, $0<\sigma$ with $2\rho<r$, $2\sigma<s$, and put $\bar\alpha = 2\pi\alpha$. Let $H = h+g+f$ on $\mathcal{D}_{r,s}$, where $g=\sum_{j\in J} g_j$ is as in Definition 7.1 (any finite decomposition), $f\in\mathcal{O}(r,s)$, and*

$$\delta := |f|_{r,s} \le \frac{\bar\alpha\,\rho\sigma}{4q}, \qquad \text{equivalently} \qquad u := \frac{q\delta}{\bar\alpha\rho\sigma} \le \frac{1}{4}.$$

Then there is $\phi\in\mathcal{O}(r,s)$ (the one of Proposition 7.4, so $|\phi|_{r,s}\le q\delta/\bar\alpha$) such that the time-one map $\Phi = X_\phi^1$ of Part 6 satisfies:

- *(a) Φ is defined and holomorphic on $\mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$ with values in $\mathcal{D}_{r-\rho,s-\sigma}\subset\mathcal{D}_{r,s}$; it is canonical: $(D\Phi)^\top J\,D\Phi = J$, $\Phi(\theta+k,I)=\Phi(\theta,I)+(k,0)$ ($k\in\mathbb{Z}^n$), and Φ is injective on $\mathcal{D}_{r-2\rho,s-2\sigma}$ with left inverse X_ϕ^{-1} (Proposition 6.13(d),(e),(b));*
- *(b) (displacement) with $\nu=\nu_{\rho,\sigma}$ of Definition 6.2: $\nu(\Phi-\mathrm{id})\le u$ on $\mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$, and $|\Phi_\theta-\theta|_\infty \le \dfrac{q\delta}{\bar\alpha\rho}$, $|\Phi_I-I|_\infty \le \dfrac{q\delta}{e\,\bar\alpha\sigma}$; on $\mathcal{D}_{r-2\rho,s-2\sigma}$: $|D\Phi-\mathbf{1}|_\nu\le 2u$;*
- *(c) (normal form) $H\circ\Phi = h+g_+ +f_+$ on $\mathcal{D}_{r-2\rho,s-2\sigma}$ with*
$$g_+ = g + P_\Lambda T_K f, \qquad |g_+-g|_{r,s}\le\delta, \qquad f_+\in\mathcal{O}(r-2\rho,s-2\sigma);$$
- *(d) (estimate of f_+)*
$$|f_+|_{r-2\rho,s-2\sigma} \le \Big(1-\frac{2q\delta}{\bar\alpha\rho\sigma}\Big)^{-1}\Big(\frac{2q}{e\,\bar\alpha\rho\sigma}\big(|g|_*+\delta\big)+e^{-2\pi K\sigma}\Big)\delta,$$
with $|g|_*$ computed with these ρ,σ;
- *(e) (reality) if $\Lambda=-\Lambda$, and h,g,f are real on real points, then ϕ, g_+, f_+ are real on real points and Φ maps real points to real points.*

Source: Pos93 p. 193 (Iterative Lemma; the hypothesis there is $|f|_{r,s}<\alpha\rho\sigma/2q$, the estimate $|f_+|\le(1-\frac{2q\delta}{\alpha\rho\sigma})^{-1}(\frac{|g|_*+|f|}{\alpha\rho\sigma/q}+e^{-K\sigma})|f|$) and Appendix A, pp. 211–213. Pöschel's $\alpha,\sigma,|W(\Phi-\mathrm{id})|\le\frac{q\delta}{\alpha\rho\sigma}$ correspond to $\bar\alpha, 2\pi\sigma$ and u (Remark 5.11). Deviation*: the smallness is $|\phi|\le\rho\sigma/4$ (so $\eta=2|\phi|/(\rho\sigma)\le\frac12$) instead of $<\rho\sigma/2$, because Theorem 6.10 proves the Lie series identity only for $\eta\le\frac12$ (Remark 6.1); the estimate has the extra factor $2/e<1$ from Lemma 5.7. Constants only change.*

Proof. **Step S7.9: The generating function.** By Proposition 7.4, ϕ,ξ exist with $\{\phi,h\}+\xi = T_K f$, $|\xi|,|T_Kf|\le\delta$, $|\phi|_{r,s}\le q\delta/\bar\alpha$. Put $\varepsilon := |\phi|_{r,s}/(\rho\sigma)\le u\le\frac14$ and $\eta := 2|\phi|_{r,s}/(\rho\sigma)\le 2u\le\frac12$.

Step S7.10: Items (a), (b). Proposition 6.13 (with $\rho<r/2$, $\sigma<s/2$, $\varepsilon\le\frac12$) gives (a), the displacement bounds of (b) with $|\phi|/(\rho\sigma)\le u$, $|\phi|/\rho\le q\delta/(\bar\alpha\rho)$, $|\phi|/(e\sigma)\le q\delta/(e\bar\alpha\sigma)$, and $|D\Phi-\mathbf{1}|_\nu\le 2\varepsilon\le 2u$.

Step S7.11: Taylor identity. Put $f_0 = g + T_K f \in \mathcal{O}(r-\rho, s-\sigma)$ (finite norm: $g = \sum g_j$ and monotonicity, $|g|_{r,s} \le \sum |g_j|$), $w = \{h, \phi\} = -\{\phi, h\} = \xi - T_K f$ (Proposition 7.4(a), antisymmetry, Definition 6.2); w has finite norm. Apply Proposition 6.11 with the data $(r, s; r', s') \to (r-\rho, s-\sigma; r, s)$ (admissible: $\rho < r - \rho$ iff $2\rho < r$, $r - \rho \le r' - \rho$; likewise for s; and $\eta \le \frac{1}{2}$): on $\mathcal{D}_{r-2\rho, s-2\sigma}$, $(h+f_0)\circ\Phi = h+w+f_0+\mathcal{R}$ with $\mathcal{R} = \int_0^1 \{(1-t)w + f_0, \phi\} \circ X_\phi^t \, dt$. Here $h+w+f_0 = h+g+\xi = h+g_+$, and $(1-t)w + f_0 = g + f_t$ with $f_t := (1-t)\xi + tT_K f$.

Step S7.12: Decomposition of $H \circ \Phi$. $H = (h + f_0) + (f - T_K f)$ with $f - T_K f \in \mathcal{O}(r, s)$, so on $\mathcal{D}_{r-2\rho, s-2\sigma}$

$$H \circ \Phi = h + g_+ + f_+, \qquad f_+ := \mathcal{R} + (f - T_K f) \circ \Phi.$$

Both terms are holomorphic and periodic (Theorem 6.10(ii), Corollary 6.12), so $f_+ \in \mathcal{O}(r-2\rho, s-2\sigma)$ once the norm below is finite. This is (c); $|g_+ - g| = |\xi| \le \delta$.

Step S7.13: The integrand. $v_t := \{g + f_t, \phi\} \in \mathcal{O}(r-\rho, s-\sigma)$ is affine in t, so $(t, z) \mapsto v_t(z)$ and $Dv_t(z)$ are continuous. By Lemma 7.5 and Lemma 5.7 (the single-term decomposition f_t, $|f_t| \le \delta$ since $|\xi|, |T_K f| \le \delta$),

$$|v_t|_{r-\rho, s-\sigma} \le \frac{2}{e\rho\sigma}\big(|g|_* + \delta\big)|\phi|_{r,s} \le \frac{2q}{e\,\bar\alpha\rho\sigma}\big(|g|_* + \delta\big)\delta =: V.$$

Step S7.14: The two terms. Corollary 6.12 (data as in Step S7.11) gives $|\mathcal{R}|_{r-2\rho, s-2\sigma} \le (1-\eta)^{-1} V$. By Lemma 5.9 and monotonicity, $|f - T_K f|_{r-\rho, s-\sigma} \le |f - T_K f|_{r,s-\sigma} \le e^{-2\pi K\sigma}\delta$, and Theorem 6.10(ii) (same data) gives $|(f - T_K f)\circ\Phi|_{r-2\rho, s-2\sigma} \le (1-\eta)^{-1} e^{-2\pi K\sigma}\delta$. Add and use $(1-\eta)^{-1} \le (1 - 2q\delta/(\bar\alpha\rho\sigma))^{-1}$ ($|\phi| \le q\delta/\bar\alpha$). This is (d).

Step S7.15: Reality. If $\Lambda = -\Lambda$ and h, g, f are real, then ω is real at real points (h is real-valued and holomorphic on the open set $V_r D$, so its partial derivatives at real points of $V_r D$ are real), and Proposition 7.4(c) gives ϕ, ξ real; Φ preserves real points (Proposition 6.13(b)); so $g_+ = g + \xi$ is real and $f_+ = H \circ \Phi - h - g_+$ is real on real points of $\mathcal{D}_{r-2\rho, s-2\sigma}$. □

Remark 7.7 (Special cases for track P). (1) *First step, $g = 0$:* $|g|_* = 0$ and (d) reads $|f_+| \le (1 - \frac{2q\delta}{\bar\alpha\rho\sigma})^{-1}\big(\frac{2q\delta}{e\bar\alpha\rho\sigma} + e^{-2\pi K\sigma}\big)\delta$. (2) *Averaging, $\Lambda = \{0\}$:* $P_\Lambda T_K f = \hat f_0 = \langle f \rangle$, the θ-average; (N1) is $|k \cdot \omega(I)| \ge \alpha$ for $0 < |k|_1 \le K$; the transformed Hamiltonian is $h + g + \langle f \rangle + f_+$. (3) *Constants:* no constant depends on f or on H beyond δ, $|g|_*$; the data are $(n, r, s, \rho, \sigma, K, \alpha, L, p)$. The hypotheses involve only δ through $u \le \frac{1}{4}$.

7.5. **Composition of the steps.**

Lemma 7.8 (Telescoping)**.** *Let $U_0 \supset U_1 \supset \cdots \supset U_{m+1}$ be subsets of $\mathbb{C}^{2n}$ and $\Phi_i : U_{i+1} \to U_i$ $(0 \le i \le m)$ maps. Put $\Psi = \Phi_0 \circ \cdots \circ \Phi_m : U_{m+1} \to U_0$. Then for every $z \in U_{m+1}$ and every seminorm μ on $\mathbb{C}^{2n}$ (e.g. $|\Pi_I \cdot|_\infty$, $|\Pi_\theta \cdot|_\infty$), $\mu(\Psi(z) - z) \le \sum_{i=0}^m \sup_{U_{i+1}} \mu(\Phi_i - \mathrm{id})$. If every Φ_i is holomorphic and $(D\Phi_i)^\top J D\Phi_i = J$, then Ψ is holomorphic, $(D\Psi)^\top J D\Psi = J$; if every Φ_i is injective, so is Ψ.*

Proof. **Step S7.16: Telescoping.** Put $z_{m+1} = z$, $z_i = \Phi_i(z_{i+1}) \in U_i$ (so $z_{i+1} \in U_{i+1}$ for the next step because $U_{i+1} \subset U_i$). Then $\Psi(z) - z = \sum_{i=0}^m (z_i - z_{i+1})$ and $z_i - z_{i+1} = (\Phi_i - \mathrm{id})(z_{i+1})$; apply the triangle inequality. The chain rule (E2) gives $D\Psi = \prod D\Phi_i(z_{i+1})$ and $(D\Psi)^\top J D\Psi = J$ by induction; injectivity is clear. □

7.6. **The Normal Form Lemma.**

Theorem 7.9 (Normal Form Lemma)**.** *Let $(D, r, K, \alpha, L, \Lambda, h)$ satisfy (N1), (N2), $K > 0$, and $KLr \le \alpha/p$ $(p > 1,\ q = p/(p-1))$. Let $s > 0$ and $H = h + f$ with $f \in \mathcal{O}(r, s)$ of finite norm $\varepsilon := |f|_{r,s}$. Assume*

$$\text{(i)}\quad \varepsilon \le \frac{1}{2^7 q}\cdot\frac{\alpha r}{K}, \qquad\qquad \text{(ii)}\quad \kappa = 2\pi K s \ge 6.$$

Put $r_* = r/2$, $s_* = s/6$, $c = 2qK/(\alpha r)$, $g_0 = P_\Lambda T_K f$. Then there are

- *a holomorphic, injective map $\Psi : \mathcal{D}_{r_*,s_*} \to \mathcal{D}_{r,s}$ with $(D\Psi)^\top J D\Psi = J$ and $\Psi(\theta + k, I) = \Psi(\theta, I) + (k, 0)$ $(k \in \mathbb{Z}^n)$;*
- *$g, f_* \in \mathcal{O}(r_*, s_*)$, with g a trigonometric polynomial in θ with frequencies in $\Lambda \cap Z_K$ (so $P_\Lambda g = g$: $h + g + f_*$ is in Λ-resonant normal form up to f_*),*

such that $H \circ \Psi = h + g + f_*$ on $\mathcal{D}_{r_*,s_*}$ and

(a) $|g - g_0|_{r_*,s_*} \le 16 c\,\varepsilon^2$ $\quad (g = g_0$ if $N = 0$ (Step S7.19) or in the second case of Step S7.18);
(b) $|f_*|_{r_*,s_*} \le e^{-2\pi K s/6}\varepsilon = e^{-\kappa/6}\varepsilon$;
(c) $|\Pi_I \Psi - I|_\infty \le qK\varepsilon/\alpha$ $\big(= \frac{1}{2} c\, r\varepsilon\big)$ and $|\Pi_\theta \Psi - \theta|_\infty \le 10\, q\varepsilon/(2\pi\alpha r)$ on $\mathcal{D}_{r_*,s_*}$;
(d) if $\Lambda = -\Lambda$ and h, f are real on real points, then g, f_* are real on real points and Ψ maps real points to real points.

The data on which Ψ, g, f_* are constructed are $(h, f, K, \alpha, L, \Lambda, p)$ and (r, s); r_*, s_*, c depend only on (r, s, K, α, q).

Source: Pos93 p. 192 (Normal Form Lemma) and pp. 193–196 (proof). Changes: (1) frozen conventions ($\bar\alpha = 2\pi\alpha$, $\kappa = 2\pi K s$ are his α, Ks; max norms); (2) $|\Pi_I\Psi - I| \le qK\varepsilon/\alpha$ in the max norm instead of $\|\Pi_I\Psi - I\| \le c\varepsilon r = 2qK\varepsilon/\alpha$ (Euclidean); (3) the smallness $\eta \le \frac{1}{2}$ of Theorem 6.10 is built into Theorem 7.6; every numerical step below was recomputed (Remark 7.12).

Source gap: Pöschel's text p. 194 ends the second case with "$\le e^{Ks/6}\varepsilon$"; the correct (and used) bound is $e^{-Ks/6}\varepsilon$, as the first line of that display shows.

Proof. Throughout $\bar\alpha = 2\pi\alpha$ and $\kappa = 2\pi K s$; note $\bar\alpha s = \alpha\kappa/K$ and $K \ge 3/(\pi s)$. *Indexing:* $V_i = \mathcal{D}_{r_i,s_i}$, $f_0 := f$, $g_{-1} := 0$; step i applies Theorem 7.6 to $H_i = h + g_{i-1} + f_i$ on V_i with losses (ρ_i, σ_i), producing $\Phi_i : V_{i+1} \to V_i$, $g_i = g_{i-1} + P_\Lambda T_K f_i$, f_{i+1} on V_{i+1}, $r_{i+1} = r_i - 2\rho_i$, $s_{i+1} = s_i - 2\sigma_i$; $\varepsilon_i := |f_i|_{V_i}$, $\varepsilon_0 = \varepsilon$. ($V_{i+1}$ stands for $\mathcal{D}_{r_i - 2\rho_i, s_i - 2\sigma_i}$, which is where Φ_i is injective and f_{i+1} lives; Φ_i maps it into $\mathcal{D}_{r_i - \frac{3}{2}\rho_i, s_i - \frac{3}{2}\sigma_i} \subset V_i$.)

Step S7.17: Step 0: hypotheses of the Iterative Lemma. Let $\rho_0 = r/8$, $\sigma_0 = s/6$, so $r_1 = \frac{3}{4}r$, $s_1 = \frac{2}{3}s$, $\rho_0\sigma_0 = rs/48$. Then $2\rho_0 < r$, $2\sigma_0 < s$, $KLr \le \alpha/p$. From (i), (ii): $\varepsilon \le \frac{\alpha r}{2^7 qK} \le \frac{\alpha r}{2^7 q}\cdot\frac{\pi s}{3} = \frac{\pi\alpha r s}{384 q} = \frac{\bar\alpha\rho_0\sigma_0}{16 q}$. Hence $y_0 := \frac{q\varepsilon}{\bar\alpha\rho_0\sigma_0} \le \frac{1}{16}$ and the hypothesis $u \le \frac{1}{4}$ of Theorem 7.6 holds. Also $\kappa/6 = 2\pi K\sigma_0 \ge 1$.

Step S7.18: Step 0: two cases. Apply Theorem 7.6(d) with $g = 0$ ($|g|_* = 0$), $\delta = \varepsilon$: $|f_1| \le (1 - 2y_0)^{-1}\big(\frac{2}{e}y_0 + e^{-\kappa/6}\big)\varepsilon$, with $(1 - 2y_0)^{-1} \le \frac{8}{7}$.

Case A: $e^{-\kappa/6} \le y_0$. Then $|f_1| \le \frac{8}{7}\big(\frac{2}{e} + 1\big)y_0\varepsilon \le \frac{16}{7}y_0\varepsilon$, so

$$\varepsilon_1 \le \frac{16}{7}\cdot\frac{48\, q\varepsilon^2}{\bar\alpha r s} \le \frac{2^7 q\varepsilon^2}{\bar\alpha r s} \le \frac{2^7 q\varepsilon}{\bar\alpha r s}\cdot\frac{\alpha r}{2^7 qK}\,\varepsilon = \frac{\varepsilon}{2\pi K s} = \frac{\varepsilon}{\kappa} \le \frac{\varepsilon}{6} \le \frac{\varepsilon}{4}. \tag{6.1}$$

Case B: $e^{-\kappa/6} > y_0$. Repeat the step with twice the losses $\rho' = r/4$, $\sigma' = s/3$ (so $2\rho' < r$, $2\sigma' < s$, $\bar\alpha\rho'\sigma' = 4\bar\alpha\rho_0\sigma_0$, the hypothesis $u' = y_0/4 \le \frac{1}{4}$ holds). Then $|f_1|_{r/2,s/3} \le (1 - y_0/2)^{-1}\big(\frac{2}{e}\cdot\frac{y_0}{4} + e^{-2\kappa/6}\big)\varepsilon$; as $e^{-2\kappa/6} \le e^{-1}e^{-\kappa/6}$ and $y_0 < e^{-\kappa/6}$, this is $\le \frac{32}{31}\big(\frac{1}{2e} + \frac{1}{e}\big)e^{-\kappa/6}\varepsilon \le e^{-\kappa/6}\varepsilon$. We stop: $\Psi = \Phi_0'$ on $\mathcal{D}_{r/2,s/3} \supset \mathcal{D}_{r_*,s_*}$, $g = g_0$, $f_* = f_1$ restricted to V_{r_*,s_*}. Items (a), (b) hold; items (c), (d) in Step S7.22. From now on *Case A*.

Step S7.19: Number of steps. Let $N = \lfloor \kappa/(12\ln 2)\rfloor \ge 0$, so $N \le \frac{\kappa}{12\ln 2} < N + 1$ and $\kappa \ge 12\ln 2\, N > 8N$ ((E1)). If $N = 0$ stop after Step 0: $\Psi = \Phi_0$, $g = g_0$, $f_* = f_1$ on $V_1 \supset \mathcal{D}_{r_*,s_*}$ and

$|f_*| \le \varepsilon_1 \le \varepsilon/4 \le e^{-\kappa/6}\varepsilon$, because $\kappa < 12\ln 2$ gives $e^{-\kappa/6} > e^{-2\ln 2} = \frac14$. Let $N \ge 1$ and, for $1 \le i \le N$,

$$\rho_i = \rho = \frac{r}{8N}, \quad \sigma_i = \sigma = \frac{s}{4N}, \qquad r_i = \tfrac34 r - 2(i-1)\rho, \quad s_i = \tfrac23 s - 2(i-1)\sigma;$$

then $r_{N+1} = \frac34 r - \frac{r}{4} = r_*$ and $s_{N+1} = \frac23 s - \frac{s}{2} = s_*$. Note $2\pi K\sigma = \kappa/(4N) \ge 3\ln 2$, i.e. $e^{-2\pi K\sigma} \le \frac18$, and $\rho\sigma = rs/(32N^2)$. Smallness constants: from (i), $\kappa > 8N$ and $K = \kappa/(2\pi s)$,

$$\varepsilon \le \frac{\alpha r \cdot 2\pi s}{2^7 q\kappa} \le \frac{\bar\alpha rs}{2^{10} qN}, \qquad \text{hence by (6.1)} \quad \varepsilon_1 \le \frac{2^7 q}{\bar\alpha rs}\Big(\frac{\bar\alpha rs}{2^{10}qN}\Big)^2 = \frac{\bar\alpha rs}{2^{13}qN^2} = \frac{\bar\alpha\rho\sigma}{2^8 q}. \tag{6.2}$$

Step S7.20: Induction. *Claim:* for $1 \le i \le N$, step i is admissible and $\varepsilon_{i+1} \le \varepsilon_i/4$. Hence $\varepsilon_i \le 4^{1-i}\varepsilon_1$ for $1 \le i \le N+1$, and $|g_i - g_{i-1}|_{V_i} = |P_\Lambda T_K f_i| \le \varepsilon_i$ (Lemma 5.9). Induct on i; $\varepsilon_1 \le \varepsilon/4$ is (6.1) (so for $i = 1$ the claim "$\varepsilon_i \le 4^{1-i}\varepsilon_1$" is trivial). Let $i \le N$ and suppose $\varepsilon_j \le 4^{1-j}\varepsilon_1$ for $1 \le j \le i$. *Admissibility.* $r_i \ge r_{N+1} = r/2 > 2\rho$; $s_i \ge s_N = s_* + 2\sigma = \frac{s}{6} + \frac{s}{2N} > 2\sigma$; $KLr_i \le KLr \le \alpha/p$; and $\varepsilon_i \le \varepsilon_1 \le \bar\alpha\rho\sigma/(2^8 q) \le \bar\alpha\rho\sigma/(4q)$ by (6.2). *Size of* $|g_{i-1}|_*$ (computed with (r_i, s_i, ρ, σ) in Definition 7.1). Decompose $g_{i-1} = g_0 + \sum_{j=1}^{i-1}(g_j - g_{j-1})$, the term $j \ge 1$ living on V_j with $r_j - r_i = 2(i-j)\rho \ge 2\rho$, $s_j - s_i \ge 2\sigma$, so its weight is $\le \frac12(\frac13 + \frac13) \le \frac12$; the term $g_0 = P_\Lambda T_K f$ lives on $V_0 = \mathcal{D}_{r,s}$ with norm $\le \varepsilon$ and $r - r_i \ge r - r_1 = \frac{r}{4}$, $s - s_i \ge \frac{s}{3}$, so its weight is $\le \frac12\big(\frac{r/8N}{r/4} + \frac{s/4N}{s/3}\big) = \frac{5}{8N} \le \frac{1}{N}$. Thus

$$|g_{i-1}|_* \le \tfrac12(\varepsilon_{i-1} + \cdots + \varepsilon_1) + \tfrac1N\varepsilon \le \tfrac12 \cdot \tfrac43\varepsilon_1 + \tfrac{\varepsilon}{N} \le \varepsilon_1 + \frac{\varepsilon}{N} \le \frac{\bar\alpha\rho\sigma}{2^8 q} + \frac{\bar\alpha\rho\sigma}{2^5 q} \le \frac{\bar\alpha\rho\sigma}{2^4 q},$$

using (6.2) and $\varepsilon/N \le \bar\alpha rs/(2^{10}qN^2) = 2^5\bar\alpha\rho\sigma/(2^{10}q)$. *The step.* Theorem 7.6(d) with $\delta = \varepsilon_i$: $\frac{2q}{\bar\alpha\rho\sigma}\varepsilon_i \le 2^{-7}$, so $(1-2^{-7})^{-1} = \frac{128}{127}$, and $\frac{2q}{e\bar\alpha\rho\sigma}(|g_{i-1}|_* + \varepsilon_i) \le \frac{2}{e}(\frac{1}{16} + \frac{1}{2^8}) \le \frac{1}{16} + \frac{1}{2^8}$, $e^{-2\pi K\sigma} \le \frac18$:

$$\varepsilon_{i+1} \le \frac{128}{127}\Big(\frac{1}{2^4} + \frac{1}{2^8} + \frac18\Big)\varepsilon_i \le \frac{128}{127} \cdot 0.1915\,\varepsilon_i < \frac14\varepsilon_i.$$

Step S7.21: Conclusion of the construction. Put $\Psi = \Phi_0 \circ \cdots \circ \Phi_N$, $g = g_N = g_0 + \sum_{i=1}^N P_\Lambda T_K f_i$, $f_* = f_{N+1}$, all restricted to $V_{N+1} = \mathcal{D}_{r_*,s_*}$. Since $H_{i+1} = H_i \circ \Phi_i$, $H \circ \Psi = H_{N+1} = h + g + f_*$. g is a finite sum of trigonometric polynomials with frequencies in $\Lambda \cap Z_K$. Lemma 7.8 with $U_i = V_i$ gives the symplectic, periodic (each Φ_i is, Theorem 7.6(a)) and injective Ψ. (b): $|f_*| = \varepsilon_{N+1} \le 4^{-N}\varepsilon_1 \le 4^{-N-1}\varepsilon \le e^{-\kappa/6}\varepsilon$, because $N + 1 > \kappa/(12\ln 2)$ gives $4^{-(N+1)} = e^{-(N+1)2\ln 2} < e^{-\kappa/6}$. (a): $g - g_0 = \sum_{i=1}^N P_\Lambda T_K f_i$, so $|g - g_0| \le \frac43\varepsilon_1 \le \frac43 \cdot \frac{2^7 q\varepsilon^2}{\bar\alpha rs}$ by (6.1); $\bar\alpha s = \alpha\kappa/K$ and $\kappa > 8N \ge 8$ give $\le \frac43 \cdot \frac{2^7}{8} \cdot \frac{qK\varepsilon^2}{\alpha r} \le 32\frac{qK\varepsilon^2}{\alpha r} = 16c\varepsilon^2$.

Step S7.22: (c) Displacement. By Lemma 7.8 and Theorem 7.6(b) ($|\Phi_I - I| \le q\delta/(e\bar\alpha\sigma)$, $|\Phi_\theta - \theta| \le q\delta/(\bar\alpha\rho)$, $\delta = \varepsilon_i$): in Case A

$$|\Pi_I\Psi - I|_\infty \le \frac{q\varepsilon}{e\bar\alpha\sigma_0} + \sum_{i=1}^N \frac{q\varepsilon_i}{e\bar\alpha\sigma} \le \frac{6q\varepsilon}{e\bar\alpha s} + \frac{q}{e\bar\alpha\sigma} \cdot \frac43\varepsilon_1.$$

Now $1/(\bar\alpha s) = K/(\alpha\kappa) \le K/(6\alpha)$; $1/(\bar\alpha\sigma) = 4NK/(\alpha\kappa) \le 4K/(12\ln 2\,\alpha)$ because $N \le \kappa/(12\ln 2)$, and $\varepsilon_1 \le \varepsilon/4$; so the right side is at most $\frac{qK\varepsilon}{e\alpha}\big(1 + \frac{4}{12\ln 2} \cdot \frac43 \cdot \frac14\big) \le \frac{qK\varepsilon}{e\alpha} \cdot 1.17 \le \frac{qK\varepsilon}{\alpha}$. Similarly $|\Pi_\theta\Psi - \theta|_\infty \le \frac{q\varepsilon}{\bar\alpha\rho_0} + \frac{q}{\bar\alpha\rho} \cdot \frac43\varepsilon_1 = \frac{8q\varepsilon}{\bar\alpha r} + \frac{8Nq}{\bar\alpha r} \cdot \frac43 \cdot \frac{\varepsilon}{\kappa} \le \frac{q\varepsilon}{\bar\alpha r}\big(8 + \frac{32}{3\cdot 12\ln 2}\big) \le \frac{10\,q\varepsilon}{\bar\alpha r}$ (using $\varepsilon_1 \le \varepsilon/\kappa$ from (6.1), $N/\kappa \le 1/(12\ln 2)$). In Case B, $\Psi = \Phi_0'$ with losses $(2\rho_0, 2\sigma_0)$ and $\delta = \varepsilon$, and the bounds are smaller by the factor $\frac12$.

Step S7.23: (d) Reality. Each step preserves reality (Theorem 7.6(e)); g_i, f_i, H_i stay real on real points and so do the Φ_i; compositions of real-point-preserving maps preserve real points. □

Corollary 7.10 (Consequence for the action variable)**.** *In Theorem 7.9, if $(\theta', I') = \Psi(\theta, I)$ with $(\theta, I) \in \mathcal{D}_{r_*,s_*}$ then $|I' - I|_\infty \le qK\varepsilon/\alpha$; in particular if $\varepsilon \le \alpha r/(2^7 qK)$ (hypothesis (i)) this is $\le r/2^7$.*
Source: Pos93 p. 196, last display.

Proof. Theorem 7.9(c) and hypothesis (i). □

7.7. **The case $2\pi Ks < 6$.**

Proposition 7.11 (No transformation for small Ks)**.** *Let $H = h + f$ on $\mathcal{D}_{r,s}$, $f \in \mathcal{O}(r,s)$ with finite norm ε and $\kappa = 2\pi Ks < 6$. Then, with $\Psi = \mathrm{id}$, $g = 0$, $f_* = f$, one has $H \circ \Psi = h + g + f_*$ on $\mathcal{D}_{r,s}$ and*

$$|\Psi - \mathrm{id}| = 0, \qquad |f_*|_{r,s} = \varepsilon \le e \cdot e^{-\kappa/6}\varepsilon,$$

and $g = 0 = P_\Lambda g$ is in Λ-normal form for every Λ. (No nonresonance, convexity or smallness hypothesis is used.)
Source: Pos93 p. 197, first line ("For $Ks \le 6$ the same reasoning applies in original coordinates, letting $g = 0$ and $f_* = f_\varepsilon$").
Source gap: The source does not say what is claimed about $|f_*|$ in this case. With $f_* = f$, the bound $|f_*| \le e^{-\kappa/6}\varepsilon$ of the Normal Form Lemma is false *for $\kappa < 6$ (take $\Psi = \mathrm{id}$, $f \ne 0$, $K \to 0$); what holds is $|f_*| \le e^{1-\kappa/6}\varepsilon$, since $\kappa/6 < 1$. Part 7 (Proposition 1, the time bound $\frac{sr}{5\varepsilon}e^{Ks/6}$) must therefore carry the factor e^{-1} in this case or use $|f_*| = \varepsilon$ directly; this changes constants only. The trivial statement is recorded here once so that Parts 7 and 9 cite it.*

Proof. **Step S7.24: Trivial.** Identity map, $g = 0$; $|f_*| = \varepsilon$ and $1 \le e^{1-\kappa/6}$ for $\kappa \le 6$. □

7.8. **Deviations, errata, uses.**

Remark 7.12 (Deviations from Pöschel and bookkeeping).
(1) *Lie-series range.* Part 6 proves $u \circ X_\phi^1 = \sum_h \frac{1}{h!}\mathrm{ad}_\phi^h u$ for $\eta = 2|\phi|/(\rho\sigma) \le \frac{1}{2}$ only. Hence the Iterative Lemma assumes $|\phi| \le \rho\sigma/4$, which is implied by $\delta \le \bar\alpha\rho\sigma/(4q)$ (Pöschel: $\delta < \alpha\rho\sigma/2q$). All hypotheses on ε in the Normal Form Lemma are already stronger than this: $\varepsilon_0 \le \bar\alpha\rho_0\sigma_0/(16q)$ and $\varepsilon_i \le \bar\alpha\rho\sigma/(2^8 q)$ (Steps S7.17, S7.20). The numerical constants 2^7, 4^{1-i}, $N = \lfloor\kappa/(12\ln 2)\rfloor$, $r_* = r/2$, $s_* = s/6$, $16c\varepsilon^2$ are therefore *the same as Pöschel's* (re-derived above), and the number of iteration steps is unchanged.
(2) *Constants gained.* The factor $2/e < 1$ in the estimate of f_+ (from $\frac{1}{e} < \frac{1}{2}$ in Lemma 5.7); where the source's arithmetic is reused we simply bound $2/e \le 1$.
(3) *Norms.* Max norms on actions and angles replace his Euclidean/sup norms; the divisor bound uses $|k \cdot v| \le |k|_1|v|_\infty$, which is why Z_K is the ℓ^1 ball and hypothesis (N2) is stated in $|\cdot|_\infty$. The Euclidean bound $\|\Pi_I\Psi - I\| \le c\varepsilon r$ of the source follows from the max-norm statement of Theorem 7.9(c) only with a factor $\sqrt{n}$.
(4) *Not proved in the source, proved here:* the θ-displacement of Ψ and the injectivity of Ψ (from Proposition 6.13(e)); the $|W_0(\Psi - \mathrm{id})|$ bound of p. 195 ("usual telescoping arguments") is replaced by the componentwise bound of Theorem 7.9(c).
Source gap: Pöschel's last display of p. 195 asserts $|W_0(D\Psi - I)W_0^{-1}| \le 2q\varepsilon_0/(\alpha\rho_0\sigma_0)$ by telescoping; the later steps contribute $\sum_{i\ge1} q\varepsilon_i/(\bar\alpha\rho\sigma)$ in the scaled norm $\nu_{\rho,\sigma}$, which is not obviously $\lesssim q\varepsilon_0/(\bar\alpha\rho_0\sigma_0)$ for large N since $\rho\sigma/(\rho_0\sigma_0) = 3/(2N^2)$. It is not used by the stability estimates (only $\Pi_I\Psi$ is), so it is not reproduced.
(5) *The second case of the first step* (his "twice the step size") is Step S7.18, Case B, and ends with the correct sign of the exponent (see the
Source gap:
of Theorem 7.9).

(6) *Domains.* The generating function lives on $\mathcal{D}_{r,s}$, the map Φ on $\mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$ and the new remainder on $\mathcal{D}_{r-2\rho,s-2\sigma}$; hence the chain $V_{i+1}=\mathcal{D}_{r_i-2\rho_i,s_i-2\sigma_i}$ and Ψ on $\mathcal{D}_{r/2,s/6}$ as in the source.

Remark 7.13 (Consumers). Part 7 uses Theorem 7.9, Corollary 7.10 and Proposition 7.11; Part 9 fixes $q=9$, $p=\frac{9}{8}$ and, with Part 8, chooses K (as $K_a\geq\sqrt{n}K_g$) and the domain D. Track P (Part 2 of its plan) takes Theorem 7.6 with $\Lambda=\{0\}$, Remark 7.7(2), the elementary constant bookkeeping being its own.

8. Stability estimates: nonresonant, convex, quasi-convex and resonant (shared by tracks N, I)

This part is Section 3 of Pöschel 1993 (pp. 196–200): the nonresonant Proposition 1, the two energy estimates near a resonance (Lemma 1 for m-convex h, Lemma 2 for ℓ,m-quasi-convex h) and the Resonant Stability Lemma (Proposition 2), which carries these estimates from normal-form coordinates back to the original ones. The idea is that in normal form the action of an orbit moves only through the non-resonant remainder f_* (exponentially small) and through the resonant part g, whose gradient lies in the span of Λ. Energy conservation and convexity of h transverse to that span then bound the motion in the resonant directions by $\sqrt{\zeta/m}$, without any further small-divisor analysis; the span itself is handled because D is close to the exact resonance space.
Conventions. Everything is in the frozen conventions of Parts 5–7: angles of period 1, $\chi_k(\theta)=e^{2\pi ik\cdot\theta}$, geometric strip width s, weight $e^{2\pi|k|_1 s}$, max norms $|\cdot|=|\cdot|_\infty$ on actions, $\kappa=2\pi Ks$, $\bar\alpha=2\pi\alpha$. The convexity argument needs an inner product: there the Euclidean norm $\|\cdot\|_2$ on $\mathbb{R}^n$ is used, with the comparisons of Lemma 2.1 ($|x|\leq\|x\|_2\leq\sqrt{n}|x|$). Time and the values of Hamiltonians are those of the frozen system $\dot\theta=\partial_I H$, $\dot I=-\partial_\theta H$ with period-1 angles; no rescaling to Pöschel's 2π-normalisation is needed, because Part 5 (Remark 5.11) shows that the quantities $s\,r$, $s\,m\,r^2/\omega$, $|\phi|/(\rho\sigma)$ are the same numbers in both normalisations (checked below where they occur, Remark 8.21). We assume $n\geq 1$.

8.1. External facts.

(E1) *Banach fixed point theorem.* A contraction of a nonempty closed subset of $\mathbb{R}^N$ into itself has a fixed point (Mathlib `ContractingWith.exists_fixedPoint`). Exists.

(E2) *Taylor, Lagrange form, order one.* If $\varphi:[0,1]\to\mathbb{R}$ is C^2, then $\varphi(1)=\varphi(0)+\varphi'(0)+\frac{1}{2}\varphi''(\xi)$ for some $\xi\in(0,1)$ (Mathlib `taylor_mean_remainder_lagrange`). Exists.

(E3) *Numerics.* $e>2.7$, so $\frac{1}{12e}<0.0307$, $\frac{6}{5e}<0.442$, $\frac{15e}{192}>\frac{1}{5}$; Young's inequality $ab\leq\frac{a^2}{2c}+\frac{cb^2}{2}$ ($c>0$); $xe^{-x}\leq e^{-1}$ for $x\geq 0$. Exists.

(E4) *Orthogonal projections.* For a subspace $E\subset\mathbb{R}^n$ with orthogonal projection Π and $\Pi^\perp=1-\Pi$: Π is self-adjoint, $\|\Pi v\|_2\leq\|v\|_2$, $\langle u,\Pi v\rangle=\langle\Pi u,v\rangle$, and $\|\Pi v\|_2=\operatorname{dist}_2(v,E^\perp)$. Exists.

Track-K and earlier-part results used: [K:`bp:p1:ext-mvt`], [K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:hamiltonian-real`], [K:`bp:p1:real-restriction`]; (E3), Lemma 2.1, Lemma 2.7; Part 4 (Definition 4.2, Lemma 4.19, Theorems 4.11, 4.13, Corollary 4.16, Lemma 4.22, Theorem 4.23); Part 5 (Definition 5.1, 5.2, Lemmas 5.3, 5.4, 5.6, 5.8, 5.9); Lemma 6.5; Part 7 (Definition 7.1, Lemma 7.2, Theorem 7.9, Proposition 7.11).

8.2. Real systems on a real neighbourhood of D.

Definition 8.1 (Real neighbourhoods, real systems, orbits)**.** Let $D\subset\mathbb{R}^n$ and $\varrho,s>0$. Put $U_\varrho D=V_\varrho D\cap\mathbb{R}^n=\{I\in\mathbb{R}^n:\exists J\in D,\ |I-J|<\varrho\}$ (Definition 5.1); it is open, and $W_\varrho:=\mathbb{R}^n\times U_\varrho D$ is open in $\mathbb{R}^{2n}$ and equals $\mathcal{D}_{\varrho,s}\cap\mathbb{R}^{2n}$ (real θ lies in T_s). For $0<\varrho\leq r$, $W_\varrho\subset W_r$. $\mathcal{O}(r,s)^{\mathbb{R}}$ denotes the set of $G\in\mathcal{O}(r,s)$ with $G(x)\in\mathbb{R}$ for $x\in W_r$ ("real on real points"). For such G the *real Hamiltonian* is $G^{\mathbb{R}}(x)=G(x)$ for $x\in W_r$ and $G^{\mathbb{R}}(x)=0$ otherwise (a total function $\mathbb{R}^{2n}\to\mathbb{R}$; its values off W_r never matter, Remark 4.1). A *G-orbit in W_ϱ on J* ($J\subset\mathbb{R}$ an interval, $0<\varrho\leq r$)

is a $z : \mathbb{R} \to \mathbb{R}^{2n}$ with $\texttt{IsSolIn}\, X_{G^{\mathbb{R}}}\, W_\varrho\, J\, z$ (Definition 4.2). We write $z = (\theta, I)$ and $I(t) = z(t)_2$. For h holomorphic on $V_r D$ and real on $U_r D$: $h_{\mathbb{R}} = h|_{U_r D}$, $\omega = \nabla h_{\mathbb{R}}$ (real, $= \partial_I h$), $Q(I) = D^2 h_{\mathbb{R}}(I)$ (symmetric).
Source: Ours (bookkeeping for [K:bp:p1:hamiltonian-real] and Part 4); Pöschel's $U_r D$, $V_r D$ (p. 189) in max norm.

Lemma 8.2 (Holomorphic real systems). *Let $G \in \mathcal{O}(r,s)^{\mathbb{R}}$, $0 < \varrho \le r$. Then (a) $G^{\mathbb{R}}$ is C^∞ on W_r and $G^{\mathbb{R}}(\theta + k, I) = G^{\mathbb{R}}(\theta, I)$; (b) at $x \in W_r$, $G^{\mathbb{R}}$ is differentiable and $X_{G^{\mathbb{R}}}(x) = X_G(x) = \big((\partial_{I_i} G(x))_i, (-\partial_{\theta_i} G(x))_i\big) \in \mathbb{R}^{2n}$ (complex partials at the real point); (c) $X_{G^{\mathbb{R}}}$ is C^1 on W_r and 1-periodic in θ; (d) Theorems 4.23, 4.11, 4.13, Corollary 4.16 and Lemma 4.19 apply to $F = X_{G^{\mathbb{R}}}$ on the open set W_ϱ; in particular $G^{\mathbb{R}}$ is constant along every G-orbit on every interval (energy conservation).*
Source: Ours; [K:bp:p1:hamiltonian-real], [K:bp:p1:holomorphic-regularity]*; the special case $G = h + f$ of N-HEAD-C is Lemma 2.3.*

Proof. **Step S8.1: Smoothness.** G is holomorphic on $\mathcal{D}_{r,s}$, hence C^∞ in the real sense ([K:bp:p1:holomorphic-regularity], (E3)); $W_r = \iota^{-1}(\mathcal{D}_{r,s})$ is open and $G^{\mathbb{R}} = G \circ \iota$ on it, so $G^{\mathbb{R}} \in C^\infty(W_r)$, with $D^k G^{\mathbb{R}} = D^k G \circ \iota$ on real vectors. Periodicity is the periodicity in $\mathcal{O}(r,s)$.

Step S8.2: Field. [K:bp:p1:hamiltonian-real] with $V = \mathcal{D}_{r,s}$ (open, G holomorphic, real at the real points of V since $V \cap \mathbb{R}^{2n} = W_r$): $G^{\mathbb{R}}$ agrees with $\operatorname{Re} G$ on the neighbourhood W_r of each real point of V, so $X_{G^{\mathbb{R}}} = X_G$ there. The field is C^1 by Lemma 4.22, periodic by (a) and Lemma 4.24.

Step S8.3: (d). Lemma 4.22 ($G^{\mathbb{R}} \in C^2$ on the open set W_ϱ); the cited results need only F C^1 on the open set W_ϱ, periodicity (for the corollary) and, for energy, differentiability of $G^{\mathbb{R}}$ at the points of the orbit. □

Lemma 8.3 (Derivative bound at real points). *Let $u \in \mathcal{O}(\varrho, s)$ have finite norm. Then for all $\theta \in \mathbb{R}^n$, $I \in U_\varrho D$: $\sum_{j=1}^n |\partial_{\theta_j} u(\theta, I)| \le |u|_{\varrho,s}/(e\,s)$; in particular $\|\partial_\theta u(\theta, I)\|_2 \le |u|_{\varrho,s}/(es)$. Also $|u(x)| \le |u|_{\varrho,s}$ for $x \in \mathcal{D}_{\varrho,s}$.*
Source: Pos93 p. 198 ("by the Cauchy estimate of Lemma B.3": $\sup\|\partial_\theta f_*\| \le \frac{1}{es}|f_*|_{r,s}$); the limit $s' \downarrow 0$ in Lemma 5.6(a) is ours.

Proof. **Step S8.4: Strip loss.** For $0 < s' < s$ and fixed I, Step S5.12 gives $\sum_j |\partial_{\theta_j} u|_{I,s'} \le |u|_{I,s}/(e(s - s'))$ (Lemma 5.6(a)). By the pointwise bound in Step S5.7 (Lemma 5.4(a)), $|v(\theta, I)| \le |v|_{I,s'}$ for $\theta \in T_{s'} \supset \mathbb{R}^n$.

Step S8.5: Limit. Let $s' \downarrow 0$; $\|v\|_2 \le \sum_j |v_j|$ (Lemma 2.1). The last claim is Lemma 5.4(a). □

Lemma 8.4 (Resonant gradients lie in the span of Λ). *Let $\Lambda \subset \mathbb{Z}^n$, $\Pi = \Pi_\Lambda$ the Euclidean orthogonal projection onto $\operatorname{span}_{\mathbb{R}} \Lambda$, $\Pi^\perp = 1 - \Pi$. If $g \in \mathcal{O}(r,s)^{\mathbb{R}}$ has finite norm and $P_\Lambda g = g$, then $\partial_\theta g(\theta, I) \in \operatorname{ran} \Pi$, i.e. $\Pi^\perp \partial_\theta g = 0$, for all $(\theta, I) \in W_r$.*
Source: Pos93 p. 198 ("the vector $Q\Delta I$ is given by the integral ... of $\partial_\theta f_*$ only, since g is in Λ-resonant normal form").

Proof. **Step S8.6: Coefficients.** The components of $u = \Pi^\perp \partial_\theta g$ are finite combinations of $\partial_{\theta_j} g \in \mathcal{O}(r,s)$ with coefficients $\hat{u}_k = 2\pi i\, \Pi^\perp k\, \hat{g}_k$ (Lemma 5.3(b)). For $k \in \Lambda$, $\Pi^\perp k = 0$; for $k \notin \Lambda$, $\hat{g}_k = 0$ because $g = P_\Lambda g$ (Lemma 5.9). So $\hat{u}_k = 0$ for all k.

Step S8.7: Vanishing. For $s' < s$, u has finite norm at width s' (Lemma 5.6(a)), so $u = \sum_k \hat{u}_k \chi_k = 0$ on $\mathcal{D}_{r,s'}$ (Lemma 5.3(d)); real points lie in it. □

8.3. **Transfer through a symplectic change of variables.**

Lemma 8.5 (Conjugation of orbits)**.** *Let $H \in \mathcal{O}(r,s)^{\mathbb{R}}$, $\tilde{H} \in \mathcal{O}(\tilde{r},\tilde{s})^{\mathbb{R}}$, and $\Psi : \mathcal{D}_{\tilde{r},\tilde{s}} \to \mathcal{D}_{r,s}$ holomorphic, real at real points, with $(D\Psi)^{\top} J D\Psi = J$ and $\tilde{H} = H \circ \Psi$. If $\tilde{z}$ is an $\tilde{H}$-orbit in $W_{\tilde{r}}$ on J, then $z = \Psi \circ \tilde{z}$ is an H-orbit in W_r on J.*
Source: Standard; Corollary 6.6(c) for a single flow map, here for an arbitrary symplectic Ψ.

Proof. **Step S8.8: Real points.** Ψ maps $W_{\tilde{r}} = \mathcal{D}_{\tilde{r},\tilde{s}} \cap \mathbb{R}^{2n}$ into $\mathcal{D}_{r,s} \cap \mathbb{R}^{2n} = W_r$; the real derivative $Y = D\Psi(x)$ at $x \in W_{\tilde{r}}$ is the (real) restriction of the complex one ([K:`bp:p1:real-restriction`]), and $Y^{\top} J Y = J$; hence $\det Y = \pm 1$ and $Y J Y^{\top} = J$ (Step S6.10).

Step S8.9: Chain rule. $\tilde{H}^{\mathbb{R}} = H^{\mathbb{R}} \circ \Psi$ on $W_{\tilde{r}}$, so $\nabla \tilde{H}^{\mathbb{R}}(x) = Y^{\top} \nabla H^{\mathbb{R}}(\Psi x)$ and, with $X_u = J\nabla u$, $\frac{d}{dt}\Psi(\tilde{z}) = Y\tilde{z}' = Y J Y^{\top} \nabla H^{\mathbb{R}}(\Psi\tilde{z}) = X_{H^{\mathbb{R}}}(\Psi\tilde{z})$ for $t \in J$ (Lemma 8.2(b)); $z(t) \in W_r$ by Step S8.8. □

Lemma 8.6 (Real preimages of a near-identity map)**.** *Let $D \subset \mathbb{R}^n$, $\tilde{r}, \tilde{s} > 0$, $\Psi : \mathcal{D}_{\tilde{r},\tilde{s}} \to \mathbb{C}^{2n}$ holomorphic, real at real points, $G = \Psi - \mathrm{id}$, with $|G_\theta| \le d_\theta$, $|G_I| \le d_I$ on $\mathcal{D}_{\tilde{r},\tilde{s}}$. Suppose there are $0 < \varrho' \le \tilde{r}$, $0 < \sigma' \le 2\tilde{s}$ with $\kappa_* := \max(d_\theta/\sigma', d_I/\varrho') \le \frac{1}{4}$. Then for every $y = (\theta_y, I_y) \in \mathbb{R}^n \times D$ there is $x = (\theta_x, I_x) \in W_{\tilde{r}}$ with $\Psi(x) = y$, $|\theta_x - \theta_y| \le d_\theta$, $|I_x - I_y| \le d_I$.*
Source: Ours. Pöschel p. 196 and p. 200 assert ("Ψ^{-1} takes any given initial position in $D \times \mathbb{T}^n$ to some initial position in $U_\rho D \times \mathbb{T}^n$") without proof; a left inverse (Proposition 6.13(e)) does not give this.

Proof. Let $\nu(a,b) = \max(|a|/\sigma', |b|/\varrho')$ on $\mathbb{C}^{2n}$ (the scaled norm of Definition 6.2), so $\nu(G) \le \kappa_*$.

Step S8.10: A closed convex set. $X = \{x \in \mathbb{R}^{2n} : \nu(x - y) \le \kappa_*\}$ is closed, convex, nonempty. For $x \in X$: $|I_x - I_y| \le \varrho'/4 < \tilde{r}$ and $I_y \in D$, so $x \in W_{\tilde{r}}$.

Step S8.11: Self-map. $T(x) = y - G(x)$ is real (Ψ real at real points) and $\nu(T(x) - y) = \nu(G(x)) \le \kappa_*$: $T : X \to X$.

Step S8.12: Contraction. For $x \in X$ the complex ν-ball $\{\nu(z - x) < \frac{1}{2}\}$ lies in $\mathcal{D}_{\tilde{r},\tilde{s}}$: $|\mathrm{Im}\,\theta'| \le |\theta' - \theta_x| < \sigma'/2 \le \tilde{s}$ and $|I' - I_y| < \varrho'/2 + \varrho'/4 < \tilde{r}$. On it G is holomorphic with $\nu(G) \le \kappa_*$, so Lemma 5.8 ($R = \frac{1}{2}$) gives $|DG(x)|_\nu \le 2\kappa_* \le \frac{1}{2}$. The real derivative of $G|_{\mathbb{R}^{2n}}$ is the restriction ((E3)); [K:`bp:p1:ext-mvt`] on the convex set X gives $\nu(T(x) - T(x')) \le \frac{1}{2}\nu(x - x')$.

Step S8.13: Fixed point. (E1) gives $x = y - G(x)$, i.e. $\Psi(x) = x + G(x) = y$, and $y - x = G(x)$ gives the displacement bounds. □

Corollary 8.7 (Preimages for the Normal Form Lemma)**.** *In Theorem 7.9 with $\Lambda = -\Lambda$, h, f real on real points, $\kappa \ge 6$: every $y \in \mathbb{R}^n \times D$ is $\Psi(x)$ for some $x \in W_{r/2}$ with $|I_x - I_y| \le qK\varepsilon/\alpha$ ($\le r/2^7$).*

Proof. **Step S8.14: Apply Lemma 8.6.** Ψ is real at real points (Theorem 7.9(d)); $\tilde{r} = r/2$, $\tilde{s} = s/6$, $d_I = qK\varepsilon/\alpha \le r/2^7$ (hypothesis (i)), $d_\theta = 10q\varepsilon/(2\pi\alpha r)$ (item (c)). Take $\varrho' = r/2$, $\sigma' = s/3 = 2\tilde{s}$: $d_I/\varrho' \le \frac{1}{64}$ and, by (i), $d_\theta/\sigma' = \frac{30q\varepsilon}{2\pi\alpha rs} \le \frac{30}{2\pi 2^7 Ks} = \frac{30}{2^7\kappa} \le \frac{30}{768} < \frac{1}{4}$. □

Lemma 8.8 (Transfer of an a priori confinement through a symplectic map)**.** *Let $D \subset \mathbb{R}^n$, $H \in \mathcal{O}(r,s)^{\mathbb{R}}$, $\tilde{H} \in \mathcal{O}(\tilde{r},\tilde{s})^{\mathbb{R}}$, $\Psi : \mathcal{D}_{\tilde{r},\tilde{s}} \to \mathcal{D}_{r,s}$ as in Lemma 8.5, and $|\cdot|_*$ a norm on $\mathbb{R}^n$ with $|\cdot| \le |\cdot|_*$. Assume*

- *(T1) $|\Pi_I \Psi(x) - I_x|_* \le d_*$ for all $x \in W_{\tilde{r}}$;*
- *(T2) every $y \in \mathbb{R}^n \times D$ equals $\Psi(x)$ for some $x \in W_{\tilde{r}}$;*
- *(T3) (*a priori confinement in the new variables*) there are $\varrho > 0$ with $d_* + \varrho < \tilde{r}$ and $T : D \to (0,\infty]$ such that for every $I_y \in D$, every interval $\tilde{J} \ni 0$ and every $\tilde{H}$-orbit $\tilde{z}$ in $W_{\tilde{r}}$ on $\tilde{J}$ with $|\tilde{I}(0) - I_y|_* \le d_*$: $|\tilde{I}(t) - \tilde{I}(0)|_* \le \varrho$ for all $t \in \tilde{J}$, $|t| \le T(I_y)$.*

Then for every interval $J \ni 0$ and every H-orbit z in W_r on J with $I_y := I(0) \in D$:

$$|I(t) - I(0)|_* \le \varrho + 2d_* \qquad (t \in J,\ |t| \le T(I_y)).$$

Source: Ours; this is the transfer step of Pöschel pp. 196, 200 ("Going back to original coordinates"), in stopped form so that Part 10 can connect it with existence (Theorem 4.34).

Proof. **Step S8.15: A normal-form orbit.** By (T2) choose $x \in W_{\tilde r}$ with $\Psi(x) = z(0)$; by (T1) $|I_x - I_y|_* \le d_*$. Let $\tilde z$ with maximal interval (T_-, T_+) be the maximal solution of $X_{\tilde H^{\mathbb R}}$ in $W_{\tilde r}$ through x (Lemma 8.2(d), Theorem 4.13).

Step S8.16: It lives long enough. (T3) applies on (T_-, T_+). The set $K = \{I : |I - I_x|_* \le \varrho\}$ is compact and, since $|I - I_y| \le |I - I_y|_* \le \varrho + d_* < \tilde r$ and $I_y \in D$, contained in $U_{\tilde r}D$. If some $t \ge 0$ with $t \le T(I_y)$ had $t \ge T_+$, then $T_+ < \infty$ and $\tilde I(t') \in K$ for all $t' \in [0, T_+)$, so $T_+ = \infty$ by Corollary 4.16 (the field is periodic, Lemma 8.2(c)): a contradiction. Likewise for $t \le 0$. Thus $\{|t| \le T(I_y)\} \subset (T_-, T_+)$.

Step S8.17: Identification. $\Psi \circ \tilde z$ is an H-orbit in W_r on (T_-, T_+) (Lemma 8.5) with value y at 0, as is z on J; Theorem 4.11 (with $F = X_{H^{\mathbb R}}$, $W = W_r$) gives $z = \Psi \circ \tilde z$ on $J \cap (T_-, T_+)$.

Step S8.18: Estimate. For $t \in J$, $|t| \le T(I_y)$: $I(t) - I(0) = (\Pi_I \Psi(\tilde z(t)) - \tilde I(t)) + (\tilde I(t) - \tilde I(0)) + (\tilde I(0) - I_y)$, of size $\le d_* + \varrho + d_*$ by (T1), (T3) and $\tilde I(0) = I_x$. □

Corollary 8.9 (Stopped form for any extension of the field)**.** *Let $H \in \mathcal O(r,s)^{\mathbb R}$, $|\cdot|_* \ge |\cdot|$ a norm, $0 < \varrho < r$, $T \in (0, \infty]$ and suppose that every H-orbit in W_r on every interval $J \ni 0$ with $I(0) \in D$ satisfies $|I(t) - I(0)|_* \le \varrho$ for $t \in J$, $|t| \le T$. Let $F : \mathbb R^{2n} \to \mathbb R^{2n}$ be any function with $F = X_{H^{\mathbb R}}$ on W_r and $z : \mathbb R \to \mathbb R^{2n}$, $J \ni 0$ an interval with $z'(t) = F(z(t))$ for $t \in J$ and $z(0) \in \mathbb R^n \times D$. Then $|I(t) - I(0)|_* \le \varrho$ for $t \in J$, $|t| \le T$; in particular $z(t) \in W_r$ there.*

Proof. **Step S8.19: The stopped interval.** $\{t \in J : z(t) \in W_r\}$ is relatively open in J and contains 0; let J' be its connected component through 0. z is an H-orbit in W_r on J', so the bound holds on $J' \cap [-T, T]$.

Step S8.20: No exit. If $t_e \in J \cap [-T, T]$ were an endpoint of J' not in J', continuity would give $|I(t_e) - I(0)|_* \le \varrho$, hence $|I(t_e) - I(0)| < r$ and $I(t_e) \in U_r D$ ($I(0) \in D$), so t_e would belong to the component: contradiction. □

Lemma 8.10 (First exit from a norm ball)**.** *Let $|\cdot|_*$ be a norm on $\mathbb R^n$, $J \ni 0$ an interval, $z : J \to \mathbb R^{2n}$ continuous, $I_0 = I(0)$, $\varrho > 0$. If $|I(t_1) - I_0|_* > \varrho$ for some $t_1 \in J$, then there is $t^* \in J$ between 0 and t_1 with $|I(t^*) - I_0|_* = \varrho$ and $|I(\tau) - I_0|_* \le \varrho$ for all τ between 0 and t^*.*

Proof. **Step S8.21: Exit set.** Apply Lemma 4.19 to $V = \mathbb R^n \times \{|I - I_0|_* < \varrho\}$ (open; $z(0) \in V$), forward if $t_1 > 0$, and to $t \mapsto z(-t)$ if $t_1 < 0$ (only continuity is used): t^* exists, $z(\tau) \in V$ before it and $z(t^*) \in \overline V \setminus V$, i.e. $|I(t^*) - I_0|_* \le \varrho$ and $\not< \varrho$. □

8.4. Proposition 1: nonresonant stability.

Proposition 8.11 (Nonresonant stability estimate)**.** *Let $n \ge 1$, $D \subset \mathbb R^n$, $r, s, K, \alpha > 0$, $L \ge 0$, $p > 1$, $q = p/(p-1)$, $\kappa = 2\pi K s$. Let h be holomorphic on $V_r D$, real on $U_r D$, $f \in \mathcal O(r,s)^{\mathbb R}$ with $\varepsilon := |f|_{r,s} < \infty$, and $H = h + f$. Assume*

*(i) (*completely α, K-nonresonant*) $|k \cdot \omega(I)| \ge \alpha$ for $I \in D$ and $k \in \mathbb Z^n$, $0 < |k|_1 \le K$;*
(ii) the Lipschitz bound (N2) for ω with constant L, and $KLr \le \alpha/p$;
(iii) $\varepsilon \le \dfrac{\alpha r}{2^7 q K}$.

Put $T_{\rm nr} = \dfrac{s\,r}{5\varepsilon}e^{\kappa/6}$ *(*$= +\infty$ *if* $\varepsilon = 0$*). Then for every interval* $J \ni 0$ *and every* H*-orbit* z *in* W_r *on* J *with* $I(0) \in D$*:*

$$|I(t) - I(0)| \le \tfrac{r}{2} \ \ (\le r) \qquad \text{for all } t \in J,\ |t| \le T_{\rm nr}.$$

No convexity is used. The constants depend only on (r, s, K, α, p) *and* ε*.*
Source: Pos93 p. 196 (Proposition 1: conclusion $\|I(t) - I_0\| \le r$ *for* $|t| \le \frac{sr}{5\varepsilon}e^{Ks/6}$*; hypotheses* $\varepsilon \le \frac{1}{2^7 q}\frac{\alpha r}{K}$*,* $r \le \min(\alpha/(pMK), r_0)$*). Deviations: the conclusion is* $r/2$ *(the proof gives it); the case* $\kappa < 6$ *(his one line p. 197) is done with* $\Psi = \mathrm{id}$ *and no loss of width, so the factor* e *of Proposition 7.11 is absorbed (Remark 8.12); the transfer to the original coordinates includes the surjectivity of* Ψ*, which the source does not prove.*

Proof. Let $\tilde H = h + g + f_*$ be the normal form below.

Step S8.22: Case A, $\kappa \ge 6$**: data.** Theorem 7.9 with $\Lambda = \{0\}$ ($\Lambda \cap Z_K = \{0\}$) applies: (N1) is (i), (N2) and $KLr \le \alpha/p$ are (ii), $\varepsilon \le \alpha r/(2^7 qK)$ is (iii). It gives $\Psi : \mathcal{D}_{r/2,s/6} \to \mathcal{D}_{r,s}$ holomorphic, symplectic, real at real points, $H \circ \Psi = h + g + f_*$ with g independent of θ, $f_* \in \mathcal{O}(r/2, s/6)^{\mathbb{R}}$, $\eta := |f_*|_{r/2,s/6} \le e^{-\kappa/6}\varepsilon$, and $|\Pi_I \Psi - I| \le d := qK\varepsilon/\alpha \le r/2^7$. Put $\tilde r = r/2$, $\tilde s = s/6$, $\varrho = \frac{15}{32}r$, $d_* = d$, $|\cdot|_* = |\cdot|$, $T = T_{\rm nr}$. (T1) is the displacement bound, (T2) is Corollary 8.7; $d_* + \varrho \le \frac{61}{128}r < \tilde r$.

Step S8.23: Case A: (T3). Let $\tilde z$ be an $\tilde H$-orbit in $W_{\tilde r}$ on $\tilde J$, $|\tilde I(0) - I_y| \le d$, $I_y \in D$, and suppose $|\tilde I(t_1) - \tilde I(0)| > \varrho$ for some $t_1 \in \tilde J$, $|t_1| \le T_{\rm nr}$. Lemma 8.10 gives t^*, $|t^*| \le |t_1|$, $|\tilde I(t^*) - \tilde I(0)| = \varrho$, with $\tilde I(\tau)$ in the closed ball B of radius ϱ about $\tilde I(0)$ for τ between 0 and t^*; $B \subset U_{\tilde r}D$ as $\varrho + d < \tilde r$. On this stretch $\dot{\tilde I} = -\partial_\theta \tilde H^{\mathbb{R}} = -\partial_\theta f_*(\tilde z)$ (h and g do not depend on θ; Lemma 8.2(b)), so by Lemma 8.3 $|\dot{\tilde I}| \le \eta/(e\tilde s) = 6\eta/(es)$ and, by the mean value inequality ([K:bp:p1:ext-mvt]),

$$\varrho = |\tilde I(t^*) - \tilde I(0)| \le \frac{6\eta|t^*|}{es} \le \frac{6e^{-\kappa/6}\varepsilon}{es} \cdot \frac{sr\,e^{\kappa/6}}{5\varepsilon} = \frac{6}{5e}r < 0.442\,r < \tfrac{15}{32}r = \varrho,$$

a contradiction ((E3); if $\varepsilon = 0$ then $\eta = 0$ and the left side is 0). So (T3) holds with $T(I_y) = T_{\rm nr}$.

Step S8.24: Case A: conclusion. Lemma 8.8: $|I(t) - I(0)| \le \frac{15}{32}r + \frac{2}{2^7}r = \frac{31}{64}r \le \frac{r}{2}$.

Step S8.25: Case B, $\kappa < 6$**.** Take $\Psi = \mathrm{id}$ on $\mathcal{D}_{r,s}$, $\tilde H = H$, $g = 0$, $f_* = f$ (Proposition 7.11), $\tilde r = r$, $\tilde s = s$, $d_* = 0$, $\varrho = r/2$: (T1), (T2) are trivial. For (T3): as in Step S8.23, with $\tilde I(\tau)$ in the ball of radius $r/2$ about $\tilde I(0) \in D$ ($\subset U_r D$), $|\dot{\tilde I}| \le \varepsilon/(es)$ and $|\tilde I(t^*) - \tilde I(0)| \le \frac{\varepsilon}{es} \cdot \frac{sr\,e^{\kappa/6}}{5\varepsilon} < \frac{r}{5} < \frac{r}{2}$ (as $e^{\kappa/6} < e$): contradiction. Lemma 8.8 gives $|I(t) - I(0)| \le r/2$. □

Remark 8.12 (The factor e in the nonresonant case). Proposition 7.11 gives only $|f_*|_{r,s} \le e \cdot e^{-\kappa/6}\varepsilon$ for $\kappa < 6$. Used with the widths $(r/2, s/6)$ of the case $\kappa \ge 6$ it would give the coefficient $31/384 < \frac{1}{5}$ in the time bound. Case B avoids this by not losing width ($\tilde s = s$, $\eta = \varepsilon$): the available time is $\frac{es\,r}{2\varepsilon} > \frac{sr}{5\varepsilon}e^{\kappa/6}$ for all $\kappa < 6$, so the stated $T_{\rm nr}$ is attained with room to spare, and no factor e is lost in the final statement.

8.5. **The energy argument.**

Definition 8.13 (Resonance data, convexity, quasi-convexity)**.** Let $D \subset \mathbb{R}^n$, $\Lambda \subset \mathbb{Z}^n$, $\Pi = \Pi_\Lambda$ (Lemma 8.4), $R_\Lambda = \{\omega \in \mathbb{R}^n : k \cdot \omega = 0\ \forall k \in \Lambda\} = \ker \Pi$. D *is* δ*-close to exact* Λ*-resonances* if $\|\Pi\omega(I)\|_2 = \mathrm{dist}_2(\omega(I), R_\Lambda) \le \delta$ for all $I \in D$. For h as in Definition 8.1 and $U \subset U_r D$: h *is* m*-convex on* U if $Q(I)[\xi,\xi] \ge m\|\xi\|_2^2$ for $I \in U$, $\xi \in \mathbb{R}^n$; ℓ, m*-quasi-convex on* U if for all $I \in U$ and $\xi \in \mathbb{R}^n$: $|\langle \omega(I), \xi\rangle| > \ell\|\xi\|_2$ or $Q(I)[\xi,\xi] \ge m\|\xi\|_2^2$ (equivalently $|\langle \omega, \xi\rangle| \le \ell\|\xi\|_2 \Rightarrow Q[\xi,\xi] \ge m\|\xi\|_2^2$, Lemma 2.7(b); never full positive definiteness, review N05). $M \ge m$ denotes a bound $\|Q(I)\|_{2\to 2} \le M$ on $U_r D$ (the normalisation $m \le M$ is always achievable by enlarging M).
Source: Pos93 pp. 190, 197.

Lemma 8.14 (Energy–convexity inequality (reusable core)). *Let $U \subset \mathbb{R}^n$ be open, $h_{\mathbb{R}} \in C^2(U)$, $g, f \in C^1(\mathbb{R}^n \times U; \mathbb{R})$, $H^{\mathbb{R}} = h_{\mathbb{R}}(I) + g + f$ on $\mathbb{R}^n \times U$ (any values elsewhere), Π an orthogonal projection of $(\mathbb{R}^n, \|\cdot\|_2)$ with $\Pi^{\perp}\partial_\theta g \equiv 0$. Let $J \ni 0$ be an interval, z with $\texttt{IsSolIn}\, X_{H^{\mathbb{R}}}(\mathbb{R}^n \times U)\, J\, z$, $t_1 \in J$, $I_0 = I(0)$, $\Delta I = I(t_1) - I_0$, $I_s = I_0 + s\Delta I$, $\omega_0 = \nabla h_{\mathbb{R}}(I_0)$, and assume $I_s \in U$ for $s \in [0,1]$. Then*

(a) (energy) $h_{\mathbb{R}}(I(t_1)) - h_{\mathbb{R}}(I_0) = -\big[(g+f)(z(t_1)) - (g+f)(z(0))\big]$;
(b) (resonant directions) $\|\Pi^{\perp}\Delta I\|_2 \le |t_1| \sup_\tau \|\partial_\theta f(z(\tau))\|_2$, τ between 0 and t_1;
(c) (Taylor) $h_{\mathbb{R}}(I(t_1)) - h_{\mathbb{R}}(I_0) - \langle \omega_0, \Delta I\rangle = \frac{1}{2} Q(I_\xi)[\Delta I, \Delta I]$ for some $\xi \in (0,1)$;
(d) if $Q(I_s)[\Delta I, \Delta I] \ge m\|\Delta I\|_2^2$ for all $s \in [0,1]$, then

$$\frac{m}{2}\|\Delta I\|_2^2 \le \big|(g+f)(z(t_1)) - (g+f)(z(0))\big| + \|\Pi\omega_0\|_2\|\Delta I\|_2 + \|\omega_0\|_2\|\Pi^{\perp}\Delta I\|_2.$$

Source: Pos93 p. 198 ("Following Lochak … expanding around I_0", $\Delta h = \langle \omega_0, \Delta I\rangle + \int_0^1 (1-s)\langle Q(I(s))\Delta I, \Delta I\rangle ds$, energy conservation, the vector $Q\Delta I$ …), with the Lagrange remainder instead of the integral one. This is the part wanted by track I (energy confinement).

Proof. **Step S8.26: Energy.** $H^{\mathbb{R}}$ is C^1 on the open set $\mathbb{R}^n \times U$, hence differentiable along the orbit, and Theorem 4.23 on the interval between 0 and t_1 gives $H^{\mathbb{R}}(z(t_1)) = H^{\mathbb{R}}(z(0))$; this is (a).

Step S8.27: Resonant projection. $\dot I = -\partial_\theta H^{\mathbb{R}}(z) = -\partial_\theta g - \partial_\theta f$ ($h_{\mathbb{R}}$ is θ-independent). Hence $t \mapsto \Pi^{\perp} I(t)$ has derivative $-\Pi^{\perp}\partial_\theta f(z(t))$, of norm $\le \|\partial_\theta f(z(t))\|_2$; the mean value inequality ([K:b p:p1:ext-mvt], $\Pi^{\perp}$ linear) gives (b).

Step S8.28: Taylor and combination. (c) is (E2) for $\varphi(s) = h_{\mathbb{R}}(I_s)$ ($\varphi'(0) = \langle \omega_0, \Delta I\rangle$, $\varphi'' = Q(I_s)[\Delta I, \Delta I]$). Under the hypothesis of (d), (c) gives $\frac{m}{2}\|\Delta I\|^2 \le h_{\mathbb{R}}(I(t_1)) - h_{\mathbb{R}}(I_0) - \langle \omega_0, \Delta I\rangle$; insert (a), and $\langle \omega_0, \Delta I\rangle = \langle \Pi\omega_0, \Delta I\rangle + \langle \omega_0, \Pi^{\perp}\Delta I\rangle$ ((E4)) with Cauchy–Schwarz. □

The resonant setting (RS).. $n \ge 1$, $D \subset \mathbb{R}^n$, $r, s > 0$, $\Lambda \subset \mathbb{Z}^n$, h holomorphic on $V_r D$ and real on $U_r D$, $0 < m \le M$ with $\|Q(I)\|_{2\to 2} \le M$ on $U_r D$, $g, f_* \in \mathcal{O}(r,s)^{\mathbb{R}}$ of finite norm with $P_\Lambda g = g$, $\zeta \ge |g|_{r,s} + |f_*|_{r,s}$, $\eta = |f_*|_{r,s}$, $H = h + g + f_*$, $\delta > 0$ with D δ-close to exact Λ-resonances, $\delta \le mr/4$, $\zeta \le mr^2/32$, and $\rho := \delta/(4M)$ $\big(\le r/16\big)$. An initial action I_0 is *admissible* if $\|I_0 - J_0\|_2 < \rho$ for some $J_0 \in D$.

Lemma 8.15 (Admissible initial actions). *In (RS), for admissible I_0 (with J_0): (a) the closed ball $\bar B = \{I : \|I - I_0\|_2 \le r - \rho\} \subset U_r D$; (b) $\|\Pi\omega(I)\|_2 \le \frac{5}{4}\delta + M\|I - I_0\|_2$ for $I \in \bar B$; in particular $\|\Pi\omega(I_0)\|_2 \le \frac{5}{4}\delta$.*

Proof. **Step S8.29: Ball.** $|I - J_0| \le \|I - J_0\|_2 < (r-\rho) + \rho = r$, so $I \in U_r D$; likewise the segment $[J_0, I_0] \subset U_r D$.

Step S8.30: Closeness. By the mean value inequality for ω in $\|\cdot\|_2$ ($D\omega = Q$, $\|Q\| \le M$, convex segments inside $U_r D$): $\|\omega(I_0) - \omega(J_0)\|_2 \le M\rho = \delta/4$, so $\|\Pi\omega(I_0)\| \le \delta + \delta/4$; and $\|\Pi\omega(I)\| \le \|\Pi\omega(I_0)\| + M\|I - I_0\|_2$. □

Lemma 8.16 (No exit under convexity along the segment). *In (RS) let z be an H-orbit in W_r on $J \ni 0$, $I_0 = I(0)$ admissible, and $t_1 \in J \setminus \{0\}$ with*

(i) $\|I(\tau) - I_0\|_2 \le r - \rho$ for τ between 0 and t_1, and $\|I(t_1) - I_0\|_2 = r - \rho$;
(ii) $Q(I_s)[\Delta I, \Delta I] \ge m\|\Delta I\|_2^2$ for all $s \in [0,1]$ ($\Delta I = I(t_1) - I_0$, $I_s = I_0 + s\Delta I$);
(iii) $|t_1|\, \|\omega(I_0)\|_2\, \eta \le s\, m\, r^2/12$.

Then a contradiction arises (so these three cannot hold together).

Source: Pos93 p. 198 (end of the proof of Lemma 1): $\frac{m}{3}\|\Delta I\|^2 \le 2\zeta + \frac{5}{2m}\delta^2 + T\|\omega_0\|\frac{1}{es}|f_*| \le \frac{mr^2}{4} < \frac{m}{3}(r-\rho)^2$; all numbers recomputed.

Proof. Put $\Delta = \|\Delta I\|_2 = r - \rho \geq \frac{15}{16} r$.

Step S8.31: Apply the core. $U = U_r D$, $\bar{B} \subset U$ (Lemma 8.15) is convex so $I_s \in U$; g, f_* are C^∞ on W_r (Lemma 8.2), $\Pi^\perp \partial_\theta g = 0$ (Lemma 8.4), the orbit is a solution of $X_{H^{\mathbb{R}}}$ in $\mathbb{R}^n \times U$. Lemma 8.14(d) with (ii): $\frac{m}{2}\Delta^2 \leq A + B + C$, $A = |\Delta(g + f_*)|$, $B = \|\Pi\omega_0\|\Delta$, $C = \|\omega_0\|\|\Pi^\perp \Delta I\|$.

Step S8.32: Three terms. $A \leq 2(|g|_{r,s} + |f_*|_{r,s}) \leq 2\zeta \leq \frac{mr^2}{16}$ (Lemma 8.3, last claim). $B \leq \frac{5}{4}\delta\Delta \leq \frac{5}{2m}\delta^2 + \frac{m}{6}\Delta^2$ (Lemma 8.15, Young), and $\frac{5}{2m}\delta^2 \leq \frac{5mr^2}{32}$. By Lemma 8.14(b), (i) and Lemma 8.3, $C \leq \|\omega_0\||t_1|\frac{\eta}{es} \leq \frac{mr^2}{12e} < 0.0307\, mr^2$ by (iii).

Step S8.33: Contradiction. Hence $\frac{m}{3}\Delta^2 \leq mr^2(\frac{1}{16} + \frac{5}{32} + \frac{1}{12e}) < \frac{mr^2}{4}$, while $\frac{m}{3}\Delta^2 \geq \frac{m}{3} \cdot \frac{225}{256} r^2 > 0.29\, mr^2$ ((E3)). □

8.6. **Lemma 1 and Lemma 2.**

Lemma 8.17 (Lemma 1: stability for m-convex h in normal form)**.** *In (RS) let h be m-convex on $U_r D$. Then for every interval $J \ni 0$ and every H-orbit z in W_r on J with $I_0 = I(0)$ admissible:*

$$\|I(t) - I_0\|_2 \leq r - \rho \qquad (t \in J,\ |t| \leq T_*), \qquad T_* = \frac{s\, m\, r^2}{12\, \|\omega(I_0)\|_2\, \eta},$$

with $T_* = +\infty$ if $\|\omega(I_0)\|_2 \eta = 0$. (In particular for $\Lambda = \mathbb{Z}^n$, $f_* = 0$: all $t \in J$.)
Source: Pos93 p. 197 (Lemma 1: $\delta \leq mr/4$, $\zeta \leq mr^2/32$, $\rho = \delta/4M$, $T_* = \frac{s}{12\|\omega(I_0)\|}\frac{mr^2}{\eta}$). Statement in stopped form ("as long as the action stays in W_r": the orbit is an orbit in W_r on J); with exit analysis by Lemma 8.10.

Proof. **Step S8.34: Exit.** Suppose $\|I(t_1) - I_0\|_2 > r - \rho$ for some $t_1 \in J$, $|t_1| \leq T_*$. Lemma 8.10 ($|\cdot|_* = \|\cdot\|_2$) gives $t^* \in J$, $|t^*| \leq |t_1|$, with (i) of Lemma 8.16 for t^*.

Step S8.35: Hypotheses (ii), (iii). $I_s \in \bar{B} \subset U_r D$ (Lemma 8.15), so (ii) is m-convexity; (iii) is $|t^*| \leq T_*$. Lemma 8.16 gives the contradiction. □

Lemma 8.18 (Lemma 2: stability for ℓ, m-quasi-convex h in normal form)**.** *In (RS) let h be ℓ, m-quasi-convex on $U_r D$ and assume*

$$\text{(R)} \qquad r \leq 2\ell/m \qquad \text{and} \qquad Mr \leq \ell/4$$

(the second is an added *hypothesis, Remark 8.20(1)). For admissible I_0 let $\Omega = \max\{\|\omega(I)\|_2 : \|I - I_0\|_2 \leq r - \rho\}$. Then for every H-orbit z in W_r on $J \ni 0$ with $I(0) = I_0$ admissible,*

$$\|I(t) - I_0\|_2 \leq r - \rho \quad (t \in J,\ |t| \leq T_*), \qquad T_* = \frac{s\, m\, r^2}{12\, \Omega\, \eta}\ (= +\infty \text{ if } \Omega\eta = 0).$$

Source: Pos93 p. 199 (Lemma 2: $r \leq 2\ell/m$, $\Omega = \sup_{\|I - I_0\| \leq r - \rho} \|\omega(I)\|$, proof by contradiction at the first exit time). The hypothesis $Mr \leq \ell/4$ repairs a gap of the proof (Remark 8.20(1)).

Proof. **Step S8.36: Exit.** As in Lemma 8.17, a violation yields t^*, $|t^*| \leq T_*$ with (i) of Lemma 8.16. Put ΔI, I_s as there, $\Delta = r - \rho \geq \frac{15}{16} r$; $I_s \in \bar{B}$ and $\|\omega(I_s)\| \leq \Omega$.

Step S8.37: (iii). $\|\omega(I_0)\| \leq \Omega$, so $|t^*|\|\omega(I_0)\|\eta \leq |t^*|\Omega\eta \leq smr^2/12$.

Step S8.38: The cone condition along the segment. Fix $s' \in [0, 1]$. Then $|\langle \omega(I_{s'}), \Delta I \rangle| \leq \|\Pi\omega(I_{s'})\|\Delta + \|\omega(I_{s'})\|\|\Pi^\perp \Delta I\|$. By Lemma 8.15(b) ($\|I_{s'} - I_0\| \leq \Delta \leq r$) the first term is $\leq (\frac{5}{4}\delta + Mr)\Delta \leq (\frac{5}{16}mr + Mr)\Delta$. By Lemma 8.14(b), Lemma 8.3 and Step S8.37 the second is $\leq \Omega|t^*|\frac{\eta}{es} \leq \frac{mr^2}{12e} \leq \frac{16}{15} \cdot \frac{mr}{12e}\Delta < 0.0328\, mr\, \Delta$. With $mr \leq 2\ell$ and $Mr \leq \ell/4$:

$$|\langle \omega(I_{s'}), \Delta I \rangle| \leq \big(0.3453 \cdot 2\ell + \tfrac{\ell}{4}\big)\Delta < 0.941\, \ell\, \|\Delta I\|_2.$$

Step S8.39: Convexity along the segment. Since $|\langle\omega(I_{s'}),\Delta I\rangle|\le\ell\|\Delta I\|_2$, the first alternative of quasi-convexity at $I_{s'}\in U_rD$ fails for $\xi=\Delta I$, so $Q(I_{s'})[\Delta I,\Delta I]\ge m\|\Delta I\|^2$: (ii) of Lemma 8.16 holds. That lemma gives the contradiction. □

8.7. **Proposition 2: the Resonant Stability Lemma.**

Proposition 8.19 (Resonant Stability Lemma)**.** *Let $n\ge1$, $D\subset\mathbb{R}^n$, $r,s,K,\alpha>0$, $1<p\le\frac32$ and $q=p/(p-1)$ (≥3), $\kappa=2\pi Ks$. Let $\Lambda\subset\mathbb{Z}^n$ be a subgroup, $0<m\le M$, $\ell,\delta>0$, h holomorphic on V_rD and real on U_rD with*

$$\|\partial_I^2h(I)\|_{2\to2}\le M\quad\text{for all } I\in V_rD\ (\text{complex Hessian, operator norm on } (\mathbb{C}^n,\|\cdot\|_2)),$$

h ℓ,m-quasi-convex on U_rD, $f\in\mathcal{O}(r,s)^{\mathbb{R}}$ with $\varepsilon=|f|_{r,s}<\infty$, $H=h+f$. Assume:

(a) D is α,K-nonresonant modulo Λ: $|k\cdot\omega(I)|\ge\alpha$ for $I\in D$, $k\in Z_K\setminus\Lambda$, and δ-close to exact Λ-resonances;

(b) $\varepsilon\le\dfrac{p}{q}\cdot\dfrac{mr^2}{2^7}$ and $\delta\le\dfrac{mr}{8}$;

(c) $r\le\min\Big(\dfrac{\alpha}{p\sqrt{n}\,MK},\ \dfrac{4\ell}{m}\Big)$ and (added) $Mr\le\ell/2$.

For $I_0\in D$ let $\Omega(I_0)=\max\{\|\omega(I)\|_2:\|I-I_0\|_2\le r/2\}$ and

$$T(I_0)=\frac{s\,m\,r^2}{288\,\Omega(I_0)\,\varepsilon}\,e^{\kappa/6}\qquad(=+\infty\ \text{if}\ \Omega\varepsilon=0).$$

Then for every interval $J\ni0$ and every H-orbit z in W_r on J with $I_0=I(0)\in D$,

$$\|I(t)-I_0\|_2\le\tfrac{r}{2}\ (\text{so } |I(t)-I_0|\le r)\qquad(t\in J,\ |t|\le T(I_0)),$$

and if $\Lambda=\mathbb{Z}^n$ this holds for all $t\in J$.

Source: Pos93 pp. 199–200 (Proposition 2: $\varepsilon\le\frac{p}{q}\frac{mr^2}{2^7}$, $\delta\le\frac{mr}{8}$, $r\le\min(\frac{\alpha}{pMK},\frac{4\ell}{m},r_0)$, $q\ge3$; $\|I(t)-I_0\|\le r$ for $|t|\le\frac{s}{288\Omega}\frac{mr^2}{\varepsilon}e^{Ks/6}$; $\Lambda=\mathbb{Z}^n$: all t). Deviations: $\sqrt{n}$ in the first bound of (c) (Euclidean Hessian bound M against the max-norm Lipschitz bound $L=\sqrt{n}M$ of Part 7); added hypothesis $Mr\le\ell/2$ and the radius $r/2$ for $\kappa<6$ (Remark 8.20(1),(2)); Ω taken over the radius $r/2$ (the proof confines to $r/2$).

Proof. **Step S8.40: Reductions.** Replace δ by $\delta'=mr/8\ge\delta$ (D is δ'-close). Then $\rho':=\delta'/(4M)=\frac{mr}{32M}$. Since $\frac{p}{q}=\frac{1}{q-1}\le\frac12$, (b) gives $\varepsilon\le mr^2/2^8$. By (c) and $m\le M$: $L:=\sqrt{n}M$ satisfies $KLr\le\alpha/p$, and $\varepsilon\le\frac{pmr^2}{q2^7}\le\frac{pMr^2}{q2^7}\le\frac{\alpha r}{2^7qK}$. The complex Hessian bound gives (N2) with $L=\sqrt{n}M$ (Lemma 7.2: $|Qv|_\infty\le\|Qv\|_2\le M\|v\|_2\le\sqrt{n}M|v|_\infty$). Put $\tilde r=r/2$; $M\tilde r\le\ell/4$, $\tilde r\le2\ell/m$, $\delta'=m\tilde r/4$.

Step S8.41: Three cases. (A) $\Lambda\ne\mathbb{Z}^n$, $\kappa\ge6$: Theorem 7.9 (conditions (N1), (N2), $K>0$, (i), (ii) hold; $\Lambda=-\Lambda$, reality) gives $\Psi:\mathcal{D}_{r/2,s/6}\to\mathcal{D}_{r,s}$ with $H\circ\Psi=h+g+f_*$, $P_\Lambda g=g$, $\tilde s=s/6$, $\eta\le e^{-\kappa/6}\varepsilon$ and $|g|_{r,s}$-type bound: $|g|_{r/2,s/6}\le|g_0|_{r,s}+16c\varepsilon^2$, where $16c\varepsilon^2=\frac{32qK\varepsilon^2}{\alpha r}\le\frac{\varepsilon}{4}$ by (i), so $|g|_{r/2,s/6}+\eta\le\frac54\varepsilon+\frac{\varepsilon}{e}<2\varepsilon=:\zeta$. (B) $\Lambda\ne\mathbb{Z}^n$, $\kappa<6$: $\Psi=\mathrm{id}$ on $\mathcal{D}_{r/2,s}$, $g=0$, $f_*=f$, $\tilde s=s$, $\eta=\varepsilon$, $\zeta=\varepsilon$. (C) $\Lambda=\mathbb{Z}^n$: $\Psi=\mathrm{id}$, $\tilde s=s$, $g=f$ ($P_\Lambda g=g$), $f_*=0$, $\eta=0$, $\zeta=\varepsilon$. In all cases $\tilde H=h+g+f_*$ on $\mathcal{D}_{\tilde r,\tilde s}$ is in (RS) with $(r,s)\to(\tilde r,\tilde s)$ and δ': $\delta'=m\tilde r/4$, and $\zeta\le2\varepsilon\le mr^2/128=m\tilde r^2/32$.

Step S8.42: (T1), (T2). Case A: $\|\Pi_I\Psi-I\|_2\le\sqrt{n}\frac{qK\varepsilon}{\alpha}\le\frac{q\varepsilon}{pMr}\le\frac{mr}{2^7M}=\frac{\rho'}{4}=:d_*$ (Theorem 7.9(c), $\alpha\ge p\sqrt{n}MKr$, (b)); (T2) is Corollary 8.7. Cases B, C: $d_*=0$, trivial. Norm $|\cdot|_*=\|\cdot\|_2$; $\varrho:=\tilde r-\rho'$; then $d_*+\varrho<\tilde r$.

Step S8.43: (T3). Let $\tilde z$ be an $\tilde H$-orbit in $W_{\tilde r}$, $\|\tilde I(0)-I_y\|_2\le d_*<\rho'$, $I_y\in D$: $\tilde I(0)$ is admissible. Lemma 8.18 (hypotheses (R) at radius $\tilde r$) gives $\|\tilde I(t)-\tilde I(0)\|_2\le\varrho$ for $|t|\le\tilde T_*=\frac{\tilde s\,m\tilde r^2}{12\tilde\Omega\eta}$, $\tilde\Omega=\max_{\|I-\tilde I(0)\|_2\le\varrho}\|\omega\|_2$. Since $\varrho+d_*\le\tilde r-\rho'+\rho'/4$, the ball lies in $\{\|I-I_y\|_2\le r/2\}$ so $\tilde\Omega\le\Omega(I_y)$.

Case A: $\tilde{T}_* \geq \frac{(s/6)(mr^2/4)}{12\Omega\, e^{-\kappa/6}\varepsilon} = T(I_y)$. Case B: $\tilde{T}_* = \frac{smr^2}{48\Omega\varepsilon} \geq T(I_y)$ because $e^{\kappa/6} < e < 6$. Case C: $\tilde{T}_* = \infty$.

Step S8.44: Conclusion. Lemma 8.8: $\|I(t) - I_0\|_2 \leq \varrho + 2d_* \leq \tilde{r} - \rho' + \rho'/2 < \frac{r}{2}$ for $|t| \leq T(I_0)$ ($t \in J$), and all t in case C. Finally $|\cdot| \leq \|\cdot\|_2$. □

8.8. **Errata, deviations, dictionary, consumers.**

Remark 8.20 (Suspected defects in Pos93 (to the ledger)). (1) *p. 199, proof of Lemma 2*: "$|\langle \omega(I(s)), P\Delta I\rangle| \leq \frac{5}{4}\delta\|\Delta I\|$". The closeness $\frac{5}{4}\delta$ was derived (p. 198) only for $U_\rho D$, but $I(s)$ ranges over the ball of radius $r-\rho$ about I_0, where the mean value theorem gives only $\|P\omega(I(s))\| \leq \frac{5}{4}\delta + M\|I(s) - I_0\|$. Without a bound on Mr the cone condition $|\langle \omega(I(s)), \Delta I\rangle| \leq \ell\|\Delta I\|$ need not hold, so Lemma 2 (and Proposition 2, which uses it) is not proved as printed. Repair: the hypothesis $Mr \leq \ell/4$ (Lemma 8.18), resp. $Mr \leq \ell/2$ in Proposition 8.19. In Theorem 1 (p. 190) one has $r \leq r_0/(AK)$, $r_0 \leq 4\ell/m$, $A = 11M/m$, so $Mr \leq \frac{4\ell}{11K}$, and Part 9 must check $Mr \leq \ell/2$ there ($K \geq 1$ suffices). The lemma may be true without it; we do not claim so.

(2) *p. 200, Proposition 2, case $Ks < 6$*: "we simply apply those two lemmata to the given hamiltonian H with $g = 0$ and $f_* = f_\varepsilon$" at radius r, but Lemma 2 needs $r \leq 2\ell/m$ while only $r \leq 4\ell/m$ is assumed. Repaired by applying them at radius $r/2$ (case B).

(3) *p. 196, Proposition 1 and p. 200*: "Ψ^{-1} takes any given initial position in $D \times \mathbb{T}^n$ to some initial position in $U_\rho D \times \mathbb{T}^n$" needs surjectivity of Ψ onto a neighbourhood of $D \times \mathbb{T}^n$; not argued (Lemma 8.6 supplies it by a contraction argument using the displacement bounds of Theorem 7.9(c)).

(4) *p. 197, first line*: "For $Ks \leq 6 \ldots f_* = f_\varepsilon$": the bound $|f_*| \leq e^{-Ks/6}\varepsilon$ is false there (Part 7, Proposition 7.11); not used (Remark 8.12).

(5) Misprint-free checks: the numbers $\frac{m}{3}\|\Delta I\|^2 \leq \frac{mr^2}{4} < \frac{m}{3}(r-\rho)^2$ (p. 198) and T_*, $\frac{s}{288\Omega}$ (p. 200) were recomputed and agree; the transfer constants in Proposition 8.19 are sharper than printed (Ψ-displacement $qK\varepsilon/\alpha$ instead of $2qK\varepsilon/\alpha$ in the max norm).

Remark 8.21 (Dictionary with Pöschel's normalisation). With $\varphi = 2\pi\theta$, $I_P = I/2\pi$ (Remark 5.11): $s_P = 2\pi s$, $r_P = r/2\pi$, $\omega_P = 2\pi\omega$, $m_P = 4\pi^2 m$, $\eta, \varepsilon, \zeta, \delta_P = \delta$ unchanged up to the Euclidean/max conventions. Then $s_P r_P = sr$, $s_P m_P r_P^2/\omega_P = s\, m\, r^2/\omega$, so T_*, T_{nr} and T have the printed form; and $\partial_\varphi f_* = \partial_\theta f_*/2\pi$ makes $|\partial_\theta f_*| \leq \eta/(es)$ the same estimate as his $|\partial_\theta f_*| \leq \eta/(es_P)$.

Remark 8.22 (Consumers). Part 9 (Theorem 1 of Pöschel) takes $q = 9$, $p = \frac{9}{8}$ ($\frac{p}{q} = \frac{1}{8}$, $q \geq 3$), Λ a maximal K-lattice, D a block of Part 8 (with $K = K_a \geq \sqrt{n}K_g$, review N06), and uses Proposition 8.11 for $\Lambda = \{0\}$ and Proposition 8.19 otherwise. With Part 1: $M = nM_{\mathbb{C}}$ (Lemma 2.5), $\ell, m = \mu$ from Lemma 2.7(b); the Hessian bound on $V_r D$ requires $V_r D \subset B(0,R)_s$ (Lemma 2.2(c)). Part 10 connects the stopped form with existence through Corollary 8.9 (take $F = X_{H_{h,f}}$, which equals $X_{H^{\mathbb{R}}}$ on W_r when $U_r D \subset \mathcal{B}_s(R)$) and Theorem 4.34. Track I takes Lemma 8.14 and Lemmas 8.16, 8.17, 8.18, which use no property of h, g, f_* beyond the stated ones.

9. Geometry of resonances (shared by tracks N and P)

This part is Section 4 of Pöschel 1993 (pp. 200–203): the volume $|\Lambda|$ of an integer lattice, maximal K-lattices, resonance zones and blocks, the covering $\mathbb{R}^n = B_\Theta \cup B_1^* \cup \cdots \cup B_n^*$, the Geometric Lemma and the Covering Lemma. It is pure linear algebra and lattice geometry: no Hamiltonian, no regularity and no holomorphy enters, the dimension $n \geq 1$ and the sublattices of $\mathbb{Z}^n$ are arbitrary, and the frequency map is only a map $\omega : P \to \mathbb{R}^n$. Track P may take every statement below as it stands.

Conventions. Throughout this part $\|\cdot\|$ is the *Euclidean* norm on $\mathbb{R}^n$ ($= |\cdot|_2$ of Part 2), $x \cdot y$ the dot product, $|k|_1 = \sum|k_i|$ and $|x|_\infty = \max|x_i|$; matrices are real, $A^\top$ is the transpose. The geometry needs orthogonal projections, hence the Euclidean norm; Part 2, Lemma 2.1 relates it to the max norm and to $|k|_1$. Since a lattice cut-off appears in two roles, we write K (or K_g) for the *geometric* cut-off, a bound on $\|k\|$, and K_a for the *analytic* cut-off of Parts 5–8, a bound on $|k|_1$, with $Z_{K_a} = \{k \in \mathbb{Z}^n : |k|_1 \le K_a\}$ (Part 7). The geometric ball is $\mathcal{E}_K = \{k \in \mathbb{Z}^n : \|k\| \le K\}$ (a finite set). The compatibility of the two cut-offs is Corollary 9.16 (finding N06 of the review of Part 0).

Remark 9.1 (Direction of the cut-off compatibility: correction to Parts 7 and 8). Parts 7 and 8 (remarks on cut-offs and "consumers") state that the analytic cut-off must satisfy $K_a \ge \sqrt{n}K_g$. This is the wrong way round for what Part 8 needs. The nonresonance *conclusion* of the Geometric Lemma holds on the Euclidean ball $\mathcal{E}_{K_g}$, and Part 7 needs it on the ℓ^1 ball Z_{K_a}, which is contained in $\mathcal{E}_{K_a}$ (not in $\mathcal{E}_{K_g}$ unless $K_a \le K_g$). So one needs $K_a \le K_g$. The factor $\sqrt{n}$ enters in the other place: the radius condition $r \le \alpha/(p\sqrt{n}\,MK_a)$ of Part 8 (Proposition 8.19(c)) is implied by the covering radius $r_\Lambda = \alpha_\Lambda/(pMK_g)$ exactly when $K_g \ge \sqrt{n}K_a$. Hence the correct compatibility is $K_g \ge \sqrt{n}\,K_a$ (Corollary 9.16), and Part 9 should take, e.g., $K_g = \sqrt{n}K_a$ (with $K_a \ge 1/\sqrt{n}$ so that $K_g \ge 1$). With $K_a \ge \sqrt{n}K_g > K_g$ the claim "$Z_{K_a} \setminus \Lambda$ is nonresonant on the blocks" would not follow (take $k = \lfloor K_a \rfloor e_1$).

9.1. **External facts.**

(E1) *Subgroups of $\mathbb{Z}^n$.* Every subgroup of $\mathbb{Z}^n$ is a free $\mathbb{Z}$-module of rank $d \le n$ (Mathlib `Submodule.basisOfPid`, applied to the standard basis `Pi.basisFun`; `LinearAlgebra/FreeModule/PID.lean`). Exists.

(E2) *Gram matrices.* If $A \in \mathbb{R}^{n\times d}$ has injective $x \mapsto Ax$ then $A^\top A$ is positive definite and $\det(A^\top A) > 0$ (Mathlib `Matrix.PosDef.conjTranspose_mul_self` and `Matrix.PosDef.det_pos`; `LinearAlgebra/Matrix/PosDef.lean`); $\det(MN) = \det M \det N$, $\det M^\top = \det M$ (`Matrix.det_mul`, `Matrix.det_transpose`). Exists.

(E3) *Schur complement.* For $G \in \mathbb{R}^{d\times d}$ invertible, $\det\left(\begin{smallmatrix} G & b \\ c^\top & \gamma \end{smallmatrix}\right) = \det G\,(\gamma - c^\top G^{-1} b)$ (Mathlib `Matrix.det_fromBlocks`$_{11}$; `LinearAlgebra/Matrix/SchurComplement.lean`). Exists.

(E4) *Unimodular matrices.* A matrix $U \in \mathbb{Z}^{d\times d}$ is invertible over $\mathbb{Z}$ iff $\det U \in \{\pm 1\}$ (`Matrix.isUnit_iff_isUnit_det`, `Int.isUnit_iff`); the change-of-basis matrix between two bases of a free module is invertible (`Basis.invertibleToMatrix`). A nonzero integer has modulus ≥ 1. Exists.

(E5) *Independence over $\mathbb{Q}$ and $\mathbb{R}$.* Columns of an integer matrix that are $\mathbb{Z}$-linearly independent are $\mathbb{Q}$-independent (`LinearIndependent.iff_fractionRing`, `RingTheory/Localization/Module.lean`); $\mathbb{Q}$-independence of the d columns of a rational matrix is the nonvanishing of one $d \times d$ minor (`Matrix.rank_eq_finrank_span_cols` and rank by minors), which does not depend on the field, so they are $\mathbb{R}$-independent. Exists (the second half as a short lemma over the field extension).

(E6) *Elementary facts.* Cauchy–Schwarz $|x \cdot y| \le \|x\|\|y\|$; a finite nonempty partially ordered set has a maximal element; finite sets of vectors in $\mathbb{Z}^n$ (the ball $\mathcal{E}_K$); a finite spanning set of a vector space contains a basis. Exists.

Part 2 (Lemma 2.1) is used for the norm comparisons; no track-K result is used.

9.2. **Lattices, dimension, volume.**

Definition 9.2 (Lattices)**.** A *lattice* is a subgroup $\Lambda \le \mathbb{Z}^n$. Its *span* is $W_\Lambda = \operatorname{span}_{\mathbb{R}} \Lambda \subset \mathbb{R}^n$ and its *dimension* is $d = \dim \Lambda := \dim_{\mathbb{R}} W_\Lambda$. The trivial lattice is $\Theta = \{0\}$ ($d = 0$). The *resonance space* of Λ is $R_\Lambda = \{\omega \in \mathbb{R}^n : k \cdot \omega = 0\ \forall k \in \Lambda\} = W_\Lambda^\perp$, of dimension $n - d$.
Source: Pos93 pp. 200–201.

Lemma 9.3 (Bases and unimodular change)**.** *Let $\Lambda \le \mathbb{Z}^n$.*

(a) Λ *has a* $\mathbb{Z}$*-basis* $a_1, \dots, a_d$ *with* $d \le n$*; every* $\mathbb{Z}$*-basis is* $\mathbb{R}$*-linearly independent, hence* $\dim_{\mathbb{R}} W_\Lambda = d = \operatorname{rank}_{\mathbb{Z}} \Lambda$*. For a basis put* $A = (a_1 | \dots | a_d) \in \mathbb{Z}^{n\times d}$ *("basis matrix").*
(b) If $\tilde{A} = AU$ *is another basis matrix then* $U \in \mathbb{Z}^{d\times d}$ *and* $\det U = \pm 1$*; conversely* AU *is a basis matrix if* $U \in \mathbb{Z}^{d\times d}$*,* $\det U = \pm 1$*.*
(c) If $\Gamma \le \Lambda$ *with* $\dim \Gamma = \dim \Lambda = d$ *and basis matrices* $\tilde{A}$ *of* Γ *and* A *of* Λ*, then* $\tilde{A} = AN$ *with* $N \in \mathbb{Z}^{d\times d}$ *and* $\det N \neq 0$*.*

Source: Pos93 p. 201 ("there is a unimodular $d \times d$*-matrix* U *such that* $\tilde{A} = AU$*").*

Proof. **Step S9.1: Basis.** (a): (E1) gives a $\mathbb{Z}$-basis with $d \le n$ elements; (E5) gives $\mathbb{R}$-independence.

Step S9.2: Unimodular. (b): each column of $\tilde{A}$ lies in Λ, so $\tilde{A} = AU$ with an integer matrix U (coordinates in the basis A) and likewise $A = \tilde{A}\tilde{U}$; then $A = AU\tilde{U}$ gives $U\tilde{U} = 1$ by injectivity of A ((E2)), so $\det U \det \tilde{U} = 1$ with both integers, i.e. $\det U = \pm 1$ ((E4)). Conversely AU has columns in Λ, and $U^{-1} \in \mathbb{Z}^{d\times d}$ shows they generate Λ and are independent.

Step S9.3: Sublattice. (c): $\tilde{A} = AN$ as in (b). If $Nx = 0$ then $\tilde{A}x = ANx = 0$, so $x = 0$ ($\tilde{A}$ injective by (a)); thus N is injective, $\det N \neq 0$. □

Definition 9.4 (Volume of a lattice)**.** For $\Lambda \neq \Theta$ with basis matrix A put $|\Lambda| := \sqrt{\det(A^\top A)}$; $|\Theta| := 1$. (The radicand is > 0 by (E2) and Lemma 9.3(a); geometrically $|\Lambda|$ is the d-dimensional volume of the parallelepiped spanned by a basis.)
Source: Pos93 p. 200–201.

Lemma 9.5 (Orthogonal projection onto a span)**.** *Let* $A \in \mathbb{R}^{n\times d}$ *(*$d \ge 1$*) have independent columns,* W *their span,* $G = A^\top A$ *(invertible, (E2)) and* $\Pi_W = AG^{-1}A^\top$*,* $Q_W = 1 - \Pi_W$*. Then:*

(a) $\Pi_W^\top = \Pi_W = \Pi_W^2$*,* $\Pi_W v \in W$ *for all* v*,* $\Pi_W v = v$ *for* $v \in W$*, and* $Q_W v \perp W$*. Hence* $\|v\|^2 = \|\Pi_W v\|^2 + \|Q_W v\|^2$*,* $\|\Pi_W v\| \le \|v\|$*,* $\|Q_W v\| \le \|v\|$*. For* $d = 0$ *put* $\Pi_{\{0\}} = 0$*.*
(b) $\Pi_W v$ *is the unique* $w \in W$ *with* $v - w \perp W$*.*
(c) $\operatorname{dist}_2(\omega, W^\perp) = \|\Pi_W \omega\|$ *for all* $\omega \in \mathbb{R}^n$ *(the minimum is attained at* $Q_W\omega$*).*
(d) If $W \subset W'$ *are two such spans, then* $\Pi_{W'}\Pi_W = \Pi_W \Pi_{W'} = \Pi_W$*.*

Consequently $R_\Lambda = W_\Lambda^\perp$ *and* $\operatorname{dist}_2(\omega, R_\Lambda) = \|\Pi_{W_\Lambda}\omega\|$*; write* $\Pi_\Lambda = \Pi_{W_\Lambda}$*,* $Q_\Lambda = Q_{W_\Lambda}$*.*
Source: Standard (Pos93 p. 202 uses these properties silently).

Proof. **Step S9.4: Algebra.** $\Pi_W^\top = AG^{-\top}A^\top = \Pi_W$ (G symmetric), $\Pi_W^2 = AG^{-1}(A^\top A)G^{-1}A^\top = \Pi_W$, $\Pi_W A = A$, so $\Pi_W v = v$ on W; $A^\top Q_W = A^\top - GG^{-1}A^\top = 0$ gives $Q_W v \perp W$. Pythagoras follows from $\Pi_W v \cdot Q_W v = v \cdot \Pi_W Q_W v = 0$.

Step S9.5: Uniqueness, distance. (b): if $w, w' \in W$ both have $v - w, v - w' \perp W$ then $w - w' \in W \cap W^\perp = 0$. (c): for $u \in W^\perp$, $\|\omega - u\|^2 = \|\Pi_W\omega\|^2 + \|Q_W\omega - u\|^2$ (as $\Pi_W\omega \in W \perp Q_W\omega - u$), with equality for $u = Q_W\omega \in W^\perp$.

Step S9.6: Nested spans. (d): $\Pi_W v \in W \subset W'$ so $\Pi_{W'}\Pi_W v = \Pi_W v$; transposing gives $\Pi_W \Pi_{W'} = \Pi_W$. □

Lemma 9.6 (Volume)**.** *Let* $n \ge 1$ *and let* $\Lambda, \Gamma \le \mathbb{Z}^n$ *be lattices.*

*(a) (*independence of the basis*)* $\det(A^\top A)$ *does not depend on the basis matrix* A *of* Λ*; so* $|\Lambda|$ *is well defined.*
(b) $|\Lambda|^2 = \det(A^\top A)$ *is a positive integer; hence* $|\Lambda| \ge 1$*.*
(c) If $\Gamma \le \Lambda$ *and* $\dim \Gamma = \dim \Lambda$ *then* $|\Gamma| = |\det N||\Lambda| \ge |\Lambda|$*,* N *as in Lemma 9.3(c).*
(d) $|\mathbb{Z}g| = \|g\|$ *for* $g \in \mathbb{Z}^n \setminus \{0\}$*;* $|\mathbb{Z}^n| = 1$*.*
*(e) (*adding a vector*) If* $k \in \mathbb{Z}^n \setminus W_\Lambda$*, then* $\Lambda + \mathbb{Z}k$ *has dimension* $\dim \Lambda + 1$ *and* $|\Lambda + \mathbb{Z}k| = |\Lambda|\,\|Q_\Lambda k\|$*, with* $Q_\Lambda k \neq 0$*.*
*(f) (*Hadamard*) If* $v_1, \dots, v_d \in \mathbb{Z}^n$ *are* $\mathbb{R}$*-independent and generate* Γ*, then* $|\Gamma| \le \prod_i \|v_i\|$*.*

Source: Pos93 p. 201 (a), p. 202 (the inequalities "the volume of Λ_+ is not bigger than the volume of Λ times the length of the component of k orthogonal to Λ"; (c),(e) make this precise).

Proof. **Step S9.7: (a),(b).** If $\tilde{A} = AU$, $\det \tilde{A}^\top \tilde{A} = \det U^\top \det(A^\top A) \det U = (\det U)^2 \det A^\top A = \det A^\top A$ by Lemma 9.3(b) and (E2). The matrix $A^\top A$ is an integer matrix, so its determinant is an integer, > 0 by (E2). For $\Lambda = \Theta$, $|\Theta| = 1$.

Step S9.8: (c). $\tilde{A}^\top \tilde{A} = N^\top (A^\top A) N$, so $|\Gamma|^2 = (\det N)^2 |\Lambda|^2$ with $\det N \in \mathbb{Z} \setminus \{0\}$ (Lemma 9.3(c)), $|\det N| \geq 1$. (For $\Lambda = \Theta$ there is nothing to prove.)

Step S9.9: (d). $A = g$ gives $\sqrt{g^\top g} = \|g\|$; $A = 1$ gives 1.

Step S9.10: (e). By Lemma 9.5(a), $k \notin W_\Lambda$ means $Q_\Lambda k \neq 0$ (else $k = \Pi_\Lambda k \in W_\Lambda$). With A a basis matrix of Λ the columns of $A' = (A|k)$ generate $\Lambda + \mathbb{Z}k$ and are $\mathbb{R}$-independent, hence form a basis, $\dim = d + 1$. Then $A'^\top A' = \left(\begin{smallmatrix} G & A^\top k \\ k^\top A & k^\top k \end{smallmatrix}\right)$, $G = A^\top A$, and (E3) gives $\det A'^\top A' = \det G\,(k^\top k - k^\top A G^{-1} A^\top k) = \det G\,(\|k\|^2 - k \cdot \Pi_\Lambda k) = \det G\,\|Q_\Lambda k\|^2$, since $k \cdot \Pi_\Lambda k = \|\Pi_\Lambda k\|^2$ (Lemma 9.5(a)) and Pythagoras. For $\Lambda = \Theta$ this reads $|\mathbb{Z}k| = \|k\|$ by (d).

Step S9.11: (f). Induction on d with (e): for $\Gamma_j = \langle v_1, \dots, v_j \rangle$, $|\Gamma_j| = |\Gamma_{j-1}| \|Q_{\Gamma_{j-1}} v_j\| \leq |\Gamma_{j-1}| \|v_j\|$ (Lemma 9.5(a)), starting from $|\Theta| = 1$. □

Remark 9.7 (Totality). Every square root and division in this part is guarded: $|\Lambda| > 0$ (Lemma 9.6(b)); G^{-1} exists by (E2); $\|Q_\Lambda k\| > 0$ when divided by (Lemma 9.6(e)); the radicands $\|\Pi_+ \omega\|^2 - \|\Pi \omega\|^2$ and $A^2 - 1$ in Lemma 9.13 are shown to be ≥ 0 (indeed > 0) before the square root is taken. Nothing is evaluated outside its domain.

9.3. Maximal K-lattices, zones, blocks, covering of frequency space.

Definition 9.8 (K-lattices)**.** Let $K \geq 0$. A lattice Λ is a *K-lattice* if it is generated (as a group) by $\Lambda \cap \mathcal{E}_K$. It is a *maximal K-lattice* if, moreover, every K-lattice $\Lambda' \supset \Lambda$ with $\dim \Lambda' = \dim \Lambda$ equals Λ. The family of maximal K-lattices is $\mathbb{M}_K$; $\mathbb{M}_K^d$ denotes those of dimension d.

Source: Pos93 p. 200 ("lattices generated by vectors in $\mathbb{Z}_K^n$ and not properly contained in any other lattice of the same dimension"). The Euclidean ball is the one used on p. 202 ("it suffices to assume $\|k\| \leq K$"). If one reads "any other lattice" as an arbitrary subgroup of $\mathbb{Z}^n$ the maximal ones are the saturated ones $W \cap \mathbb{Z}^n$; these belong to our $\mathbb{M}_K$ (Lemma 9.9(a)), and all statements below hold for the larger family $\mathbb{M}_K$.

Lemma 9.9 (Maximal K-lattices)**.** *Let $K \geq 0$ and Λ a K-lattice.*

- *(a) $\Lambda \in \mathbb{M}_K$ iff $\mathcal{E}_K \cap W_\Lambda \subset \Lambda$ (i.e. every integer vector of norm $\leq K$ in the real span of Λ lies in Λ). In particular $\Theta \in \mathbb{M}_K$, and a saturated K-lattice is maximal.*
- *(b) Every K-lattice Γ is contained in some $\Lambda \in \mathbb{M}_K$ with $\dim \Lambda = \dim \Gamma$. In particular, for $k \in \mathcal{E}_K \setminus \{0\}$ there is $\Lambda \in \mathbb{M}_K^1$ with $k \in \Lambda$.*
- *(c) If $\Lambda \in \mathbb{M}_K$ and $k \in \mathcal{E}_K \setminus \Lambda$, then $k \notin W_\Lambda$, so $\dim \Lambda < n$ and $\Lambda + \mathbb{Z}k$ is a K-lattice of dimension $\dim \Lambda + 1$.*
- *(d) If $\Lambda \in \mathbb{M}_K$ and $\dim \Lambda = n$ then $\mathcal{E}_K \subset \Lambda$; if $K \geq 1$ then $\Lambda = \mathbb{Z}^n$, and $\mathbb{Z}^n \in \mathbb{M}_K$. If $K < 1$ then $\mathbb{M}_K = \{\Theta\}$.*
- *(e) If $\Lambda \in \mathbb{M}_K$ has dimension d then $1 \leq |\Lambda| \leq K^d$.*

Source: Pos93 pp. 200–202 (the properties used in the proofs of the Geometric Lemma and of the optimality remark); the verification is ours.

Proof. **Step S9.12: Characterisation.** ($\Rightarrow$) Let $k \in \mathcal{E}_K \cap W_\Lambda$. Then $\Lambda + \mathbb{Z}k$ is generated by $(\Lambda \cap \mathcal{E}_K) \cup \{k\} \subset \mathcal{E}_K$, so it is a K-lattice; $k \in W_\Lambda$ gives $W_{\Lambda + \mathbb{Z}k} = W_\Lambda$, same dimension; maximality gives $\Lambda + \mathbb{Z}k = \Lambda$, $k \in \Lambda$. ($\Leftarrow$) Let $\Lambda' \supset \Lambda$ be a K-lattice of the same dimension. Then $W_\Lambda \subset W_{\Lambda'}$ and the dimensions agree, so $W_{\Lambda'} = W_\Lambda$; Λ' is generated by $\Lambda' \cap \mathcal{E}_K \subset \mathcal{E}_K \cap W_\Lambda \subset \Lambda$, so $\Lambda' \subset \Lambda$. Saturated: $\Lambda = W_\Lambda \cap \mathbb{Z}^n \supset \mathcal{E}_K \cap W_\Lambda$. Θ: $W = \{0\}$.

Step S9.13: Existence. (b): the set $\mathcal{S}$ of K-lattices $\Lambda \supset \Gamma$ with $\dim\Lambda = \dim\Gamma$ is finite (each is generated by a subset of the finite set $\mathcal{E}_K$), nonempty and ordered by inclusion; take an inclusion-maximal element Λ. Any K-lattice $\Lambda' \supset \Lambda$ of the same dimension is in $\mathcal{S}$, hence equals Λ. For $k \neq 0$ apply this to $\Gamma = \mathbb{Z}k$ (K-lattice of dimension 1).

Step S9.14: (c). By (a), $k \in \mathcal{E}_K \cap W_\Lambda$ would give $k \in \Lambda$; so $k \notin W_\Lambda$, and $\dim\Lambda + 1 = \dim(W_\Lambda + \mathbb{R}k) \leq n$. $\Lambda + \mathbb{Z}k$ is a K-lattice as in Step S9.12, of dimension $\dim\Lambda + 1$.

Step S9.15: (d). $W_\Lambda = \mathbb{R}^n$, so $\mathcal{E}_K \subset \Lambda$ by (a). If $K \geq 1$ then $e_i \in \mathcal{E}_K$, so $\Lambda = \mathbb{Z}^n$; and $\mathbb{Z}^n$ is generated by $\mathcal{E}_K \cap \mathbb{Z}^n$ and is saturated, so it is in $\mathbb{M}_K$ by (a). If $K < 1$: $\mathcal{E}_K = \{0\}$, so the only K-lattice is Θ.

Step S9.16: (e). The lower bound is Lemma 9.6(b). $\Lambda \cap \mathcal{E}_K$ spans W_Λ ($\dim d$), so it contains d independent vectors $v_1, \dots, v_d$ ((E6)); they generate $\Gamma \leq \Lambda$ with $\dim\Gamma = d$. By Lemma 9.6(c),(f): $|\Lambda| \leq |\Gamma| \leq \prod\|v_i\| \leq K^d$. For $d = 0$ both sides are 1. □

Definition 9.10 (Zones, blocks)**.** Fix $K \geq 0$ and positive numbers $\lambda_1, \dots, \lambda_n$. For $\Lambda \in \mathbb{M}_K$ of dimension $d \geq 1$ put

$$\delta_\Lambda = \frac{\lambda_d}{|\Lambda|}, \qquad Z_\Lambda = \{\omega \in \mathbb{R}^n : \operatorname{dist}_2(\omega, R_\Lambda) < \delta_\Lambda\} = \{\omega : \|\Pi_\Lambda \omega\| < \delta_\Lambda\}$$

(Lemma 9.5(c)); $Z_\Theta = \mathbb{R}^n$. Put $Z_d^* = \bigcup_{\Lambda \in \mathbb{M}_K^d} Z_\Lambda$ for $1 \leq d \leq n$ and $Z_{n+1}^* = \emptyset$, and

$$B_\Lambda = Z_\Lambda \setminus Z_{d+1}^* \quad (\Lambda \in \mathbb{M}_K^d,\ d \geq 1), \qquad B_\Theta = \mathbb{R}^n \setminus Z_1^*, \qquad B_d^* = \bigcup_{\Lambda \in \mathbb{M}_K^d} B_\Lambda \ (1 \leq d \leq n).$$

(The zones depend on K and on $\lambda = (\lambda_1, \dots, \lambda_n)$; Z_d^* is empty if $\mathbb{M}_K^d$ is.)
Source: Pos93 p. 201.

Proposition 9.11 (Covering of frequency space)**.** *For every $K \geq 0$ and $\lambda_1, \dots, \lambda_n > 0$: $\mathbb{R}^n = B_\Theta \cup B_1^* \cup \dots \cup B_n^*$, i.e. every $\omega \in \mathbb{R}^n$ lies in B_Λ for some $\Lambda \in \mathbb{M}_K$ (a finite family).*
Source: Pos93 p. 201 (the chain $\mathbb{R}^n = Z_\Theta = B_\Theta \cup Z_1^* = B_\Theta \cup B_1^* \cup Z_2^* = \dots$).

Proof. **Step S9.17: Finite family.** $\mathbb{M}_K$ consists of subgroups generated by subsets of the finite set $\mathcal{E}_K$, so it is finite.

Step S9.18: Top dimension. Fix ω. Let $\mathcal{D} = \{d \in \{1, \dots, n\} : \omega \in Z_d^*\}$. If $\mathcal{D} = \emptyset$ then $\omega \notin Z_1^*$, so $\omega \in B_\Theta$. Otherwise let $d = \max\mathcal{D}$ and choose $\Lambda \in \mathbb{M}_K^d$ with $\omega \in Z_\Lambda$. Since $d+1 \notin \mathcal{D}$ (or $d = n$, $Z_{n+1}^* = \emptyset$), $\omega \notin Z_{d+1}^*$, so $\omega \in Z_\Lambda \setminus Z_{d+1}^* = B_\Lambda \subset B_d^*$. □

9.4. **The Geometric Lemma.**

Lemma 9.12 (A numerical inequality)**.** *If $A \geq \sqrt{2}$ then $A^2 - 1 \geq 1 > 0$ and $\sqrt{A^2-1} - 1 \geq A - \sqrt{2}$.*
Source: Pos93 p. 202 ("since $\sqrt{A^2-1} - 1 \geq A - \sqrt{2} \geq E$").

Proof. $A^2 - 1 \geq 1$ is clear. Put $c = \sqrt{2} - 1 \in (0, 1)$; $A - c \geq 1 > 0$. Then $(A-c)^2 \leq A^2 - 1 \iff -2cA + c^2 \leq -1 \iff A \geq \frac{1+c^2}{2c} = \frac{4-2\sqrt{2}}{2(\sqrt{2}-1)} = \sqrt{2}$ (using $c^2 = 3 - 2\sqrt{2}$). Both sides being nonnegative, $\sqrt{A^2-1} \geq A - c$. □

Lemma 9.13 (Geometric Lemma)**.** *Let $n \geq 1$, $K > 0$, $E > 0$, $A \geq E + \sqrt{2}$, $\lambda_1, \dots, \lambda_n > 0$ with*

$$\frac{\lambda_{d+1}}{\lambda_d} \geq AK \qquad (1 \leq d < n),$$

and let $\mathbb{M}_K$, δ_Λ, B_Λ be as in Definition 9.10.

(a) (B_Θ) For $\omega \in B_\Theta$ and $k \in \mathbb{Z}^n$ with $0 < \|k\| \leq K$: $|k \cdot \omega| \geq \lambda_1$.

(b) (B_Λ) Let $\Lambda \in \mathbb{M}_K$ with $d = \dim \Lambda \geq 1$, $\omega \in B_\Lambda$ and $k \in \mathbb{Z}^n \setminus \Lambda$ with $\|k\| \leq K$. Then $|k \cdot \omega| \geq E\,K\,\delta_\Lambda$. Also $\operatorname{dist}_2(\omega, R_\Lambda) < \delta_\Lambda$. (For $d = n$ there is no such k, Lemma 9.9(d).)

So B_Θ is completely K-nonresonant with constant $\alpha_\Theta = \lambda_1$, and B_Λ is K-nonresonant modulo Λ with constant $\alpha_\Lambda \geq EK\delta_\Lambda$, in the Euclidean cut-off $\|k\| \leq K$.

Source: Pos93 pp. 201–202 (Geometric Lemma and its proof). Deviation: the proof is organised with the explicit formulas of Lemma 9.14, which replace the chain "$k \cdot \omega = Qk \cdot QP_+\omega + Pk \cdot P\omega \ldots$" (same content, with all projections made explicit); all inequalities have been recomputed and agree.

Lemma 9.14 (The projection formula)**.** *Let $W \subset \mathbb{R}^n$ be the span of a basis matrix of a lattice Λ (or $W = \{0\}$), $k \in \mathbb{R}^n \setminus W$, $W_+ = W + \mathbb{R}k$, $\Pi = \Pi_W$, $Q = Q_W$, $\Pi_+ = \Pi_{W_+}$. Then $Qk \neq 0$ and for every $\omega \in \mathbb{R}^n$*

$$\Pi_+\omega - \Pi\omega = \frac{Qk \cdot \omega}{\|Qk\|^2}\, Qk, \qquad k \cdot \omega = \Pi k \cdot \Pi\omega + Qk \cdot \omega, \qquad |Qk \cdot \omega| = \|Qk\|\,\|\Pi_+\omega - \Pi\omega\|,$$

and $\|\Pi_+\omega\|^2 = \|\Pi\omega\|^2 + \|\Pi_+\omega - \Pi\omega\|^2$.

Proof. $Qk \neq 0$ as in Lemma 9.6(e). Put $c = Qk \cdot \omega/\|Qk\|^2$ and $v = \Pi\omega + c\,Qk$. Now $Qk = k - \Pi k \in W_+$, so $v \in W_+$; W_+ is the span of $d+1$ independent vectors (a basis of W and k), so Lemma 9.5 applies to it. Further $\omega - v = Q\omega - c\,Qk \perp W$ (as $Q\omega, Qk \perp W$) and $(\omega - v) \cdot Qk = Q\omega \cdot Qk - c\|Qk\|^2 = \omega \cdot Qk - c\|Qk\|^2 = 0$ (since $Q\omega \cdot Qk = \omega \cdot Q^2k$, $Q^2 = Q$). As $W_+ = W \oplus \mathbb{R}Qk$, $\omega - v \perp W_+$, and Lemma 9.5(b) gives $\Pi_+\omega = v$. The first formula follows. Next, $k = \Pi k + Qk$ and $\Pi k \cdot \omega = \Pi k \cdot \Pi\omega$ (symmetry of Π, $\Pi^2 = \Pi$) give the second. The third: $\|\Pi_+\omega - \Pi\omega\| = |c|\|Qk\| = |Qk \cdot \omega|/\|Qk\|$. The last: $\Pi\omega \in W \perp Qk$ (Pythagoras). □

Proof of Lemma 9.13. **Step S9.19: (a): reduction to dimension one.** Let $\omega \in B_\Theta$ and $0 < \|k\| \leq K$. By Lemma 9.9(b) there is $\Lambda_k \in \mathbb{M}^1_K$ containing k; $\Lambda_k = \mathbb{Z}g$ with $g \in \mathbb{Z}^n \setminus \{0\}$ (a rank-one subgroup of $\mathbb{Z}^n$, Lemma 9.3(a)), $k = jg$ with $j \in \mathbb{Z} \setminus \{0\}$, so $\|g\| \leq \|k\|$ and $|\Lambda_k| = \|g\|$ (Lemma 9.6(d)). Hence $\delta_{\Lambda_k} = \lambda_1/\|g\| \geq \lambda_1/\|k\|$.

Step S9.20: (a): conclusion. As $\omega \notin Z_1^* \supset Z_{\Lambda_k}$: $\operatorname{dist}_2(\omega, R_{\Lambda_k}) \geq \delta_{\Lambda_k}$. Here $R_{\Lambda_k} = g^\perp = k^\perp$ and $\operatorname{dist}_2(\omega, k^\perp) = |k \cdot \omega|/\|k\|$ (Lemma 9.5(c), $\|\Pi_{\mathbb{R}k}\omega\| = |k \cdot \omega|/\|k\|$). So $|k \cdot \omega| \geq \|k\|\,\delta_{\Lambda_k} \geq \lambda_1$.

Step S9.21: (b): setting. Let Λ, d, ω, k be as in (b); write $V = |\Lambda|$, $\lambda = \lambda_d$, $\delta = \delta_\Lambda = \lambda/V$, $\Pi = \Pi_\Lambda$, $Q = Q_\Lambda$. The last assertion is $\omega \in Z_\Lambda$. By Lemma 9.9(c), $k \notin W_\Lambda$, $d < n$, and $\Lambda + \mathbb{Z}k$ is a K-lattice of dimension $d+1$; by Lemma 9.9(b) there is $\Lambda_+ \in \mathbb{M}^{d+1}_K$ with $\Lambda_+ \supset \Lambda + \mathbb{Z}k$. Then $W_{\Lambda_+} \supset W_\Lambda + \mathbb{R}k =: W_+$ and both have dimension $d+1$, so $W_{\Lambda_+} = W_+$ and $\Pi_{\Lambda_+} = \Pi_+$ of Lemma 9.14. Put $V_+ = |\Lambda_+|$, $\lambda_+ = \lambda_{d+1}$.

Step S9.22: Volumes. By Lemma 9.6(c),(e): $V_+ \leq |\Lambda + \mathbb{Z}k| = V\|Qk\|$, i.e. $\|Qk\| \geq V_+/V$; and $\|Qk\| \leq \|k\| \leq K$ gives $V_+ \leq KV$.

Step S9.23: The two projections. As $\omega \in B_\Lambda \subset Z_\Lambda$: $\|\Pi\omega\| < \delta = \lambda/V \leq K\lambda/V_+$ (Step S9.22). As $\omega \notin Z^*_{d+1} \supset Z_{\Lambda_+}$: $\|\Pi_+\omega\| \geq \delta_{\Lambda_+} = \lambda_+/V_+ \geq AK\lambda/V_+$ (hypothesis on the λ's). With $A^2 - 1 > 0$ (Lemma 9.12, as $A \geq E + \sqrt{2} > \sqrt{2}$):

$$\|\Pi_+\omega - \Pi\omega\|^2 = \|\Pi_+\omega\|^2 - \|\Pi\omega\|^2 \geq (A^2 - 1)\Big(\frac{K\lambda}{V_+}\Big)^2 \quad \text{(Lemma 9.14)}.$$

Hence the radicand is positive and $\|\Pi_+\omega - \Pi\omega\| \geq \frac{K\lambda}{V_+}\sqrt{A^2 - 1}$.

Step S9.24: The divisor. By Lemma 9.14: $k \cdot \omega = \Pi k \cdot \Pi\omega + Qk \cdot \omega$ and $|Qk \cdot \omega| = \|Qk\|\|\Pi_+\omega - \Pi\omega\| \geq \frac{V_+}{V} \cdot \frac{K\lambda}{V_+}\sqrt{A^2 - 1} = K\delta\sqrt{A^2 - 1}$; while $|\Pi k \cdot \Pi\omega| \leq \|k\|\|\Pi\omega\| \leq K\delta$. So

$$|k \cdot \omega| \geq K\delta\big(\sqrt{A^2 - 1} - 1\big) \geq K\delta(A - \sqrt{2}) \geq EK\delta$$

by Lemma 9.12 and $A - \sqrt{2} \geq E$. □

9.5. **The Covering Lemma in action space.**

Lemma 9.15 (Covering Lemma)**.** *Let $n \geq 1$; $p, b, r_0, M, m > 0$; $K \geq 1$; $A \geq \frac{pM}{bm} + \sqrt{2}$. Let P be any set and $\omega : P \to \mathbb{R}^n$ any map (no regularity). Put*

$$\lambda_d = \frac{b\,m\,r_0}{A^n K^n} A^d K^d \ (1 \leq d \leq n), \qquad r_\Lambda = \frac{r_0}{A^n K^n} \frac{A^d K^d}{|\Lambda|} \ (d = \dim \Lambda), \qquad \alpha_\Lambda = pMK\, r_\Lambda, \qquad \delta_\Lambda = b\,m\,r_\Lambda$$

for $\Lambda \in \mathbb{M}_K$ (for $\Lambda = \Theta$: $r_\Theta = r_0/(AK)^n$, $\alpha_\Theta = pMK r_\Theta$), with the blocks B_Λ of Definition 9.10 for these K, λ, and $D_\Lambda = \{I \in P : \omega(I) \in B_\Lambda\}$. Then:

- *(a) (*covering*) $P = \bigcup_{\Lambda \in \mathbb{M}_K} D_\Lambda$, a finite union, and the D_Λ may be discarded when empty.*
- *(b) (*consistency*) $\delta_\Lambda = \lambda_d/|\Lambda|$, i.e. the number $\delta_\Lambda = bmr_\Lambda$ is the radius of the zone Z_Λ of Definition 9.10.*
- *(c) (*nonresonance*) If $I \in D_\Theta$ and $k \in \mathbb{Z}^n$, $0 < \|k\| \leq K$: $|k \cdot \omega(I)| \geq \lambda_1 \geq \alpha_\Theta$. If $\Lambda \in \mathbb{M}_K$, $\dim \Lambda \geq 1$, $I \in D_\Lambda$, $k \in \mathbb{Z}^n \setminus \Lambda$, $\|k\| \leq K$: $|k \cdot \omega(I)| \geq \alpha_\Lambda$, and $\mathrm{dist}_2(\omega(I), R_\Lambda) < \delta_\Lambda = bmr_\Lambda$.*
- *(d) (*radii*) $r_\Theta = \frac{r_0}{(AK)^n} \leq r_\Lambda \leq \frac{r_0}{AK}$ for every $\Lambda \in \mathbb{M}_K$ with $1 \leq \dim \Lambda < n$; $r_{\mathbb{Z}^n} = r_0$ (the only lattice of dimension n). Hence $r_\Theta \leq r_\Lambda \leq r_0$ for all $\Lambda \in \mathbb{M}_K$.*

Source: Pos93 p. 203 (Covering Lemma with $\alpha_\Lambda = pMKr_\Lambda$, $\delta_\Lambda = bmr_\Lambda$) and p. 204 ($b = \frac{1}{8}$; bounds $r_\Theta \leq r_\Lambda \leq r_{\mathbb{Z}^n} = r_0$, and $r_\Lambda \leq r_0/(AK)$ otherwise). The parameters m, M enter only through $E = pM/(bm)$ and play no other role, as Pos93 notes. Pos93's pull-back "setting $D_\Lambda = \{I \in P : \omega(I) \in B_\Lambda\}$ and discarding..." is (a).

Proof. **Step S9.25: Parameters of the Geometric Lemma.** $\lambda_{d+1}/\lambda_d = AK$ $(1 \leq d < n)$, $\lambda_d > 0$. Put $E = pM/(bm) > 0$; then $A \geq E + \sqrt{2}$, and $K \geq 1 > 0$. Lemma 9.13 applies. Also $\delta_\Lambda = bmr_\Lambda = \frac{bmr_0 A^d K^d}{A^n K^n |\Lambda|} = \lambda_d/|\Lambda|$ (b), and $EK\delta_\Lambda = \frac{pM}{bm} K\, bm\, r_\Lambda = pMKr_\Lambda = \alpha_\Lambda$.

Step S9.26: Covering. Let $I \in P$. Proposition 9.11 applied to the point $\omega(I) \in \mathbb{R}^n$ gives $\Lambda \in \mathbb{M}_K$ with $\omega(I) \in B_\Lambda$, i.e. $I \in D_\Lambda$. Finiteness: Proposition 9.11.

Step S9.27: Nonresonance. For $I \in D_\Lambda$, $\omega(I) \in B_\Lambda$; apply Lemma 9.13(b), (a) to the point $\omega(I)$; the constants are $EK\delta_\Lambda = \alpha_\Lambda$ above and, for Θ, $\lambda_1 = bm\, r_0 AK/(AK)^n = bmA \cdot K\, r_\Theta \geq pMK\, r_\Theta = \alpha_\Theta$ since $bmA \geq pM$.

Step S9.28: Radii. $r_\Lambda = r_0(AK)^{d-n}/|\Lambda|$. For $d < n$: $|\Lambda| \geq 1$ and $AK \geq \sqrt{2} > 1$ give $r_\Lambda \leq r_0(AK)^{-1}$. Lower bound: $|\Lambda| \leq K^d$ (Lemma 9.9(e)) gives $r_\Lambda \geq r_0(AK)^{d-n}K^{-d} = r_0 A^d K^{-n} A^{-n} \geq r_0/(AK)^n$ as $A \geq 1$ (for $d = 0$ equality). $d = n$: $\Lambda = \mathbb{Z}^n$ (Lemma 9.9(d); $K \geq 1$), $|\mathbb{Z}^n| = 1$, $r_{\mathbb{Z}^n} = r_0$; and $r_\Lambda \leq r_0/(AK) \leq r_0$ for $1 \leq d < n$, $r_\Theta \leq r_0$. □

Corollary 9.16 (Euclidean versus ℓ^1 cut-offs; interface for Parts 7, 8 and 9)**.** *In the setting of Lemma 9.15 write $K = K_g$ for the geometric cut-off, and let $K_a \geq 0$ be an analytic cut-off.*

- *(a) (*inclusions with the dimension factor*) $Z_{K_a} \subset \mathcal{E}_{K_a} \subset Z_{\sqrt{n}K_a}$, i.e. $\|k\| \leq |k|_1 \leq \sqrt{n}\|k\|$ (Lemma 2.1(b)).*
- *(b) If $K_a \leq K_g$, then for every $\Lambda \in \mathbb{M}_{K_g}$ and $I \in D_\Lambda$ the point I is (α_Λ, K_a)-nonresonant modulo Λ in the sense of Definition 7.1(N1): $|k \cdot \omega(I)| \geq \alpha_\Lambda$ for all $k \in Z_{K_a} \setminus \Lambda$ (for $\Lambda = \Theta$, $k \in Z_{K_a} \setminus \{0\}$; the constant is $\alpha_\Theta \leq \lambda_1$). Also $\mathrm{dist}_2(\omega(I), R_\Lambda) < \delta_\Lambda$, i.e. D_Λ is δ_Λ-close to exact Λ-resonances (Definition 8.13).*
- *(c) If $K_g \geq \sqrt{n}K_a > 0$ then $r_\Lambda = \frac{\alpha_\Lambda}{pMK_g} \leq \frac{\alpha_\Lambda}{p\sqrt{n}\,MK_a}$, the radius condition of Proposition 8.19(c); and (b) holds because $\sqrt{n}K_a \geq K_a$. A choice satisfying all requirements is $K_g = \sqrt{n}K_a$ with $K_a \geq 1/\sqrt{n}$.*

Source: Ours (review N06 and Remark 9.1); Pos93 p. 202 only says "it suffices to assume $\|k\| \leq K$ rather than the stronger $|k| \leq K$".

Proof. **Step S9.29: Inclusions.** (a) is Lemma 2.1(b). (b): $k \in Z_{K_a} \setminus \Lambda$ has $\|k\| \le |k|_1 \le K_a \le K_g$, so Lemma 9.15(c) applies; closeness is the last assertion there. (c): $K_g \ge \sqrt{n}K_a$ gives $\frac{1}{pMK_g} \le \frac{1}{p\sqrt{n}MK_a}$; $\alpha_\Lambda = pMK_g r_\Lambda$ by definition. □

9.6. **Errata, deviations, consumers.**

Remark 9.17 (Suspected defects in Pos93 and deviations).

(1) *p. 200, definition of maximal K-lattices*: "lattices that are generated by vectors in $\mathbb{Z}^n_K = \{k : |k| \le K\}$ and are not properly contained in any other lattice of the same dimension". The proof of the Geometric Lemma (p. 202) uses the Euclidean ball ("it suffices to assume $\|k\| \le K$") and from maximality only the consequence "$k \in \mathbb{Z}^n$, $\|k\| \le K$, $k \notin \Lambda \Rightarrow k \notin W_\Lambda$". We use the Euclidean ball and the weaker maximality of Definition 9.8, which is exactly this consequence (Lemma 9.9(a)); the definition is ambiguous as printed ("any other lattice": any subgroup, or any K-lattice).

(2) *p. 202, case $d \ge 1$*: "the maximal K-lattice Λ_+ generated by Λ and k together has dimension $d+1 \le n$" and "the volume of Λ_+ is not bigger than the volume of Λ times the length of the component of k orthogonal to Λ": true, but each needs an argument (Lemma 9.9(b),(c) for existence and dimension, Lemma 9.6(c),(e) for the volume). The bound $\|P\omega\| \le \lambda/V \le K\lambda/V_+$ also uses $V_+ \le KV$, which is not stated (Step S9.22).

(3) *p. 202, case B_Θ*: "let Λ be the unique one-dimensional maximal lattice containing k. Then $|\Lambda| \le \|k\|$ …". Uniqueness is true (such a Λ is $\mathbb{Z}k_0$, k_0 the primitive vector on $\mathbb{R}k$, since $k_0 \in \mathcal{E}_K \cap \mathbb{R}k$) but is not argued and is not needed; we only use existence (Lemma 9.9(b)) and $|\Lambda| = \|g\| \le \|k\|$ (Step S9.19).

(4) *p. 203, "the covering lemma"* is stated with $A \ge \frac{pM}{bm} + \sqrt{2}$ and uses $\lambda_d = \frac{bmr_0}{A^n K^n} A^d K^d$: all checks (Lemma 9.15) agree, including $\alpha_\Lambda = EK\delta_\Lambda$. The claim "$r_\Theta \le r_\Lambda \le r_{\mathbb{Z}^n} = r_0$" (p. 204) needs $|\Lambda| \le K^d$ and $|\Lambda| \ge 1$ (Lemma 9.9(e)), not argued in the source.

(5) *The $\sqrt{n}$ direction* between $\|k\|$ and $|k|_1$ is Remark 9.1 (it concerns Parts 7, 8 and the PLAN, not Pos93).

(6) No misprint in the numbers of pp. 200–204 concerning the geometry was found; $\frac{p}{q}$-type constants belong to Parts 8 and 9.

Remark 9.18 (Consumers). Part 9 chooses q, p, b, $A = 11M/m \ge pM/(bm) + \sqrt{2}$ (to be checked there with $M = nM_{\mathbb{C}}$), $K = K_g = \sqrt{n}K_a$, applies Lemma 9.15 to $P = B(0,R)$ (or the relevant action ball) and $\omega = \omega_h$, and uses Corollary 9.16 for the hypotheses (a), (c) of Proposition 8.19 and of Proposition 8.11 (with $\Lambda = \Theta$). The radii in Lemma 9.15(d) are the ones used on p. 204 ($r_\Theta \le r_\Lambda \le r_0/(AK)$ for $\Lambda \ne \mathbb{Z}^n$, $r_{\mathbb{Z}^n} = r_0$). Track P takes Definitions 9.2–9.10, Lemmas 9.6, 9.9, Proposition 9.11 and Lemma 9.13 for any n, any K and any $\lambda_1, \dots, \lambda_n > 0$ as they stand.